# MATH 714: PROBLEMS

YOUR NAME HERE

Abstract. Math 714 is the graduate course on *The Arithmetic Of Elliptic Curves* taught at UMass Amherst. I taught this class in the Fall of 2026. The students read ahead, and we used the lecture time to work on the problems in this document. Each part below is one problem set (corresponding to one meeting), with its summaries, definitions, results and problems. Each problem is preceded by the definitions and results it uses. Write your solution to each item in the Solution block under it.

DRAFT — last edited November 11, 2026

**How to use this file.**

- Fill in your name above, and compile with pdfLaTeX. Run it twice, so that the references and the table of contents are up to date. The file is self-contained: it needs no other file.
- Write your solutions inside the solution blocks, and keep the problem text.
- Problems are numbered $1, 2, \ldots$ through the whole document: cite "Problem $n$" in a submission, and use `\cref` or `\ref` with labels for your own references. Definitions and results are numbered by section.

**An example.** A problem is a framed box with a title. Each item of the problem is followed by an empty solution block, as in the example below (it is not part of the problem set: it is printed as it appears in the file).

```
\begin{problem}[A toy problem]
  \label{prob:toy}
  \begin{enumerate}
    \item\label{it:toy-sum} Compute $1 + 2 + \dots + n$.
\begin{solution}
% Write your solution here.
\end{solution}
    \item Use \ref{it:toy-sum} to compute $1 + 2 + \dots + 100$.
\begin{solution}
% Write your solution here.
\end{solution}
  \end{enumerate}
\end{problem}
```

You write between `\begin{solution}` and `\end{solution}`, and these two lines must stay alone on their lines. A finished item looks like this:

```
\begin{solution}
\ans{$n(n+1)/2$.} Add the sum to itself in reverse order. Each of the
$n$ columns adds up to $n + 1$, so twice the sum is $n(n+1)$:
\[
  2\sum_{k=1}^{n} k = n(n+1).
\]
\begin{solremark}
```

*Date*: November 11, 2026.

```
Gauss is said to have found this as a schoolboy.
\end{solremark}
\end{solution}
```

Here `\ans{...}` states the answer in bold, in one line, before the argument. `\begin{solremark}` sets a side comment apart from the argument. Refer to a problem or to one of its items by label, as in `\Cref{prob:toy}` (which prints "Problem $n$") or `\ref{it:toy-sum}`, and never type a number by hand. Give your own labels a prefix such as `my:`, so that they do not clash with the labels of the problem sets.

**A note on how this was made.** The problems were chosen and curated by Santiago Arango-Piñeros. The first draft typesetting and the solutions were written by Claude, and then reviewed by the author. Corrections and recommendations are welcome: send them to

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## Part 1. Selmer elliptic curves

The problems below study a single family of plane cubics and use it to reconstruct, by hand, some of the main results of [Sil09, §§III.1–III.4]: Weierstrass equations, the group law, the Riemann–Roch construction of a Weierstrass model, and a first endomorphism. Section 4 is a set of worked `Magma` examples to read and re-run afterwards, with three short variations to try at the end.

**Standing notation.**

- A field $k$ with $\operatorname{char} k \neq 3$, an algebraic closure $\overline{k}$ of $k$, and $d \in k^\times$.
- The cubic form $F := X^3 + Y^3 - dZ^3$, the plane cubic $C_d := \{F = 0\} \subset \mathbb{P}^2_k$, and its rational point $O := (1 : -1 : 0)$.
- A primitive cube root of unity $\zeta \in \overline{k}$.
- A Selmer elliptic curve: a pair $(C_d, O)$ of this shape, after [Sel51]. Section 5 explains the name.
- Projective coordinates $X, Y, Z$, as usual, and lowercase letters for affine coordinates.
- The base change $\overline{C}_d := C_d \times_k \overline{k}$.

### 1. Warm-up: a plane cubic and a rational point on it

**Definition 1.1** (Elliptic curves [Sil09, §III.3])**.** An elliptic curve over $k$ is a pair $(E, O)$ with $E$ a smooth, projective, geometrically irreducible curve of genus one over $k$ and $O \in E(k)$ a fixed rational point.

**Problem 1** (It is an elliptic curve)**.**

(1) Check that $O \in C_d(k)$.

Solution.

(2) Compute the three partial derivatives of $F$ and deduce that $C_d$ is smooth, using $\operatorname{char} k \neq 3$ and $d \neq 0$. Where is each hypothesis used?

Solution.

(3) Suppose instead that $\operatorname{char} k = 3$ (only for this problem). Show that over $\overline{k}$ the curve $C_d$ is a *triple line*.[a]

Solution.

(4) Show that a smooth plane curve is geometrically irreducible.[b]

Solution.

(5) Conclude from the degree–genus formula that $(C_d, O)$ is an elliptic curve over $k$.

Solution.

[a] *Hint:* Frobenius.
[b] *Hint:* Bézout.

The group law in Problem 2(4) is the chord–tangent construction of [Sil09, §III.2].

**Proposition 1.2** (The group law and $\mathrm{Pic}^0$ [Sil09, Proposition III.3.4])**.** *The map* $P \mapsto [P - O]$ *identifies the chord–tangent group law on* $C_d(\overline{k})$ *with addition in* $\mathrm{Pic}^0 \overline{C}_d$.

**Problem 2** (The base point is an inflection point)**.**

(1) Show that the line $Z = 0$ meets $C_d$ in three *distinct* points of $C_d(\overline{k})$, and write them down. For which $k$ are all three in $C_d(k)$?

Solution.

(2) Compute the tangent line $L$ to $C_d$ at $O$, and show that $L \cap C_d = 3O$. (So $O$ is an inflection point, or flex.)

Solution.

(3) In a Weierstrass model, which line plays the role of $L$? What is different here?

Solution.

(4) Let $\oplus$ denote the group law on $C_d$ with identity $O$, defined by the chord–tangent construction. Using (2), show that for $P, Q, R \in C_d(\overline{k})$,

$$P \oplus Q \oplus R = O \iff P + Q + R \text{ is cut out on } C_d \text{ by a line.}$$

Solution.

## 2. The group law

**Problem 3** (Negation, and 2-torsion)**.**

(1) Show that $\ominus(X_0 : Y_0 : Z_0) = (Y_0 : X_0 : Z_0)$ for every point $(X_0 : Y_0 : Z_0) \in C_d(\overline{k})$. There are two things to check: that the right-hand side lies on $C_d$, and that it is the third intersection point of $C_d$ with the line through $O$ and $(X_0 : Y_0 : Z_0)$.

Solution.

(2) Deduce that the nontrivial 2-torsion of $C_d$ is cut out by the line $X = Y$, and that $\#C_d[2](\overline{k}) = 4$ when $\mathrm{char}\, k \neq 2$.

Solution.

(3) Take $k = \mathbb{Q}$. For which $d$ does $C_d$ have a nontrivial rational 2-torsion point? Write it down, and give the smallest positive integer $d$ for which this happens.

Solution.

Part (2) of Problem 4 is its heart. Parts (4) and (5) are then mechanical. If time is short, take (2) on faith and come back to its proof, but do not skip (3): it is what makes (5) painless. The identity of (2) is the polarization of a cubic form. For a diagonal cubic the trilinear form can be written down by hand.

**Problem 4** (The trilinear form, and the addition formulas)**.**
For vectors $P = (X_P, Y_P, Z_P)$, $Q$ and $R$ in $\overline{k}^3$, define

$$\langle P, Q, R\rangle \coloneqq X_P X_Q X_R + Y_P Y_Q Y_R - d\, Z_P Z_Q Z_R. \tag{2.1}$$

(1) Check that (2.1) is symmetric in $P, Q, R$, linear in each argument separately, and satisfies $\langle P, P, P\rangle = F(P)$.

Solution.

(2) Prove the *expansion identity*: for scalars $s, t$,

$$F(sP + tQ) = s^3\langle P, P, P\rangle + 3s^2 t\,\langle P, P, Q\rangle + 3st^2\,\langle P, Q, Q\rangle + t^3\langle Q, Q, Q\rangle. \tag{2.2}$$

Where do the 3's come from?[a]

Solution.

(3) Prove that $3\langle P, P, Q\rangle = \nabla F(P)\cdot Q$, and deduce that for $P \in C_d(\overline{k})$,

$$\langle P, P, Q\rangle = 0 \iff Q \text{ lies on the tangent line to } C_d \text{ at } P.$$

Solution.

(4) Let $P_1 = (X_1 : Y_1 : Z_1)$ and $P_2 = (X_2 : Y_2 : Z_2)$ be distinct points of $C_d(\overline{k})$, and parametrize the line through them as $sP_1 + tP_2$. Using (2), show that the third intersection point is at $(s : t) = (B : -A)$ where $A = \langle P_1, P_1, P_2\rangle$ and $B = \langle P_1, P_2, P_2\rangle$, and hence that

$$P_1 \oplus P_2 = (BY_1 - AY_2 : BX_1 - AX_2 : BZ_1 - AZ_2). \tag{2.3}$$

Why can $A$ and $B$ not both vanish?

Solution.

(5) Let $P = (X_0 : Y_0 : Z_0) \in C_d(\overline{k})$ and put $Q \coloneqq (Y_0^2 : -X_0^2 : 0)$. Verify that $Q$ lies on the tangent to $C_d$ at $P$, and use (2) and (3) to derive

$$[2]P = \big(-Y_0(2X_0^3 + Y_0^3) : X_0(X_0^3 + 2Y_0^3) : Z_0(X_0^3 - Y_0^3)\big). \tag{2.4}$$

Solution.

(6) Check (2.4) against Problem 3: what does it give on the line $X = Y$? (The line $X = 0$ is taken up in Problem 6.)

Solution.

(7) Where does the whole method break down if $\operatorname{char} k = 3$? Compare your answer with Problem 1(3).

Solution.

[a] *Hint:* expand $\langle sP + tQ, sP + tQ, sP + tQ\rangle$ by trilinearity into eight terms and collect using symmetry.

**Problem 5** (Arithmetic on $C_7$)**.**
Let $d = 7$, so that $P \coloneqq (2 : -1 : 1) \in C_7(\mathbb{Q})$ because $2^3 + (-1)^3 = 7$.

(1) Compute $[2]P$ using (2.4), and simplify. Verify your answer by exhibiting an identity between cubes of integers.

Solution.

(2) Compute $[3]P$ using (2.3) applied to $P$ and $[2]P$, and verify it the same way.

Solution.

(3) Compute $[4]P$.

Solution.

(4) The three rational points you have found have numerators and denominators of rapidly increasing size. What does this suggest about the order of $P$ in $C_7(\mathbb{Q})$? (You are not asked to prove it.)

Solution.

In Problem 6, parts (1)–(3) are the general theory and (5)–(7) the payoff for our family. If time is short, take (3) on faith and go to (5)–(7). Part (4) is the most satisfying: two completely different counts turn out to agree. Part (1) uses Bézout's theorem [Har77, §I.7].

**Proposition 2.1** (The Hessian criterion)**.** *Let $C = V(F) \subset \mathbb{P}^2_{\overline{k}}$ be a smooth plane curve of degree $m$ over a field of characteristic 0 or $> 3$, and let*

$$H(F) = \det\left(\frac{\partial^2 F}{\partial X_i\, \partial X_j}\right)_{0\leqslant i,j\leqslant 2}.$$

*Then a point $P \in C(\overline{k})$ is an inflection point if and only if $H(F)(P) = 0$. This is classical. Characteristic three is taken up in Problem 6(8) below.*

**Problem 6** (Inflection points and 3-torsion)**.**

(1) Let $C = V(F)$ be as in Proposition 2.1. Show that $H(F)$ is a form of degree $3(m-2)$, so of degree 3 when $m = 3$. Assuming $C$ is a smooth plane cubic, explain why $C \not\subseteq V(H(F))$, and conclude from Bézout's theorem that $C$ has exactly 9 inflection points *counted with multiplicity*.

Solution.

(2) Now forget the Hessian. Let $C$ be a smooth plane cubic over $\overline{k}$ and suppose $O \in C(\overline{k})$ is a flex, so that $(C, O)$ is an elliptic curve with the group law of Problem 2(4). Show that

$$P \text{ is a flex } \iff [3]P = O,$$

so that the set of flexes is exactly $C[3](\overline{k})$.

Solution.

(3) Show that a smooth plane cubic over $\overline{k}$ always *has* a flex, without using the Hessian.[a] Deduce that, with no flex singled out, the set of flexes is a torsor under $\mathrm{Pic}^0(C)[3]$. In particular the number of flexes is $\#C[3](\overline{k})$.

Solution.

(4) Combine (2) and (3) with the structure of $C[3]$ in characteristic $\neq 3$ to conclude that a smooth plane cubic has exactly 9 flexes. What does comparing this with (1) tell you about the intersection $C \cap V(H(F))$?

Solution.

(5) Specialize to $C_d$. Compute $H(F)$ for $F = X^3 + Y^3 - dZ^3$ and deduce that, as subsets of $C_d(\overline{k})$,

$$C_d[3](\overline{k}) = \left(C_d \cap \{XYZ = 0\}\right)(\overline{k}).$$

Write down all nine flexes.

Solution.

(6) Check (5) against the group law directly: use (2.4) to show that every point $P = (0 : Y_0 : Z_0) \in C_d(\overline{k})$ satisfies $[2]P = \ominus P$.

Solution.

(7) Which of the nine flexes lie in $C_d(k)$? Treat the three lines separately, and say what happens for $k = \mathbb{Q}$.

Solution.

(8) ($\star$) In characteristic three, $\#C[3](\overline{k})$ is 3 or 1 according as $C$ is ordinary or supersingular, so a smooth plane cubic has 3 or 1 flexes. How is that compatible with (1)?

Solution.

[a] *Hint:* for any $P_0 \in C(\overline{k})$ the class $\mathcal{O}_C(1) - 3[P_0]$ has degree 0, and multiplication by 3 is surjective on $\mathrm{Pic}^0 C$.

In Problem 7 you may use the following two results. Part (4) asks what the classification of automorphism groups in [Sil09, §III.10] predicts.

**Theorem 2.2** (Morphisms fixing $O$ [Sil09, Theorem III.4.8])**.** *A morphism of elliptic curves taking $O$ to $O$ is a group homomorphism.*

**Proposition 2.3** (Endomorphism rings [Sil09, §III.4])**.** *The endomorphism ring of an elliptic curve has no zero divisors.*

**Problem 7** (Complex multiplication, $\star$)**.**
Let $\alpha(X : Y : Z) = (X : Y : \zeta Z)$, a map defined over $k(\zeta)$.

(1) Show that $\alpha$ maps $C_d$ to itself and fixes $O$.

Solution.

(2) Conclude that $\alpha$ is a group homomorphism, hence an automorphism of the elliptic curve $(C_d, O)$ over $k(\zeta)$.

Solution.

(3) Show that $\alpha^2 + \alpha + 1 = 0$ in $\mathrm{End}(C_d \times_k k(\zeta))$, the ring of endomorphisms of $C_d$ defined over $k(\zeta)$.[a]

Solution.

(4) Deduce that $\mathbb{Z}[\zeta] \subseteq \mathrm{End}(C_d \times_k k(\zeta))$, so every Selmer curve has complex multiplication. What does the classification of automorphism groups then predict for $j(C_d)$ and for $\#\operatorname{Aut}(C_d)$ and $\#\operatorname{Aut}(C_d \times_k k(\zeta))$?

Solution.

[a] *Hint:* $\alpha^3 = 1$, and the endomorphism ring of an elliptic curve has no zero divisors.

## 3. From Riemann–Roch to a Weierstrass equation

This section reproduces, for our family, the proof of [Sil09, Proposition III.3.1]: every elliptic curve is isomorphic to a curve in Weierstrass form.

Problem 8 is the longest problem here. Parts (1)–(3) go quickly once you see that Proposition 3.2 converts "pole order at $O$" into "multiplicity of $O$ in $h \cap C_d$". Part (5) is a two-line computation with

a surprising answer, and (8) is worth coming back to. Weierstrass equations and their coefficients $a_1, \dots, a_6$ are as in [Sil09, §III.1].

**Definition 3.1** (Riemann–Roch spaces)**.** For $D \in \operatorname{Div} C_d$,

$$\mathcal{L}(D) = \left\{ f \in k(C_d)^\times : \operatorname{div}(f) \geqslant -D \right\} \cup \{0\}, \qquad \operatorname{div}(f) \in \operatorname{Div} C_d.$$

This is a space of functions defined *over* $k$.

You may use the following two results without proof.

**Proposition 3.2** (Bézout)**.** *For a nonzero linear form $h$, the scheme-theoretic intersection $h \cap C_d$ is an effective divisor of degree 3 in $\operatorname{Div} C_d$. Over $\overline{k}$ it becomes a sum of three points of $C_d(\overline{k})$ counted with multiplicity. Consequently, since $X + Y$ cuts out $3O$ by Problem 2(2),*

$$\operatorname{div}\left(\frac{h}{X+Y}\right) = (h \cap C_d) - 3O.$$

**Proposition 3.3** (Riemann–Roch for genus one [Sil09, §II.5 and proof of Proposition III.3.1])**.** *For every $n \geqslant 1$ we have $\dim_k \mathcal{L}(nO) = n$. Since $C_d$ is geometrically integral, $\mathcal{L}(D) \otimes_k \overline{k} = \mathcal{L}(D_{\overline{k}})$, so the dimension is the same whether computed over $k$ or over $\overline{k}$.*

**Problem 8** (The Weierstrass model of $C_d$)**.**
Consider the rational functions in $k(C_d)$

$$w \coloneqq \frac{Z}{X+Y}, \qquad v \coloneqq \frac{X-Y}{X+Y}.$$

For parts (2)–(6) and (8) assume $\operatorname{char} k \neq 2$. Part (7) replaces $v$ by $X/(X+Y) = (1+v)/2$, which also makes sense in characteristic 2, and reaches a model valid whenever $\operatorname{char} k \neq 3$.

(1) Using Problem 2(1) and Proposition 3.2, compute $\operatorname{div}(w)$. Deduce that $w$ has a pole of order exactly 2 at $O$ and no other pole, so $w \in \mathcal{L}(2O)$. Where are its zeros, and what are they in the language of Problem 6?

Solution.

(2) Show that the line $X - Y = 0$ does not pass through $O$. Deduce that $v$ has a pole of order exactly 3 at $O$, so $v \in \mathcal{L}(3O)$. Where are its zeros?

Solution.

(3) Using Proposition 3.3, explain why $\{1, w\}$ is a basis of $\mathcal{L}(2O)$ and $\{1, w, v\}$ is a basis of $\mathcal{L}(3O)$.

Solution.

(4) The seven functions $1, w, v, w^2, wv, w^3, v^2$ all lie in $\mathcal{L}(6O)$. Why must they satisfy a non-trivial linear relation over $k$? List their pole orders at $O$. Which two collide, and why does that force the relation to involve both?

Solution.

(5) Find the relation.[a]

Solution.

(6) Let $E_1 \colon y_1^2 = x_1^3 - 432d^2$. Show that the relation of (5) says that there is a morphism $\varphi_1 \colon C_d \to E_1$ with $\varphi_1^* x_1 = 12dw$ and $\varphi_1^* y_1 = 36dv$. Write down $\varphi_1$ and its inverse explicitly.

Solution.

(7) Where exactly did you use $\operatorname{char} k \neq 2$? Show that there is instead a morphism $\varphi_2$ with $\varphi_2^* x_2 = 3dw$ and $\varphi_2^* y_2 = 9dX/(X+Y)$ to

$$E_2 \colon y_2^2 - 9d\, y_2 = x_2^3 - 27d^2, \tag{3.1}$$

which is a Weierstrass equation valid whenever $\operatorname{char} k \neq 3$. Which of $a_1, \dots, a_6$ are nonzero?

SOLUTION.

(8) ($\star$) The relation in (5) has no $w^2$, $wv$, $w$ or $v$ term, although a general Weierstrass equation allows all four. Explain why, using the automorphisms $\sigma = \ominus$ of Problem 3 and $\alpha$ of Problem 7.[b]

SOLUTION.

[a] *Hint:* work on the affine chart $X + Y = 1$, where $X = (1+v)/2$ and $Y = (1-v)/2$, and expand $X^3 + Y^3$.
[b] *Hint:* compute $\sigma^* w$, $\sigma^* v$, $\alpha^* w$, $\alpha^* v$. Then use (4) to argue that the relation spans a one-dimensional space, so it is a simultaneous eigenvector.

## 4. Worked examples in Magma

Most of you have never opened `Magma`, so this section is written as worked examples rather than exercises: read them, then re-run them yourself and change things. Every line of output below is reproduced from `Magma` V2.28-8 [BCP97], with whitespace tidied and one long type name elided. Three short variations to try on your own are collected at the end.

**Warning 4.1** (Closed points in `Magma`)**.** A closed point $P$ of $C_d$ is a point of the scheme $C_d$. Its residue field $k(P)$ is a finite extension of $k$, and $\deg P := [k(P) : k]$. The closed points of degree one are the rational points, and a closed point $P$ with $k(P)/k$ separable corresponds to an orbit of $\deg P$ points of $C_d(\overline{k})$ under $\operatorname{Gal}(k^{\mathrm{sep}}/k)$. The divisor group $\operatorname{Div} C_d$ is the free abelian group on the closed points [Poo06, §1.9], whereas [Sil09, §II.3] works with points of $C_d(\overline{k})$ and Galois-invariant divisors on $\overline{C}_d$. `Magma` follows the first convention: `Places` returns closed points, `Divisor` builds elements of $\operatorname{Div} C_d$, and `RiemannRochSpace` returns $\mathcal{L}(D)$ as a $k$-vector space.

Each example continues the previous one. Begin with

```
QQ := Rationals();
P2<X,Y,Z> := ProjectiveSpace(QQ,2);
d := 7;
C := Curve(P2, X^3 + Y^3 - d*Z^3);
O := C![1,-1,0];
```

The notation `P2<X,Y,Z>` names the coordinates so that we can write equations in them, and `C!` coerces a list of three numbers into a rational point of `C`. Typing `O := (1:-1:0);` does not work.

**Example 4.2** (Setting up, and a trap)**.** First, Problem 1:

```
IsNonSingular(C);  // true
Genus(C);          // 1
O;                 // (-1 : 1 : 0)
```

The last line is worth a pause. We asked for $(1:-1:0)$ and got $(-1:1:0)$: a point of $\mathbb{P}^2(\mathbb{Q})$ is defined only up to scaling, and `Magma` normalizes the representative it stores, here by $-1$.

Now ask for the Weierstrass model. The second return value `mp` is the isomorphism from `C` to `E`.

```
E, mp := EllipticCurve(C, 0);
E;
// Elliptic Curve defined by y^2 - 9/49*y = x^3 - 27/2401 over Rational Field
jInvariant(E);                          // 0
E eq EllipticCurve([0,0,0,0,-432*d^2]);   // false
```

That `false` looks alarming, since Problem 8(6) says the answer should be $E_1\colon y_1^2 = x_1^3 - 432d^2$. It is not a contradiction: `eq` compares Weierstrass *coefficients*, and by [Sil09, Proposition III.3.1(b)] a Weierstrass model is unique only up to the changes of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$. The right question is whether the curves are isomorphic, and against which model:

```
Eours := EllipticCurve([0,0,-9*d,0,-27*d^2]);
Eours;                          // y^2 - 63*y = x^3 - 1323
bool, iso := IsIsomorphic(Eours, E);
bool;                           // true
IsomorphismData(iso);           // [ 0, 0, 0, 1/7 ]
```

So $r = s = t = 0$ and $u = 1/7 = 1/d$: `Magma` found exactly the model $E_2$ of Problem 8(7), with $x_2$ and $y_2$ divided by $d^2$ and $d^3$. The two maps agree as well,

```
DefiningEquations(mp);   // [ 3/7*Z, 9/49*X, X + Y ]
```

against our $(3dZ : 9dX : X + Y)$, the ratios being $d^2$ and $d^3$ exactly as $u = 1/d$ demands. The moral: for elliptic curves, `eq` is almost never the question you mean to ask.

We can even watch this happen for all $d$ at once, by working over the function field $\mathbb{Q}(d)$ instead of over $\mathbb{Q}$:

```
Fd<D> := FunctionField(Rationals());
P2d<Xd,Yd,Zd> := ProjectiveSpace(Fd,2);
Cd := Curve(P2d, Xd^3 + Yd^3 - D*Zd^3);
Ed, mpd := EllipticCurve(Cd, Cd![1,-1,0]);
Ed;
// Elliptic Curve defined by y^2 - 9/D^2*y = x^3 - 27/D^4
DefiningEquations(mpd);
// [ 3/D*Zd, 9/D^2*Xd, Xd + Yd ]
```

which is (3.1) rescaled by $u = d$, as an identity in $d$.

Finally, the predictions of Problem 7(4):

```
#AutomorphismGroup(E);                  // 2
K<zeta> := CyclotomicField(3);
#AutomorphismGroup(BaseChange(E,K));    // 6
```

**Example 4.3** (The addition formulas, checked)**.** A `Magma function` takes arguments and returns a value. Here both are sequences of three coordinates, so that (2.3) and (2.4) can be typed almost verbatim.

```
dbl := function(P)
  x := P[1]; y := P[2]; z := P[3];
  return [-y*(2*x^3+y^3), x*(x^3+2*y^3), z*(x^3-y^3)];
```

```
end function;

add := function(P1,P2)
  A := P1[1]^2*P2[1] + P1[2]^2*P2[2] - d*P1[3]^2*P2[3];
  B := P1[1]*P2[1]^2 + P1[2]*P2[2]^2 - d*P1[3]*P2[3]^2;
  return [B*P1[2]-A*P2[2], B*P1[1]-A*P2[1], B*P1[3]-A*P2[3]];
end function;

P := [2,-1,1];
Q := dbl(P);      C!Q;   //  (5/3 : 4/3 : 1)
R := dbl(Q);      C!R;   //  (-1256/183 : 1265/183 : 1)
S := add(P,Q);    C!S;   //  (-17/38 : 73/38 : 1)
```

These are the rational points of Problem 5. Coercing through `C!` normalizes them, which is why $(15 : 12 : 9)$ prints as $(5/3 : 4/3 : 1)$. Now transport everything to $E$ through `mp` and compare with Magma's own group law:

```
EP := mp(C!P);
mp(C!Q) eq 2*EP;          // true
mp(C!S) eq 3*EP;          // true
mp(C!R) eq 4*EP;          // true
```

Three `true`s: formulas derived from nothing but (2.1) and a $3 \times 3$ determinant agree with a general-purpose implementation.

**Example 4.4** (Riemann–Roch, mechanically)**.** Magma can carry out Problem 8 without being told the answer. The generators `x` and `y` of `FunctionField(C)` are the affine coordinates $X/Z$ and $Y/Z$ of $C_d$ (not Weierstrass coordinates), so our two functions are built from those.

```
FF<x,y> := FunctionField(C);   // x = X/Z,  y = Y/Z
w := 1/(x+y);                  // Z/(X+Y)
v := (x-y)/(x+y);              // (X-Y)/(X+Y)
pl0 := Place(0);
Valuation(w, pl0);             // -2
Valuation(v, pl0);             // -3
```

Those are the pole orders of Problem 8(1), (2). A valuation of $-2$ is a pole of order 2. As in Warning 4.1, `Place(0)` is the closed point $O$ and `Divisor` builds an element of $\operatorname{Div} C_d$, so everything here happens over $\mathbb{Q}$. Next, Proposition 3.3 itself:

```
D := Divisor(pl0);
for n in [1..6] do print n, Dimension(RiemannRochSpace(n*D)); end for;
// 1 1
// 2 2
// 3 3
// 4 4
// 5 5
// 6 6
```

Now ask for a basis of $\mathcal{L}(2O)$ without suggesting one:

```
V2, m2 := RiemannRochSpace(2*D);
```

```
[m2(b) : b in Basis(V2)];      // [ x^2 - y*x + y^2, 1 ]
```

which looks nothing like our $w$. But $X^3+Y^3=(X+Y)(X^2-XY+Y^2)$, and on $C_d$ the left side is $dZ^3$, so

$$\frac{X^2-XY+Y^2}{Z^2}=\frac{dZ^3}{Z^2(X+Y)}=d\,\frac{Z}{X+Y}=d\,w,$$

and indeed

```
(x^2 - x*y + y^2) eq d*w;      // true
```

Finally the relation. We list the seven functions of Problem 8(4), express each in a basis of $\mathcal{L}(6O)$ (that is what `@@` does, applying the inverse of the map `mV`), and take the kernel of the resulting matrix.

```
V, mV := RiemannRochSpace(6*D);
fns := [FF!1, w, v, w^2, w*v, w^3, v^2];
[-Valuation(f, pl0) : f in fns];   // [ 0, 2, 3, 4, 5, 6, 6 ]
M := Matrix([Eltseq(f @@ mV) : f in fns]);
K := Kernel(M);
Dimension(K);                      // 1
Basis(K);                          // [ ( 1  0  0  0  0  -28  3) ]
```

The pole orders are $0,2,3,4,5,6,6$: seven functions, six available orders, the collision at 6 forcing the relation exactly as in Problem 8(4). And the kernel is one-dimensional, spanned by the vector reading

$$1-28w^3+3v^2=0,$$

which is $3v^2+1=4dw^3$ at $d=7$. We got it by hand from a factorization. `Magma` got it from linear algebra in a Riemann–Roch space.

**Now try these.** Each one is a small edit to Example 4.2 or Example 4.3, and in each case the worksheet already tells you what the answer must be.

(1) Set `d := 2`. Problem 3(3) predicts a rational point of order two. Find it, verify with the group law that it has order two, and compute its image under $\varphi_1$ by hand.

Solution.

(2) Set `d := 8`. Problem 6(7) predicts rational 3-torsion. Find the two rational 3-torsion points other than $O$, use your `dbl` function to check that $[2]P=\ominus P$, and compute $\varphi_1(P)$ by hand.

Solution.

(3) Use `IsIsomorphic` to decide whether $C_1\simeq C_8$ and whether $C_1\simeq C_2$ over $\mathbb{Q}$. Pass it the *Weierstrass* models: on the plane curves themselves `Magma` answers `For genus 0 or 1 curves the basefield must be finite`. Reconcile the answers with the cubic-twist criterion of Section 5.

Solution.

## 5. Where the name comes from

Selmer [Sel51] studied the general diagonal cubic $aX^3+bY^3+cZ^3=0$, and produced the standard counterexample to the Hasse principle: the curve $3X^3+4Y^3+5Z^3=0$ has real points and $\mathbb{Q}_p$-points

for every prime $p$, but no rational point. Such a curve is a torsor under its Jacobian [Poo17, §5.12], and that Jacobian is a Selmer curve in our sense:

$$\mathrm{Jac}(aX^3 + bY^3 + cZ^3) \simeq \big(y^2 = x^3 - 432(abc)^2\big).$$

Two ways to test this in `Magma`: the curve $X^3 + 2Y^3 + 3Z^3 = 0$ *does* have the rational point $(1:1:-1)$, so it is its own Jacobian and should be isomorphic to $C_6$, that is to $y^2 = x^3 - 432 \cdot 6^2$; and for Selmer's own curve one can call `Jacobian(GenusOneModel(3*X^3 + 4*Y^3 + 5*Z^3))` and compare with $y^2 = x^3 - 432 \cdot 60^2$. Both come out `true`.

Finally, by Problem 8(6) together with the classification of twists of $y^2 = x^3 + A$ [Sil09, §X.5], for $\mathrm{char}\, k \neq 2, 3$ the curves $C_d$ and $C_{d'}$ are isomorphic over $k$ exactly when $d/d' \in (k^\times)^3$, so the family is parametrized by $k^\times/(k^\times)^3$. These are the cubic twists of $X^3 + Y^3 = Z^3$, and they are the natural first examples for the descent machinery of [Sil09, §X] and [Poo17, Ch. 8].

## Part 2. The Weil pairing, endomorphisms and automorphisms

Section 6 shows that the determinant of the mod $N$ Galois representation is the cyclotomic character. Section 7 exhibits an endomorphism ring that is an order in a definite quaternion algebra, the third case of the classification [Sil09, Theorem III.9.3], on a single curve over $\mathbb{F}_9$. Section 8 computes the automorphism groups of order 12 and 24 in characteristic 3 and 2, the two cases of [Sil09, Theorem III.10.1] in which the group is not cyclic.

**Reading.** [Sil09, §§III.8–III.10], with [Sil09, Appendix A.1] for the automorphism groups in characteristic 2 and 3. For the quaternion background: [Voi21, Definition 2.2.1] for the algebra $(a, b \mid \mathbb{Q})$, [Voi21, §§3.2–3.3] for the standard involution, the reduced trace and the reduced norm, [Voi21, Main Theorem 5.4.4 and §14.1] for split and definite algebras, and [Voi21, Definitions 10.2.1 and 10.4.1] for orders and maximal orders. Proposition 7.6, Theorem 7.7 and Proposition 7.8 are the results we actually use.

**`Magma`.** If you have no local installation, every block below runs in the free online calculator, which caps how long a single job may take.

**Standing notation.**

- A field $k$, with an algebraic closure $\overline{k}$ and the separable closure $k^{\mathrm{sep}} \subseteq \overline{k}$.
- Multiplication by $m$, written $[m]$, and the dual $\hat{\varphi}$ of an isogeny $\varphi$.
- The $q$-power Frobenius endomorphism $\mathsf{F} \in \mathrm{End}(E)$ of a curve over $\mathbb{F}_q$. Beware that $\varphi_\ell$ already means the map induced by $\varphi$ on $T_\ell(E)$, so the Frobenius *endomorphism* is $\mathsf{F}$ and never $\varphi_q$.

### 6. The Weil pairing and the cyclotomic character

Fix $N \geqslant 2$ with $\mathrm{char}\, k \nmid N$, so that $E[N](k^{\mathrm{sep}}) \cong (\mathbb{Z}/N\mathbb{Z})^2$.

**Definition 6.1** (The Weil pairing)**.** We write

$$e_N \colon E[N](k^{\mathrm{sep}}) \times E[N](k^{\mathrm{sep}}) \to \mu_N(k^{\mathrm{sep}})$$

for the Weil pairing.

**Definition 6.2** (Galois representations)**.** We write

$$\bar{\rho}_{E,N} \colon \mathrm{Gal}_k \to \mathrm{Aut}\big(E[N](k^{\mathrm{sep}})\big) \cong \mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$$

for the mod $N$ Galois representation, the isomorphism depending on a choice of basis, and $\rho_{E,\ell}$ for the $\ell$-adic representation on $T_\ell(E)$.

**Proposition 6.3** (Properties of the Weil pairing [Sil09, Propositions III.8.1 and III.8.2])**.** *The Weil pairing $e_N$ is*

*(1) bilinear;*
*(2) alternating: $e_N(T,T) = 1$, so $e_N(S,T) = e_N(T,S)^{-1}$;*
*(3) non-degenerate: if $e_N(S,T) = 1$ for all $S \in E[N](k^{\mathrm{sep}})$, then $T = O$;*
*(4) Galois equivariant: $\sigma\big(e_N(S,T)\big) = e_N(\sigma(S),\sigma(T))$ for $\sigma \in \mathrm{Gal}_k$;*
*(5) compatible: $e_{NN'}(S,T) = e_N([N']S,T)$;*
*(6) adjoint for isogenies: $e_N(\varphi S,T) = e_N(S,\hat{\varphi}T)$.*

*These properties are.*

**Proposition 6.4** (The determinant of $\varphi_\ell$ [Sil09, Proposition III.8.6])**.** *Let $\varphi \in \mathrm{End}(E)$ and let $\ell \neq \mathrm{char}\, k$ be a prime. Then $\det \varphi_\ell = \deg \varphi$.*

**Problem 9** (The determinant is the cyclotomic character)**.**

(1) Let $\zeta \in \mu_N(k^{\mathrm{sep}})$ be primitive. Show that for each $\sigma \in \mathrm{Gal}_k$ there is a unique $\chi_N(\sigma) \in \mathbb{Z}/N\mathbb{Z}$ with $\sigma(\zeta) = \zeta^{\chi_N(\sigma)}$, that $\chi_N(\sigma)$ is a unit, and that the resulting map $\chi_N\colon \mathrm{Gal}_k \to (\mathbb{Z}/N\mathbb{Z})^\times$ is a homomorphism independent of the choice of $\zeta$. This is the mod $N$ cyclotomic character.

Solution.

(2) Let $S,T$ be a basis of $E[N](k^{\mathrm{sep}})$ and put $\zeta \coloneqq e_N(S,T)$. Show that $\zeta$ is a *primitive* $N$-th root of unity, using only parts (1)–(3) of Proposition 6.3.

Solution.

(3) Write $\sigma(S) = aS + cT$ and $\sigma(T) = bS + dT$, so that $\bar{\rho}_{E,N}(\sigma) = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ in the basis $S,T$. Expand $e_N(\sigma(S),\sigma(T))$ and deduce

$$\det \bar{\rho}_{E,N} = \chi_N.$$

Solution.

(4) Take $k = \mathbb{Q}$. Show that $\det \bar{\rho}_{E,N}$ is surjective for *every* elliptic curve $E$ over $\mathbb{Q}$, that $\mathbb{Q}(\zeta_N) \subseteq \mathbb{Q}\big(E[N]\big)$, and that $\det \bar{\rho}_{E,N}(c) = -1$ for $c$ a complex conjugation: the representation is odd.

Solution.

(5) Take $k = \mathbb{F}_q$ with $q$ prime to $N$. Show that $\det \bar{\rho}_{E,N}(\mathsf{F}) = q$, and check this against the identity $\det \varphi_\ell = \deg \varphi$.

Solution.

(6) Check (5) numerically in `Magma`, taking $N = 3$, $q = 5$ and $E\colon y^2 = x^3 + 1$ over $\mathbb{F}_5$.

```
q := 5;  N := 3;
E := EllipticCurve([GF(q) | 0, 1]);      // y^2 = x^3 + 1
#E;                                      // 6
Ek := ChangeRing(E, GF(q^2));
#Ek, AbelianGroup(Ek);                   // 36, Z/6 + Z/6
T3 := [P : P in Points(Ek) | N*P eq Ek!0];
#T3;                                     // 9
S := [P : P in T3 | P ne Ek!0][1];
T := [P : P in T3 | P ne Ek!0 and P ne S and P ne -S][1];
zeta := WeilPairing(S, T, N);  Order(zeta);    // 3: a primitive cube root
Frob := func<P | P eq Ek!0 select Ek!0 else Ek![P[1]^q, P[2]^q]>;
```

```
WeilPairing(Frob(S), Frob(T), N) eq zeta^q;    // true
```

Two lines carry the mathematics. That `Order(zeta)` is 3 is (2). That the last line is `true` is $\det \bar{\rho}_{E,3}(\mathsf{F}) = q$, which is (5) on this curve. Note that `Frob` raises both coordinates to the $q$-th power: that is the $q$-power Frobenius, an endomorphism because $E$ is defined over $\mathbb{F}_q$. Now write the matrix of $\mathsf{F}$ in the basis $S, T$ and read its determinant off:

```
[ <a, b> : a, b in [0..2] | a*S + b*T eq Frob(S) ];   // [ <1, 0> ]
[ <a, b> : a, b in [0..2] | a*S + b*T eq Frob(T) ];   // [ <0, 2> ]
GF(3)!q;                                              // 2
```

Check that a different choice of basis moves the matrix but not the determinant.

Solution.

**Remark 6.5.** Part (3) of Problem 9 gives back [Sil09, Corollary III.8.1.1]: if $E[N](k^{\mathrm{sep}}) \subseteq E(k)$, then $\bar{\rho}_{E,N}$ is trivial, so $\chi_N$ is trivial, so every $\sigma \in \mathrm{Gal}_k$ fixes a primitive $N$-th root of unity, and $\mu_N(k^{\mathrm{sep}}) \subseteq k$.

A word, for the curious, on the example in Problem 9(6): choosing it was half of that part. We want $q \not\equiv 1 \pmod{N}$, so that $\chi_N$ is *not* trivial and the check has content. By Remark 6.5 that already forces $E[N](k^{\mathrm{sep}}) \not\subseteq E(\mathbb{F}_q)$, so the $N$-torsion becomes rational only over an extension. The curve $E\colon y^2 = x^3 + 1$ over $\mathbb{F}_5$ has $\#E(\mathbb{F}_5) = 6$ and $E(\mathbb{F}_{25}) \cong (\mathbb{Z}/6\mathbb{Z})^2$, so all of $E[3](\overline{\mathbb{F}}_5)$ is already rational over the quadratic extension: small enough to list, large enough to have something to say.

## 7. A quaternionic endomorphism ring

The classification of [Sil09, Theorem III.9.3] puts the endomorphism ring of an elliptic curve in one of three cases: $\mathbb{Z}$, an order in an imaginary quadratic field, or an order in a definite quaternion algebra over $\mathbb{Q}$. The first two are easy to exhibit: a generic curve has endomorphism ring $\mathbb{Z}$, and $y^2 = x^3 + x$ has $\mathbb{Z}[i]$ inside its endomorphism ring. Here is an example of the third, on the smallest curve that has one.

**Definition 7.1** (The quaternion algebra $D$ [Voi21, Definition 2.2.1])**.** Let $D := (-1, -3 \mid \mathbb{Q})$, the quaternion algebra with $i^2 = -1$, $j^2 = -3$ and $ij = -ji$. Its reduced discriminant is the product of the primes $p$ at which $D \otimes_{\mathbb{Q}} \mathbb{Q}_p$ is not split.

**Definition 7.2** (Definite quaternion algebras [Voi21, Main Theorem 5.4.4 and §14.1])**.** A quaternion algebra $B$ over $\mathbb{Q}$ is definite when $B \otimes_{\mathbb{Q}} \mathbb{R}$ is Hamilton's $\mathbb{H}$ rather than $M_2(\mathbb{R})$, equivalently when its reduced norm is a positive definite quadratic form in the four coordinates. Otherwise $B$ is indefinite.

**Definition 7.3** (Orders and the reduced discriminant [Voi21, §15.2])**.** Orders are as in [Voi21, Definition 10.2.1]. The reduced discriminant $\mathrm{discrd}(\mathcal{O})$ of an order $\mathcal{O} \subseteq D$ is the positive integer with $\mathrm{discrd}(\mathcal{O})^2 = \left|\det\big(\mathrm{trd}(\alpha_k \alpha_l)\big)_{k,l}\right|$ for a $\mathbb{Z}$-basis $\alpha_1, \ldots, \alpha_4$ of $\mathcal{O}$.

**Proposition 7.4** (The characteristic polynomial of Frobenius [Sil09, §V.2])**.** *Let $E$ be an elliptic curve over $\mathbb{F}_q$ with $\mathsf{F}$ its $q$-power Frobenius, $a := q + 1 - \#E(\mathbb{F}_q)$, and $\ell \nmid q$ a prime. Then $\mathsf{F}$ satisfies $\mathsf{F}^2 - [a]\mathsf{F} + [q] = 0$ in $\mathrm{End}(E)$: by [Sil09, Proposition III.8.6] the characteristic polynomial of $\mathsf{F}_\ell$ is $T^2 - aT + q$, and every $\varphi \in \mathrm{End}(E)$ satisfies $\varphi^2 - [\mathrm{tr}\,\varphi_\ell]\varphi + [\deg \varphi] = 0$ in $\mathrm{End}(E)$, not merely on $T_\ell(E)$.*

**Corollary 7.5** (Factoring through a separable isogeny [Sil09, Corollary III.4.11]). *If $\varphi\colon E_1 \to E_2$ is separable and $\psi\colon E_1 \to E_3$ is an isogeny with $\ker\varphi \subseteq \ker\psi$, then $\psi = \lambda\circ\varphi$ for a unique isogeny $\lambda\colon E_2 \to E_3$.*

**Proposition 7.6** (Where $D$ is split [Voi21, Exercise 11.11 and §14.1]). *The algebra $D\otimes_{\mathbb{Q}}\mathbb{Q}_p$ is split for every prime $p \neq 3$, and $D\otimes_{\mathbb{Q}}\mathbb{Q}_3$ is not. So $D$ is a division algebra, and its reduced discriminant is 3.*

**Theorem 7.7** (Maximal orders [Voi21, Theorem 15.5.5]). *An order $\mathcal{O} \subseteq D$ is maximal if and only if $\operatorname{discrd}(\mathcal{O}) = 3$.*

**Proposition 7.8** (The index and the discriminant [Voi21, §15.2]). *If $\mathcal{O} \subseteq \mathcal{O}'$ are orders of $D$, then $\operatorname{discrd}(\mathcal{O}) = [\mathcal{O}' : \mathcal{O}]\operatorname{discrd}(\mathcal{O}')$.*

For the rest of this section

$$E\colon y^2 = x^3 - x \quad \text{over } \mathbb{F}_3,$$

so that the generic $E$ of Proposition 7.4 is this curve, with $q = 3$.

**Problem 10** ($\operatorname{End}(E_{\mathbb{F}_9})$ is a maximal order in a quaternion algebra).
Every part below has a hand computation and a check to run in `Magma`. The curve is small enough that $[i]$ and $\mathsf{F}$ can be verified on all sixteen points of $E(\mathbb{F}_9)$.

(1) Check that $E$ is an elliptic curve over $\mathbb{F}_3$, list $E(\mathbb{F}_3)$, and deduce that $a = 0$ and that

$$\mathsf{F}^2 = [-3].$$

Record also that every point of $E[2](\overline{\mathbb{F}}_3)$ lies in $E(\mathbb{F}_3)$: this is what makes part (5) work.

```
E3 := EllipticCurve([GF(3) | -1, 0]);    // prints as y^2 = x^3 + 2*x
Points(E3);
#E3;                                     // 4
3 + 1 - #E3;                             // 0        (so a = 0)
jInvariant(E3);                          // 0
```

`Magma` hands you the point count and the $j$-invariant. The identity $\mathsf{F}^2 = [-3]$ is yours to deduce, from Proposition 7.4. Read $E[2](\overline{\mathbb{F}}_3) \subseteq E(\mathbb{F}_3)$ off the list of rational points — which points of $y^2 = f(x)$ have order 2?

Solution.

(2) Show that $x^2 + 1$ is irreducible over $\mathbb{F}_3$, so that $\mathbb{F}_9 = \mathbb{F}_3(i)$ with $i^2 = -1$. Show that

$$[i](x, y) \coloneqq (-x, iy)$$

defines an endomorphism of $E_{\mathbb{F}_9}$, and that $[i]^2 = [-1]$. Why is a morphism $E \to E$ fixing $O$ automatically a group homomorphism? Mind Warning 7.9 before you start typing.

```
F9 := GF(9);  i := Sqrt(F9!(-1));
i^2, MinimalPolynomial(i);               // 2 ($.1^2 + 1), and 2 = -1
E9 := ChangeRing(E3, F9);  #E9;          // 16
imap := func<P | P eq E9!0 select E9!0 else E9![-P[1], i*P[2]]>;
forall{P : P in Points(E9) | imap(P) in Points(E9)};        // [i] is well defined
forall{<P,Q> : P, Q in Points(E9) | imap(P+Q) eq imap(P)+imap(Q)};  // additive
forall{P : P in Points(E9) | imap(imap(P)) eq -P};          // [i]^2 = [-1]
// all true
```

The second `forall` is additivity, checked on all $16^2$ pairs. That is a proof only because the curve is finite, and it is no answer to the question above: additivity is not a coincidence to be confirmed point by point but a consequence of fixing $O$.

Solution.

(3) Show that $\mathsf{F}[i] = -[i]\mathsf{F}$. Conclude that $\mathrm{End}(E_{\mathbb{F}_9})$ is not commutative, so that $E_{\mathbb{F}_9}$ falls into the third case of the classification.

```
Frob := func<P | P eq E9!0 select E9!0 else E9![P[1]^3, P[2]^3]>;
forall{P : P in Points(E9) | Frob(Frob(P)) eq -3*P};            // Frob^2 = [-3]
forall{P : P in Points(E9) | Frob(imap(P)) eq -imap(Frob(P))};   // Frob[i] = -[i]Frob
// both true
```

The first line verifies the $\mathsf{F}^2 = [-3]$ you deduced in (1) from the characteristic polynomial, which is worth having independently. For the second, do the computation by hand as well: it comes down to a single fact about $i^3$.

Solution.

(4) Deduce that

$$\mathrm{End}(E_{\mathbb{F}_9}) \otimes_{\mathbb{Z}} \mathbb{Q} \cong D = (-1, -3 \mid \mathbb{Q}).$$

Write down the reduced norm of $D$ and check that $D$ is definite. How does the reduced norm compare with the degree of an endomorphism?

```
D<ii,jj> := QuaternionAlgebra<Rationals() | -1, -3>;
RamifiedPrimes(D), Discriminant(D);      // [ 3 ] 3
IsDefinite(D);                           // true
Trace(jj), Norm(jj);                     // 0 3
```

The two arguments are the squares of the generators, so the algebra `Magma` builds is the $D$ of Definition 7.1. Here `Trace` and `Norm` on a quaternion algebra are the reduced trace and the reduced norm, and the pair they return for `jj` should be recognizable as $(\mathrm{tr}\,\mathsf{F}, \deg \mathsf{F})$. All of this is about $D$ alone. That $D$ is the *whole* of $\mathrm{End}(E_{\mathbb{F}_9}) \otimes \mathbb{Q}$ is the one claim here `Magma` cannot be asked, and it needs a rank count.

Solution.

(5) Show that $\frac{1+\mathsf{F}}{2}$ and $\frac{i+i\mathsf{F}}{2}$ lie in $\mathrm{End}(E_{\mathbb{F}_9})$.[a] Then show that

$$\mathcal{O} := \mathbb{Z} + \mathbb{Z}i + \mathbb{Z}\,\frac{1+\mathsf{F}}{2} + \mathbb{Z}\,\frac{i+i\mathsf{F}}{2}$$

is an order of $D$, compute $\mathrm{discrd}(\mathcal{O})$, and conclude with Theorem 7.7 that $\mathrm{End}(E_{\mathbb{F}_9}) = \mathcal{O}$ is a maximal order.

```
O0 := QuaternionOrder([D!1, ii, jj, ii*jj]);
Discriminant(O0);                        // 12
O  := QuaternionOrder([D!1, ii, (1+jj)/2, (ii+ii*jj)/2]);
Discriminant(O), IsMaximal(O);           // 3 true
```

`QuaternionOrder` takes a $\mathbb{Z}$-basis and `Discriminant` of an order is its reduced discriminant, so the two lists above are the $\mathbb{Z}$-bases of $\mathcal{O}_0$ and $\mathcal{O}$. Do not reach for `Index` to compare the two orders:

`Magma` has no index for quaternion orders, and answers `Runtime error in 'Index': Bad argument types`. Recover $[\mathcal{O} : \mathcal{O}_0]$ from the two discriminants instead, by Proposition 7.8, and check it against the determinant of the change of basis.

The first sentence of this part is where the real work is, and none of the code touches it: $\frac{1+\mathsf{F}}{2}$ is a quotient in a ring with no division, and only the kernel argument puts it in $\mathrm{End}(E_{\mathbb{F}_9})$.

Solution.

[a] *Hint:* use Corollary 7.5 with $\varphi = [2]$.

**Warning 7.9** ($\mathbb{F}_9$ does not come with $i$)**.** `GF(9)` does *not* come with $i$. `Magma`'s default generator of $\mathbb{F}_9$ is a root of the Conway polynomial $x^2 + 2x + 2$, so writing `F9<i> := GF(9)` gives an $i$ with $i^2 = i + 1$, not $i^2 = -1$, and every check in Problem 10 would silently fail. Construct the square root of $-1$ explicitly, and confirm it: `MinimalPolynomial` of the element you build must be $x^2 + 1$, and `i^2` must print as 2, which is $-1$ in $\mathbb{F}_9$.

## 8. Automorphisms in characteristic 2 and 3

For $\operatorname{char} k = 2, 3$ the geometric automorphism group of an elliptic curve may be non-abelian, which is the subject of Problem 11. In the following statements, $E$ is an elliptic curve over $k$, given by a Weierstrass equation with coefficients $a_1, a_2, a_3, a_4, a_6$.

**Definition 8.1** (Changes of variables)**.** A change of variables preserving a Weierstrass equation is a substitution

$$x = u^2 x' + r, \qquad y = u^3 y' + u^2 s x' + t, \qquad u \in \overline{k}^\times,\ r, s, t \in \overline{k}. \tag{8.1}$$

For such $u, r, s, t$ we write

$$\Psi_{u,r,s,t}(x, y) \coloneqq (u^2 x + r,\ u^3 y + u^2 s x + t).$$

**Theorem 8.2** (The automorphism group [Sil09, Theorem III.10.1])**.** *The group* $\mathrm{Aut}(E_{\overline{k}})$ *is finite, of order*

$$\#\operatorname{Aut}(E_{\overline{k}}) = \begin{cases} 2 & \text{if } j(E) \neq 0, 1728, \\ 4 & \text{if } j(E) = 1728 \text{ and } \operatorname{char} k \neq 2, 3, \\ 6 & \text{if } j(E) = 0 \text{ and } \operatorname{char} k \neq 2, 3, \\ 12 & \text{if } j(E) = 0 = 1728 \text{ and } \operatorname{char} k = 3, \\ 24 & \text{if } j(E) = 0 = 1728 \text{ and } \operatorname{char} k = 2. \end{cases}$$

**Corollary 8.3** (The cyclic cases [Sil09, Corollary III.10.2])**.** *If* $\operatorname{char} k \neq 2, 3$*, then* $\mathrm{Aut}(E_{k^{\mathrm{sep}}}) \cong \mu_n(k^{\mathrm{sep}})$ *with* $n = \#\operatorname{Aut}(E_{\overline{k}})$*, via* $[\zeta](x, y) \coloneqq (\zeta^2 x, \zeta^3 y)$*. In particular, the geometric automorphism group is cyclic in that case.*

**Proposition 8.4** (The action on the coefficients [Sil09, Proposition III.3.1(b) and Table 3.1])**.** *Under a change of variables* (8.1) *the coefficients transform by*

$$\begin{aligned} u a_1' &= a_1 + 2s, \\ u^2 a_2' &= a_2 - s a_1 + 3r - s^2, \\ u^3 a_3' &= a_3 + r a_1 + 2t, \\ u^4 a_4' &= a_4 - s a_3 + 2r a_2 - (t + rs) a_1 + 3r^2 - 2st, \\ u^6 a_6' &= a_6 + r a_4 + r^2 a_2 + r^3 - t a_3 - t^2 - rt a_1. \end{aligned} \tag{8.2}$$

*A substitution (8.1) carries $E$ to the curve with coefficients $a_i'$, so it is a change of coordinates from $E$ to itself exactly when $a_i' = a_i$ for every $i$, and then $\Psi_{u,r,s,t}$ is the automorphism of $E$ that it defines. Setting $a_i' = a_i$ in (8.2) therefore turns those five lines into five equations in the four unknowns $u, r, s, t$, whose solutions are exactly the elements of* $\mathrm{Aut}(E_{\overline{k}})$.

The whole point of this section is visible in (8.2): observe the 2's and 3's in the coefficients. The characteristic enters through the terms $2s$, $2t$ and $3r$. Away from 2 and 3 they are invertible multiples of $s$, $t$ and $r$, which determines those three and leaves $u$ as the only free parameter: that is why the group is cyclic there. In characteristic 3 the term $3r$ vanishes, in characteristic 2 the terms $2s$ and $2t$ do, and the parameters they no longer pin down are what make the two groups of Problem 11 large, and non-cyclic.

**Problem 11** (The two non-cyclic cases)**.**
As in Problem 10, each part pairs a hand computation with a check in `Magma`. Here the machine is best at naming the group, which is the trickiest thing to do by hand.

(1) Let $\mathrm{char}\, k = 3$. Show that every short Weierstrass equation $y^2 = x^3 + Ax + B$ has $j = 0$, and explain why a curve with $j \neq 0$ in characteristic 3 needs an $a_2x^2$ term. So the case "$j = 0 = 1728$, $\mathrm{char}\, k = 3$" of Theorem 8.2 is exactly the short-Weierstrass one. The first claim is a one-line sweep over the whole family:

```
{ jInvariant(EllipticCurve([GF(3) | A, B])) : A, B in GF(3) | A ne 0 };  // { 0 }
jInvariant(EllipticCurve([GF(3) | 0, 1, 0, 0, -1]));   // 1
```

The condition `A ne 0` is what keeps the equation non-singular: work out what $\Delta \neq 0$ becomes for $y^2 = x^3 + Ax + B$ once $27 = 0$, and you will find that $B$ drops out. The set that comes back has one element, which is the first claim. The second line is a curve with an $a_2x^2$ term, and its $j$-invariant is not in that set.

Solution.

(2) Let $\mathrm{char}\, k = 3$ and $E\colon y^2 = x^3 - x$. Using (8.2), show first that $s = t = 0$, and then that (8.1) is an automorphism of $E$ if and only if

$$u^4 = 1 \qquad \text{and} \qquad r^3 - r = 0,$$

i.e. $u \in \mu_4(\overline{\mathbb{F}}_3) \subseteq \mathbb{F}_9^\times$ and $r \in \mathbb{F}_3$. Conclude $\#\,\mathrm{Aut}(E_{\overline{\mathbb{F}}_3}) = 12$, and that all 12 are already defined over $\mathbb{F}_9$. Where did $r$ become free, and what would have forced $r = 0$ if $\mathrm{char}\, k \neq 3$? Two counts settle both claims at once:

```
E3 := EllipticCurve([GF(3) | -1, 0]);
#AutomorphismGroup(E3), #AutomorphismGroup(ChangeRing(E3, GF(9)));   // 6 12
```

Predict both numbers from your conditions on $u$ and $r$ before running it. Over $\mathbb{F}_3$ the fourth roots of unity are only $\pm 1$, while $r$ already ranges over all of $\mathbb{F}_3$. Over $\mathbb{F}_9$ all four roots of unity appear. The second number being 12, and being reached over $\mathbb{F}_9$ rather than over some larger field, is exactly the claim that all twelve automorphisms are defined there.

Solution.

(3) Write $\varphi_{u,r} \coloneqq \Psi_{u,r,0,0}$, so that $\varphi_{u,r}(x, y) = (u^2x + r, u^3y)$. Compute $\varphi_{u',r'} \circ \varphi_{u,r}$ and deduce

$$\mathrm{Aut}(E_{\overline{\mathbb{F}}_3}) \cong \mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z},$$

where $\mathbb{Z}/3\mathbb{Z} = \mathbb{F}_3$ is the group of translations $r$, where $\mathbb{Z}/4\mathbb{Z} = \mu_4(\overline{\mathbb{F}}_3)$ is the group of the $u$, and where $u$ acts on $r$ by $r \mapsto u^2 r$. Since $u^4 = 1$ forces $u^2 = \pm 1$, that action is not faithful: it factors through the squaring map $\mu_4(\overline{\mathbb{F}}_3) \twoheadrightarrow \{\pm 1\} = \mathrm{Aut}(\mathbb{Z}/3\mathbb{Z})$, so $\pm 1$ act trivially and $\pm i$ act by $r \mapsto -r$. Show this group is non-abelian with a *unique* element of order 2, and deduce that it is neither cyclic, nor dihedral, nor $A_4$: it is the dicyclic group of order 12. Which automorphism is the unique involution? `AutomorphismGroup` returns a polycyclic group and `IdentifyGroup` gives its label in the small-groups database.

```
G3 := AutomorphismGroup(ChangeRing(E3, GF(9)));
IdentifyGroup(G3), GroupName(G3);        // <12, 1>  C3:C4
IsAbelian(G3), IsCyclic(G3);             // false false
{* Order(g) : g in G3 *};                // {* 1, 2, 3^^2, 4^^6, 6^^2 *}
```

The last line is the multiset of element orders. Say how it rules out the dihedral group and $A_4$ — how many involutions does each of those have? — and match its single element of order 2 against the automorphism you named.

Solution.

(4) Let $\mathrm{char}\, k = 2$ and $E\colon y^2 + y = x^3$. Using (8.2), show that $a_1' = a_1$ automatically, that $a_2' = 0$ forces $r = s^2$, and then that (8.1) is an automorphism if and only if

$$u^3 = 1, \qquad s^4 = s, \qquad t^2 + t = s^6.$$

Deduce that $s \in \mathbb{F}_4$, count the solutions, and conclude $\#\,\mathrm{Aut}(E_{\overline{\mathbb{F}}_2}) = 24$, all defined over $\mathbb{F}_4$. The same two counts as in (2):

```
E2 := EllipticCurve([GF(2) | 0, 0, 1, 0, 0]);          // y^2 + y = x^3
#AutomorphismGroup(E2), #AutomorphismGroup(ChangeRing(E2, GF(4)));   // 2 24
```

Predict both first. Over $\mathbb{F}_2$ the three conditions bite hard: $u^3 = 1$ leaves only $u = 1$, and of the values $s \in \mathbb{F}_2$ one of them makes $t^2 + t = s^6$ unsolvable in $\mathbb{F}_2$. The gap between the two numbers here is much wider than in characteristic 3, and that gap is the content of "all defined over $\mathbb{F}_4$".

Solution.

(5) Write $\psi_{s,t} \coloneqq \Psi_{1,s^2,s,t}$ for the automorphisms with $u = 1$, so that $\psi_{s,t}(x, y) = (x + s^2, y + sx + t)$. Show that

$$\psi_{s',t'} \circ \psi_{s,t} = \psi_{s+s',\, t+t'+s's^2}, \qquad \psi_{s,t}^2 = \psi_{0,s^3},$$

and deduce that these 8 automorphisms form a copy of the quaternion group $Q_8$. For the remaining factor, let $\omega \in \mathbb{F}_4$ be a primitive cube root of unity and $\varphi_\omega \coloneqq \Psi_{\omega,0,0,0}$, which has order 3. Check that conjugation by it is

$$\varphi_\omega \circ \psi_{s,t} \circ \varphi_\omega^{-1} = \psi_{\omega s,\, t},$$

so $\varphi_\omega$ normalizes $Q_8$, fixes its center $\langle \psi_{0,1} \rangle$, and cycles the three pairs $\pm i, \pm j, \pm k$ — the order-3 automorphism of $Q_8$. Since $\langle \varphi_\omega \rangle \cap Q_8 = 1$, conclude that

$$\mathrm{Aut}(E_{\overline{\mathbb{F}}_2}) \cong Q_8 \rtimes \mathbb{Z}/3\mathbb{Z} \cong \mathrm{SL}_2(\mathbb{F}_3).$$

As before:

```
G2 := AutomorphismGroup(ChangeRing(E2, GF(4)));
IdentifyGroup(G2), GroupName(G2);        // <24, 3>  SL(2,3)
{* Order(g) : g in G2 *};                  // {* 1, 2, 3^^8, 4^^6, 6^^8 *}
```

Exactly one element of order 2 again. Explain why that alone pins down $Q_8$ among the groups of order 8, and why it does *not* by itself pin down $\mathrm{SL}_2(\mathbb{F}_3)$ among the groups of order 24 — what extra fact did you use?

Solution.

**Remark 8.5** (Both groups are unit groups)**.** Automorphisms are the units of the endomorphism ring, $\mathrm{Aut}(E_{\overline{k}}) = \mathrm{End}(E_{\overline{k}})^\times$. In characteristic 2 and 3 the curve with $j = 0$ has $\mathrm{End}(E_{\overline{k}})$ a maximal order in the quaternion algebra over $\mathbb{Q}$ that is not split exactly at $p$ and $\infty$ [Sil09, Theorem V.3.1], [Voi21, Theorem 42.1.9]. For $p = 3$ that order is the one computed in Problem 10(5), and its unit group is the dicyclic group of order 12 of Problem 11(3). For $p = 2$ it is the Hurwitz order $\mathbb{Z} + \mathbb{Z}i + \mathbb{Z}j + \mathbb{Z}\frac{1+i+j+ij}{2}$ in $(-1,-1 \mid \mathbb{Q})$ [Voi21, §11.1], whose unit group is the binary tetrahedral group of order 24 of Problem 11(5). This explains the shape of Theorem 8.2: when $\mathrm{End}(E_{\overline{k}})$ is commutative it is $\mathbb{Z}$ or an imaginary quadratic order, whose unit groups are the cyclic groups of orders 2, 4 and 6, giving its first three lines, while a non-commutative $\mathrm{End}(E_{\overline{k}})$ is a quaternion order, and its unit group need not be, and here is not, cyclic.

Both counts can be recovered from the orders alone, with no elliptic curve anywhere in the computation:

```
for d in [-1, -3] do
  Dd := QuaternionAlgebra<Rationals() | -1, d>;
  Od := MaximalOrder(Dd);
  print RamifiedPrimes(Dd), #UnitGroup(Od);
end for;
// [ 2 ] 24     <- Hurwitz order, characteristic 2
// [ 3 ] 12     <- the endomorphism ring of Problem 10, characteristic 3
```

So the 24 and the 12 of Theorem 8.2 are the unit counts of the two maximal quaternion orders.

**Looking ahead.** Curves whose Frobenius has $a = 0$, like the one of Section 7, have a name and a theory of their own. They come later in the course, with Hasse's bound and the structure of End over $\mathbb{F}_q$. In characteristic 0 the third case of the classification never occurs, and $\mathrm{End}(E_{\overline{k}}) \neq \mathbb{Z}$ becomes complex multiplication. The oddness of $\rho_{E,\ell}$ from Problem 9(4) is the first hint of modularity.

## Part 3. Differentials, dual isogenies, and the Tate module

Here are three computations by hand, on the invariant differential, the dual isogeny and the Tate module. Each compares the multiplicative group $\mathbb{G}_\mathrm{m}$ with the elliptic curve

$$E\colon y^2 = x^3 - x, \qquad \omega_E \coloneqq \frac{\mathrm{d}x}{2y}.$$

Throughout, $k$ is a field with $\mathrm{char}\, k \neq 2$, $\overline{k}$ an algebraic closure and $k^{\mathrm{sep}} \subseteq \overline{k}$ the separable closure. Here $\mathbb{G}_\mathrm{m} = \mathbb{P}^1_k \smallsetminus \{0, \infty\}$ has coordinate $t$ and invariant differential $\omega_{\mathbb{G}_\mathrm{m}} \coloneqq \mathrm{d}t/t$.

**Reading.** [Sil09, §§III.4–III.7], and [Sil09, §III.8] for the Weil pairing of Proposition 11.1.

**A formula.** We use the following duplication formula on $E$, derived in Problem 12(5): if $x \in k(E)$ is the $x$-coordinate and $[2]\colon E \to E$ is multiplication by 2, then

$$[2]^*x \coloneqq x \circ [2] = \frac{(x^2+1)^2}{4(x^3-x)}. \tag{8.3}$$

## 9. Invariant differentials

**Problem 12** (Poles and pullbacks)**.**

(1) Rewrite $\omega_{\mathbb{G}_m}$ in the coordinate $s \coloneqq 1/t$ near $\infty$, and compute $\operatorname{div}(\omega_{\mathbb{G}_m})$ on $\mathbb{P}^1_k$. Compute $[m]^*\omega_{\mathbb{G}_m}$ for $[m](t) = t^m$, $m \in \mathbb{Z}$.

Solution.

(2) Show that $\omega_E = \mathrm{d}y/(3x^2-1)$, and that $y$ and $3x^2-1$ have no common zero on $E$. Deduce that $\omega_E$ has no zeros or poles on $E \smallsetminus \{O\}$.

Solution.

(3) Using $\operatorname{ord}_O x = -2$ and $\operatorname{ord}_O y = -3$, show that $\operatorname{ord}_O \omega_E = 0$.

Solution.

(4) Compute $[-1]^*\omega_E$ from $[-1](x,y) = (x,-y)$, and compare with $[-1]^*\omega_{\mathbb{G}_m}$.

Solution.

(5) We derive (8.3) by hand. Here $x, y \in k(E)$ are functions. For a point $P \in E(\overline{k})$ write $x_P \coloneqq x(P)$ and $y_P \coloneqq y(P)$ for its coordinates.

(a) Let $P \in E(\overline{k})$ with $y_P \neq 0$. Show that the tangent line to $E$ at $P$ has slope $\lambda_P = (3x_P^2-1)/(2y_P)$, and use that its third point of intersection with $E$ is $-[2]P$ to show that $x([2]P) = \lambda_P^2 - 2x_P$.

Solution.

(b) Let $\lambda \coloneqq (3x^2-1)/(2y) \in k(E)$, so that $\lambda(P) = \lambda_P$. Explain why (5)(a) gives the identity $[2]^*x = \lambda^2 - 2x$ of functions in $k(E)$, and simplify it using $y^2 = x^3 - x$.

Solution.

(6) Solve this part in `Magma`, working in the function field $\mathbb{Q}(E) = \mathbb{Q}(x)[y]/(y^2 - x^3 + x)$. Compute $[2]^*x$ and $[2]^*y$ from the function $\lambda$ of (5), check (8.3), and show that

$$[2]^*x - e = \frac{g_e(x)^2}{4(x^3-x)}, \qquad g_0 = x^2+1, \quad g_1 = x^2-2x-1, \quad g_{-1} = x^2+2x-1,$$

for $e \in \{0, 1, -1\}$. Show that $[2]^*y = \pm(x^2+1)(x^4-6x^2+1)/(8y^3)$ and $[2]^*\omega_E = \pm 2\omega_E$. Which sign is the correct one?

Solution.

## 10. Dual isogenies

Consider the elliptic curves

$$E_1\colon y_1^2 = x_1^3 - x_1, \qquad E_2\colon y_2^2 = x_2^3 + 4x_2, \qquad E_3\colon y_3^2 = x_3^3 - 16x_3,$$

so that $E_1 = E$, and write $\omega_i \coloneqq \mathrm{d}x_i/(2y_i)$ for the invariant differential of $E_i$. Consider the maps

$$\begin{aligned}
\varphi\colon E_1 &\to E_2, \quad (x_1, y_1) \mapsto \left(\frac{x_1^2 - 1}{x_1}, \frac{y_1(x_1^2+1)}{x_1^2}\right), \\
\psi\colon E_2 &\to E_3, \quad (x_2, y_2) \mapsto \left(\frac{y_2^2}{x_2^2}, \frac{y_2(x_2^2-4)}{x_2^2}\right), \\
\iota\colon E_3 &\to E_1, \quad (x_3, y_3) \mapsto \left(\frac{x_3}{4}, \frac{y_3}{8}\right).
\end{aligned}$$

The maps $\varphi$ and $\psi$ are special cases of [Sil09, Example III.4.5], which treats $y^2 = x^3 + ax^2 + bx$ in general, so they are isogenies. Also $\iota$ is an isomorphism. That example computes the kernels and the dual, but not the constants $c_\alpha$ of Definition 10.1: that is part (2) of Problem 13.

**Definition 10.1** (The constant $c_\alpha$)**.** For an isogeny $\alpha\colon E_i \to E_j$, define $c_\alpha$ by $\alpha^*\omega_j = c_\alpha\,\omega_i$.

**Problem 13** (A 2-isogeny and its dual)**.**

(1) Check that $\varphi$ lands on $E_2$. Show that $\ker\varphi = \{O_1, (0,0)\}$, that $\varphi(\pm 1, 0) = (0,0)$, and that $\deg\varphi = 2$.[a]

Solution.

(2) Compute $c_\varphi$.

Solution.

(3) Check that $\psi$ lands on $E_3$ and that $\iota$ lands on $E_1$, and compute $c_\psi$ and $c_\iota$.

Solution.

(4) Show that $(\iota\circ\psi\circ\varphi)^* x_1 = [2]^* x_1$, so that $\iota\circ\psi\circ\varphi = \pm[2]$. Use (2) and (3) to decide the sign, and conclude that $\hat\varphi = \iota\circ\psi$, that $c_\varphi c_{\hat\varphi} = \deg\varphi$, and that $\ker\hat\varphi = \varphi(E_1[2](\overline{k}))$.

Solution.

(5) Is there an isogeny $\mathbb{G}_\mathrm{m} \to \mathbb{G}_\mathrm{m}$ of degree 2? If so, does it have a dual?

Solution.

[a] *Hint:* write $y_1$ in terms of $\varphi^* y_2$ and $x_1$, and find a quadratic equation for $x_1$ over $\varphi^* k(E_2)$.

## 11. The Tate module

We first describe the Tate module of $\mathbb{G}_\mathrm{m}$, the familiar case, and then repeat each step for $E$. What links the two is the Weil pairing, which you may use without proof.

**Proposition 11.1** (The Weil pairing [Sil09, Proposition III.8.1])**.** *Let $m \geqslant 1$ with $\operatorname{char} k \nmid m$. There is a pairing*

$$e_m\colon E[m](k^{\mathrm{sep}}) \times E[m](k^{\mathrm{sep}}) \to \mu_m(k^{\mathrm{sep}})$$

*that is bilinear, alternating ($e_m(P,P) = 1$), non-degenerate, and Galois equivariant: $\sigma\big(e_m(P,Q)\big) = e_m(\sigma(P), \sigma(Q))$ for every $\sigma \in \mathrm{Gal}_k$. Here $E[m](k^{\mathrm{sep}}) = E[m](\overline{k})$, since $\operatorname{char} k \nmid m$. It follows that if $(P,Q)$ is a basis of $E[m](k^{\mathrm{sep}})$ over $\mathbb{Z}/m\mathbb{Z}$, then $e_m(P,Q)$ is a primitive $m$-th root of unity, and if $\sigma$ acts on this basis by the matrix $A \in \mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z})$, then $\sigma\big(e_m(P,Q)\big) = e_m(P,Q)^{\det A}$.*

**Problem 14** (The 2-power torsion of $y^2 = x^3 - x$)**.**
Regard $E$ as a curve over $\mathbb{Q}$, fix $\overline{\mathbb{Q}} \subset \mathbb{C}$, and recall that $E[m](\overline{\mathbb{Q}})$ is the kernel of $[m]$ on $E(\overline{\mathbb{Q}})$. Let $\sigma$ be complex conjugation, acting on $\overline{\mathbb{Q}}$ and on $E(\overline{\mathbb{Q}})$ coordinatewise.

(1) Recall that $T_2(\mathbb{G}_m) := \varprojlim_n \mu_{2^n}(\overline{\mathbb{Q}})$, with transition maps $\zeta \mapsto \zeta^2$. Find a $\mathbb{Z}_2$-basis of $T_2(\mathbb{G}_m)$, and write down its first three terms. How does $\sigma$ act on $\mu_4(\overline{\mathbb{Q}})$, and on $T_2(\mathbb{G}_m)$?

Solution.

(2) List $E[2](\overline{\mathbb{Q}})$. Using Problem 12(6), find the $x$-coordinates of all $P \in E(\overline{\mathbb{Q}})$ with $[2]P \in E[2](\overline{\mathbb{Q}}) \smallsetminus \{O\}$, and conclude that $\#E[4](\overline{\mathbb{Q}}) = 16$.

Solution.

(3) Show that $P_2 := (i, 1-i)$ and $Q_2 := (1+\sqrt{2}, 2+\sqrt{2})$ are points of $E(\overline{\mathbb{Q}})$, with $[2]P_2 = (0,0)$ and $[2]Q_2 = (1,0)$. Show that $(P_2, Q_2)$ is a basis of $E[4](\overline{\mathbb{Q}})$ over $\mathbb{Z}/4\mathbb{Z}$, and that $E[4](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q}(\zeta_8))$. Compare with $\mu_8(\overline{\mathbb{Q}})$ and Proposition 11.1.

Solution.

(4) Show that $\sigma$ restricts to a group automorphism of $E[4](\overline{\mathbb{Q}})$. Compute $P_2 + (1,0)$, and write $\sigma(P_2)$ and $\sigma(Q_2)$ in the basis $(P_2, Q_2)$. Compare the determinant of the resulting matrix in $\mathrm{GL}_2(\mathbb{Z}/4\mathbb{Z})$ with the action of $\sigma$ on $\mu_4(\overline{\mathbb{Q}})$ found in (1), and explain the agreement using Proposition 11.1.

Solution.

(5) Let $(P_n)_n$ and $(Q_n)_n$ be compatible sequences, $[2]P_{n+1} = P_n$ and $[2]Q_{n+1} = Q_n$, starting with $P_1 = (0,0)$, $Q_1 = (1,0)$ and $P_2$, $Q_2$ as above. Explain why they form a $\mathbb{Z}_2$-basis of $T_2(E)$, and compare with (1).

Solution.

## 12. Magma examples

**Example 12.1** (The isogenies of Problem 13)**.** We check in the function field of $E_1$ that $\varphi$ and $\psi$ land where they should, that $\iota \circ \psi \circ \varphi = [2]$, and compare with Magma's own 2-isogeny.

```
E := EllipticCurve([-1,0]);                    // y^2 = x^3 - x
Kx<x> := FunctionField(Rationals());
R<t> := PolynomialRing(Kx);
F<y> := ext<Kx | t^2 - (x^3 - x)>;                          // x = x_1, y = y_1
x2 := F!((x^2 - 1)/x);  y2 := y*(x^2 + 1)/x^2;              // phi^*x_2, phi^*y_2
y2^2 eq x2^3 + 4*x2;                                        // true
x3 := y2^2/x2^2;  y3 := y2*(x2^2 - 4)/x2^2;                 // (psi o phi)^*x_3, ^*y_3
y3^2 eq x3^3 - 16*x3;                                       // true
lam := (3*x^2 - 1)/(2*y);  xd := lam^2 - 2*x;  yd := -(y + lam*(xd - x));
x3/4 eq xd and y3/8 eq yd;                                  // true: [2]
phi := TwoIsogeny(E![0,0]);
aInvariants(Codomain(phi));                                 // [ 0, 0, 0, 4, 0 ]
phi(E![1,0]);                                               // (0 : 0 : 1)
DualIsogeny(phi);
// ... taking (x : y : 1) to ((1/4*x^2 + 1) / x : (1/8*x^2*y - 1/2*y) / x^2 : 1)
```

Magma's dual is $\iota \circ \psi$ written out in the coordinates $x_2, y_2$ of $E_2$: $\big((x_2^2+4)/(4x_2),\ y_2(x_2^2-4)/(8x_2^2)\big)$.

**Example 12.2** (Division polynomials and $E[4](\overline{\mathbb{Q}})$)**.** For $m \geqslant 2$, `DivisionPolynomial(E,m)` vanishes exactly at the $x$-coordinates of $E[m](\overline{\mathbb{Q}}) \smallsetminus \{O\}$. Its degree is $(m^2-1)/2$ for odd $m$ and $(m^2+2)/2$ for even $m$, and its factors for $m = 4$ are $x$, $x \pm 1$ and the $g_e$ of Problem 12(6).

```
[ Degree(DivisionPolynomial(E,m)) : m in [2..8] ];
// [ 3, 4, 9, 12, 19, 24, 33 ]
Factorization(DivisionPolynomial(E,4));
// x - 1, x, x + 1, x^2 - 2*x - 1, x^2 + 1, x^2 + 2*x - 1
Q8<z> := CyclotomicField(8);  i := z^2;  s := z + z^7;  // s^2 = 2
E8 := BaseChange(E, Q8);
P2 := E8![i, 1-i];  Q2 := E8![1+s, 2+s];
2*P2, 2*Q2;                          // (0 : 0 : 1) (1 : 0 : 1)
#DivisionPoints(E8!0, 4);            // 16
E8![-i, 1+i] eq 3*P2 + 2*Q2;         // true: complex conjugation
```

**Example 12.3** (Characteristic $p$)**.** Reducing modulo $p$, the $p$-division polynomial collapses, because $\#E[p](\overline{\mathbb{F}}_p) \in \{1, p\}$.

```
for p in [3,5,7,13] do
  Ep  := ChangeRing(E, GF(p));
  psi := DivisionPolynomial(Ep, p);
  print p, IsSupersingular(Ep), Degree(psi), Factorization(psi);
end for;
//  3 true   0  []
//  5 false 10  [ <x^2 + 2, 5> ]
//  7 true   0  []
// 13 false 78  [ <x^6 + 11*x^4 + 5*x^2 + 11, 13> ]
```

For $p = 3, 7$ the curve is supersingular and $E[p](\overline{\mathbb{F}}_p) = 0$. In general, $y^2 = x^3 - x$ is supersingular exactly when $p \equiv 3 \pmod 4$. For $p = 5, 13$ the division polynomial is the $p$-th power of a polynomial of degree $(p-1)/2$, whose roots are the $x$-coordinates of the $p-1$ non-zero points of $E[p](\overline{\mathbb{F}}_p) \cong \mathbb{Z}/p\mathbb{Z}$. The exponent $p$ is the inseparable degree of $[p]$. The same computation for $\mathbb{G}_{\mathrm{m}}$ is $t^p - 1 = (t-1)^p$.

**Looking ahead.** The Weil pairing of Proposition 11.1 also gives $\det \varphi_\ell = \deg \varphi$ for the map $\varphi_\ell$ induced by an isogeny on Tate modules ([Sil09, Proposition III.8.6]). Formal groups make $c_\varphi$ an honest derivative. And Frobenius factors $[p]$ in characteristic $p$.

## Part 4. The zeta function of an elliptic curve

Section 13 defines the zeta function of a curve over $\mathbb{F}_q$ as a generating function for its numbers of rational points, and shows that it is also an Euler product over the closed points of the curve. Section 14 computes the zeta function of an elliptic curve $E$ from the action of Frobenius on the Tate module, proves the Riemann hypothesis for it, and deduces Hasse's bound $|\#E(\mathbb{F}_q) - q - 1| \leqslant 2\sqrt{q}$. Section 15 shows that the zeta function, equivalently the single number $\#E(\mathbb{F}_q)$, is a complete isogeny invariant.

**Reading.** [Sil09, §V.2], with [Sil09, §§III.7–III.8] for the Tate module, and [Poo06, §§2.2.1 and 3.4–3.5] for closed points and the Euler product. Three of the problems are exercises from [Sil09, Chapter V]: Problem 16 is Exercise 5.3, Problem 19 is Exercise 5.13 and Problem 21 is Exercise 5.4.

`Magma`**.** If you have no local installation, every block below runs in the free online calculator.

**Standing notation.**

- A power $q = p^r$ of a prime $p$, and the Galois group $\mathrm{Gal}_{\mathbb{F}_q} = \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$.
- The automorphism $\sigma \in \mathrm{Gal}_{\mathbb{F}_q}$, $t \mapsto t^q$, of $\overline{\mathbb{F}}_q$, which generates $\mathrm{Gal}_{\mathbb{F}_q}$ topologically. Its fixed field in $\overline{\mathbb{F}}_q$ is $\mathbb{F}_q$, and that of $\sigma^n$ is $\mathbb{F}_{q^n}$.
- In Sections 14 and 15, an elliptic curve $E$ over $\mathbb{F}_q$, given by a Weierstrass equation with coefficients in $\mathbb{F}_q$.
- The ring $\mathrm{End}(E)$ of the endomorphisms of $E$ *defined over* $\mathbb{F}_q$.
- The $q$-power Frobenius endomorphism
$$\mathsf{F}\colon E \to E, \qquad (x, y) \mapsto (x^q, y^q).$$
- The integers
$$a \coloneqq q + 1 - \#E(\mathbb{F}_q), \qquad a_n \coloneqq q^n + 1 - \#E(\mathbb{F}_{q^n}) \quad (n \geqslant 1), \qquad a_0 \coloneqq 2,$$
so that $a_1 = a$.
- For a prime $\ell \neq p$, the Tate module $T_\ell(E) = \varprojlim_n E[\ell^n](\overline{\mathbb{F}}_q)$, a free $\mathbb{Z}_\ell$-module of rank 2, and the map $\psi_\ell \in \mathrm{End}\big(T_\ell(E)\big)$ induced by $\psi \in \mathrm{End}(E)$.

## 13. Zeta functions and closed points

Let $C$ be a smooth projective curve over $\mathbb{F}_q$. The zeta function of $C$ (Definition 13.2) packages the numbers of rational points over all the extensions of $\mathbb{F}_q$ into one power series. The exponential in it looks odd: the more obvious series is $\sum_n \#C(\mathbb{F}_{q^n})T^n$. Problem 15 explains the choice. It is [Poo06, Proposition 3.4.3 and Exercise 3.7]. In [Poo06] the ground field is perfect throughout, as $\mathbb{F}_q$ is.

**Definition 13.1** (Closed points [Poo06, Definition 2.2.1])**.** A closed point $P$ of $C$ is an integral subvariety of dimension 0. Its residue field $\mathbb{F}_q(P)$ is a finite extension of $\mathbb{F}_q$, and its degree is $\deg P \coloneqq [\mathbb{F}_q(P) : \mathbb{F}_q]$.

**Definition 13.2** (The zeta function)**.** The zeta function of $C$ is
$$Z(C/\mathbb{F}_q; T) \coloneqq \exp\Big(\sum_{n \geqslant 1} \#C(\mathbb{F}_{q^n}) \frac{T^n}{n}\Big) \in \mathbb{Q}[[T]],$$
so that $\#C(\mathbb{F}_{q^n})$ is the coefficient of $T^{n-1}$ in $\frac{d}{dT} \log Z(C/\mathbb{F}_q; T)$.

**Proposition 13.3** (Closed points are Galois orbits [Poo06, Remark 2.2.2])**.** *Sending a point of $C(\overline{\mathbb{F}}_q)$ to its image is a surjection from $C(\overline{\mathbb{F}}_q)$ onto the set of closed points of $C$, whose fibers are the $\mathrm{Gal}_{\mathbb{F}_q}$-orbits. The orbit over a closed point $P$ has $\deg P$ elements: a point of $C(\overline{\mathbb{F}}_q)$ over $P$ is an $\mathbb{F}_q$-embedding $\mathbb{F}_q(P) \hookrightarrow \overline{\mathbb{F}}_q$, and there are $\deg P$ of these.*

**Problem 15** (The Euler product)**.**
For $d \geqslant 1$, let $N_d$ be the number of closed points of $C$ of degree $d$.

(1) Show that a point of $C(\overline{\mathbb{F}}_q)$ lies in $C(\mathbb{F}_{q^n})$ if and only if $\sigma^n$ fixes it. Deduce that a $\mathrm{Gal}_{\mathbb{F}_q}$-orbit of size $d$ is contained in $C(\mathbb{F}_{q^n})$ if $d \mid n$, and is disjoint from it otherwise.

Solution.

(2) Show that $N_d$ is finite, and that
$$\#C(\mathbb{F}_{q^n}) = \sum_{d \mid n} d N_d \qquad \text{for every } n \geqslant 1.$$

Solution.

(3) Deduce the identity of formal power series

$$Z(C/\mathbb{F}_q;T) = \prod_P \bigl(1 - T^{\deg P}\bigr)^{-1} = \prod_{d\geqslant 1} (1-T^d)^{-N_d},$$

the first product running over the closed points $P$ of $C$. Why is the product well defined? Conclude that $Z(C/\mathbb{F}_q;T) \in 1 + T\mathbb{Z}[[T]]$, which is not obvious from the exponential.

Solution.

(4) Invert (2): show that $N_d = \frac{1}{d}\sum_{e|d} \mu(d/e)\, \#C(\mathbb{F}_{q^e})$, where $\mu$ is the Möbius function. So the numbers of rational points over the extensions of $\mathbb{F}_q$ and the numbers of closed points of each degree determine each other.

Solution.

(5) Take $C = \mathbb{P}^1$. Show that $Z(\mathbb{P}^1/\mathbb{F}_q;T) = 1/\bigl((1-T)(1-qT)\bigr)$. The closed points of $\mathbb{P}^1$ are the point at infinity, of degree 1, and the closed points of $\mathbb{A}^1$, which are the monic irreducible polynomials $f \in \mathbb{F}_q[t]$, with $\deg P = \deg f$. Deduce that the number of monic irreducible polynomials of degree $d$ over $\mathbb{F}_q$ is $\frac{1}{d}\sum_{e|d}\mu(d/e)q^e$. How many are there of degree 2 over $\mathbb{F}_2$, and which are they?

Solution.

(6) Put $\zeta_C(s) \coloneqq Z(C/\mathbb{F}_q;q^{-s})$ and $NP \coloneqq \#\mathbb{F}_q(P)$. Show that $\zeta_C(s) = \prod_P \bigl(1 - NP^{-s}\bigr)^{-1}$, and compare with the Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ of the Riemann zeta function.

Solution.

**Remark 13.4** (The Weil conjectures for curves)**.** Let $C$ be a smooth projective curve of genus $g$ over $\mathbb{F}_q$. The Weil conjectures (proved for curves by Weil) say:

(1) (Rationality) $Z(C/\mathbb{F}_q;T) = P(T)/\bigl((1-T)(1-qT)\bigr)$ with $P \in \mathbb{Z}[T]$ of degree $2g$ and $P(0) = 1$.
(2) (Functional equation) $Z\bigl(C/\mathbb{F}_q;1/(qT)\bigr) = q^{1-g}T^{2-2g}Z(C/\mathbb{F}_q;T)$.
(3) (Riemann hypothesis) $P(T) = \prod_{i=1}^{2g}(1-\alpha_i T)$ with $|\alpha_i| = \sqrt{q}$ for every $i$.

For $C = \mathbb{P}^1$, of genus 0, Problem 15(5) gives all three with $P = 1$. We prove them for $g = 1$ in Section 14.

## 14. The zeta function of an elliptic curve

Problem 16 is [Sil09, Exercise 5.3].

**Problem 16** (Traces and determinants)**.**
Let $A$ be a square matrix over a field $K$ of characteristic 0. We prove, in two ways, that

$$\exp\Bigl(\sum_{n\geqslant 1} \operatorname{tr}(A^n)\frac{T^n}{n}\Bigr) = \frac{1}{\det(1-AT)} \qquad \text{in } K[[T]]. \tag{14.1}$$

(1) Let $f, g \in 1 + TK[[T]]$. Show that $f$ and $g$ are units of $K[[T]]$, so that their logarithmic derivatives $f'/f$ and $g'/g$ are defined, and that if $f'/f = g'/g$, then $f = g$.[a] Where do you use that $K$ has characteristic 0? Give a counterexample in characteristic $p$.

Solution.

(2) Show that both sides of (14.1) lie in $1 + TK[[T]]$, compute their logarithmic derivatives, and conclude with (1).[b]

Solution.

(3) Give a second proof: triangularize $A$ over an algebraic closure $\bar{K}$ of $K$, reduce (14.1) to the case of a $1 \times 1$ matrix $A = (\lambda)$, and prove that case with (1), applied over $\bar{K}$.

Solution.

[a] *Hint:* compute the derivative of $f/g$.
[b] *Hint:* for $M(T)$ an invertible matrix over $K[[T]]$, $\frac{d}{dT} \det M(T) = \det M(T)\ \mathrm{tr}\big(M(T)^{-1} M'(T)\big)$.

Problem 17 is the proof of [Sil09, Theorems V.2.3.1 and V.2.4], arranged around Problem 16. In the following results, $E$ is an elliptic curve over $\mathbb{F}_q$ and $\ell \neq p$ is a prime.

**Proposition 14.1** (Frobenius is purely inseparable [Sil09, Proposition II.2.11])**.** *The Frobenius endomorphism* $\mathsf{F}$ *is purely inseparable of degree* $q$*.*

**Corollary 14.2** ($1 - \mathsf{F}$ is separable [Sil09, Corollary III.5.5])**.** *The endomorphism* $1 - \mathsf{F}$ *is separable.*

**Theorem 14.3** (Kernels of separable isogenies [Sil09, Theorem III.4.10(c)])**.** *A nonzero separable isogeny* $\psi$ *satisfies* $\#\ker\psi = \deg\psi$*, where* $\ker\psi$ *is taken in* $E(\overline{\mathbb{F}}_q)$*.*

**Proposition 14.4** (Determinant and trace on $T_\ell(E)$ [Sil09, Proposition III.8.6])**.** *For* $\psi \in \mathrm{End}(E)$*,* $\det\psi_\ell = \deg\psi$ *and* $\mathrm{tr}\,\psi_\ell = 1 + \deg\psi - \deg(1-\psi)$*.*

**Theorem 14.5** (Endomorphisms act faithfully on $T_\ell(E)$ [Sil09, Theorem III.7.4])**.** *The map* $\psi \mapsto \psi_\ell$ *is an injective ring homomorphism* $\mathrm{End}(E) \to \mathrm{End}\big(T_\ell(E)\big)$*.*

**Problem 17** (The zeta function of $E$)**.**

(1) Show that $\sigma(Q) = \mathsf{F}(Q)$ for every $Q \in E(\overline{\mathbb{F}}_q)$. Deduce that the closed points of $E$ of degree $d$ are the $\mathsf{F}$-orbits of size $d$ in $E(\overline{\mathbb{F}}_q)$, and that $E(\mathbb{F}_{q^n}) = \ker(1 - \mathsf{F}^n)$.

Solution.

(2) Show that $\mathsf{F}^n$ is the $q^n$-power Frobenius endomorphism of $E$ over $\mathbb{F}_{q^n}$, and deduce that

$$\#E(\mathbb{F}_{q^n}) = \deg(1 - \mathsf{F}^n) = \det(1 - \mathsf{F}_\ell^n) = 1 - \mathrm{tr}(\mathsf{F}_\ell^n) + q^n.$$

Where does the argument use that $1 - \mathsf{F}^n$ is separable?

Solution.

(3) Taking $n = 1$, show that the characteristic polynomial of $\mathsf{F}_\ell$ is $T^2 - aT + q$. In particular it has integer coefficients and does not depend on $\ell$.

Solution.

(4) Use Problem 16 to show that

$$Z(E/\mathbb{F}_q; T) = \frac{\det(1 - \mathsf{F}_\ell T)}{(1-T)(1-qT)} = \frac{1 - aT + qT^2}{(1-T)(1-qT)}.$$

By Problem 15, this is also $\prod_P (1 - T^{\deg P})^{-1}$ over the closed points $P$ of $E$.

Solution.

(5) Verify the functional equation $Z\big(E/\mathbb{F}_q; 1/(qT)\big) = Z(E/\mathbb{F}_q; T)$ of Remark 13.4.

Solution.

**Problem 18** (The Riemann hypothesis and Hasse's bound)**.**
Let $\alpha, \beta \in \mathbb{C}$ be the roots of $T^2 - aT + q$, so that $1 - aT + qT^2 = (1 - \alpha T)(1 - \beta T)$.

(1) For $r, s \in \mathbb{Z}$ with $s \neq 0$, show that $\det(r/s - \mathsf{F}_\ell) = \deg(r - s\mathsf{F})/s^2 \geqslant 0$. Deduce that $T^2 - aT + q$ is nonnegative on $\mathbb{R}$, and hence that $\beta = \bar{\alpha}$ and $|\alpha| = |\beta| = \sqrt{q}$. This is the Riemann hypothesis of Remark 13.4. With $\zeta_E$ as in Problem 15(6), show that every zero of $\zeta_E(s)$ satisfies $\mathrm{Re}(s) = 1/2$, and that $\zeta_E(s) = \zeta_E(1 - s)$.

Solution.

(2) Show that $a_n = \alpha^n + \beta^n$ for every $n \geqslant 0$. Deduce Hasse's bound

$$|\#E(\mathbb{F}_q) - q - 1| \leqslant 2\sqrt{q},$$

and more generally $|a_n| \leqslant 2q^{n/2}$ for every $n \geqslant 1$.

Solution.

(3) Conversely, suppose only that there is a constant $c$ with $|a_n| \leqslant c\, q^{n/2}$ for all $n \geqslant 1$. Show that the power series $\sum_{n \geqslant 1} a_n T^n = \frac{\alpha T}{1 - \alpha T} + \frac{\beta T}{1 - \beta T}$ converges for $|T| < q^{-1/2}$, and deduce that $|\alpha| = |\beta| = \sqrt{q}$. So the Riemann hypothesis is equivalent to a Hasse bound over every extension of $\mathbb{F}_q$, even with a worse constant.

Solution.

Problem 19 is [Sil09, Exercise 5.13].

**Problem 19** (A recurrence)**.**
(1) Show that the single number $\#E(\mathbb{F}_q)$ determines $\#E(\mathbb{F}_{q^n})$ for every $n \geqslant 1$.

Solution.

(2) Prove that

$$a_{n+2} = a_1 a_{n+1} - q a_n \qquad \text{for all } n \geqslant 0.$$

(Recall the convention $a_0 = 2$.)

Solution.

(3) Here is the recurrence on $y^2 = x^3 + x + 1$ over $\mathbb{F}_{101}$:

```
q := 101;  E := EllipticCurve([GF(q) | 1, 1]);          // y^2 = x^3 + x + 1
a := [TraceOfFrobenius(ChangeRing(E, GF(q^n))) : n in [1..6]];
a;                                  // [ -3, -193, 882, 16847, -139623, -1282678 ]
b := [2, a[1]];
for n in [1..5] do Append(~b, a[1]*b[n+1] - q*b[n]); end for;
b[2..7] eq a;                       // true
```

Here `TraceOfFrobenius(E)` is $a$. Compute $a_2$ by hand from $a_1 = -3$. Check Hasse's bound for $a_6$ over $\mathbb{F}_{101^6}$.

Solution.

**Problem 20** (A zeta function by hand)**.**
Let $E$ be the elliptic curve $y^2 + xy = x^3 + 1$ over $\mathbb{F}_2$ (its discriminant is 1).

(1) List the rational points of $E(\mathbb{F}_2)$, and find $a$.

Solution.

(2) Write down $Z(E/\mathbb{F}_2; T)$, and check the Riemann hypothesis for it directly.

Solution.

(3) Use Problem 19 to predict $\#E(\mathbb{F}_4)$ and $\#E(\mathbb{F}_8)$. Check the first by listing the rational points of $E(\mathbb{F}_4)$, where $\mathbb{F}_4 = \{0, 1, \omega, \omega^2\}$ with $\omega^2 = \omega + 1$.[a]

Solution.

(4) Use Problem 15(4) to find the numbers $N_1$, $N_2$, $N_3$ of closed points of $E$ of degree 1, 2 and 3. Deduce that every $\mathbb{F}_8$-rational point of $E$ is $\mathbb{F}_2$-rational. Is $\#E(\mathbb{F}_8)$ close to the edge of the Hasse interval?

Solution.

(5) Check the Euler product of Problem 15(3) against (2) up to the term in $T^2$.

Solution.

(6) Check, where `Places(E, d)` lists the closed points of degree $d$:

```
E := EllipticCurve([GF(2) | 1, 0, 0, 0, 1]);            // y^2 + xy = x^3 + 1
[#ChangeRing(E, GF(2^n)) : n in [1..8]];     // [ 4, 8, 4, 16, 44, 56, 116, 288 ]
[#Places(E, d) : d in [1..6]];               // [ 4, 2, 0, 2, 8, 8 ]
ZetaFunction(E);              // (2*$.1^2 + $.1 + 1)/(2*$.1^2 - 3*$.1 + 1)
```

Solution.

[a] *Hint:* for $x \neq 0$, substitute $y = xz$.

## 15. Isogenous curves

Problem 21 is [Sil09, Exercise 5.4]. Let $E_1$ and $E_2$ be elliptic curves over $\mathbb{F}_q$, with $q$-power Frobenius endomorphisms $\mathsf{F}_1$ and $\mathsf{F}_2$ and traces $a(E_1)$, $a(E_2)$.

**Definition 15.1** (Isogenies and Galois-equivariant maps)**.** We write $\mathrm{Hom}(E_1, E_2)$ for the isogenies $E_1 \to E_2$ *defined over* $\mathbb{F}_q$. The Galois group $\mathrm{Gal}_{\mathbb{F}_q}$ acts on each Tate module $T_\ell(E_i)$, and we write $\mathrm{Hom}_{\mathrm{Gal}_{\mathbb{F}_q}}\big(T_\ell(E_1), T_\ell(E_2)\big)$ for the $\mathbb{Z}_\ell$-linear maps that commute with this action.

**Theorem 15.2** (Tate's isogeny theorem [Sil09, Theorem III.7.7(a)])**.** *Let $\ell \neq p$ be prime. The natural map*

$$\mathrm{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \to \mathrm{Hom}_{\mathrm{Gal}_{\mathbb{F}_q}}\big(T_\ell(E_1), T_\ell(E_2)\big)$$

*is an isomorphism.*

**Corollary 15.3** (Isogenies form a free module [Sil09, Corollary III.7.5])**.** *The group* $\mathrm{Hom}(E_1, E_2)$ *is a free $\mathbb{Z}$-module of finite rank, being a subgroup of the free module of.*

**Problem 21** (Isogenous curves have the same number of rational points)**.**

(1) Let $\varphi\colon E_1 \to E_2$ be a nonzero isogeny defined over $\mathbb{F}_q$. Show that $\varphi \circ \mathsf{F}_1 = \mathsf{F}_2 \circ \varphi$, and deduce that $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$.

SOLUTION.

(2) Deduce that $Z(E_1/\mathbb{F}_q;T) = Z(E_2/\mathbb{F}_q;T)$.

SOLUTION.

For the converse, suppose that $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$, so that $\mathsf{F}_{1,\ell}$ and $\mathsf{F}_{2,\ell}$ have the same characteristic polynomial $f(T) = T^2 - aT + q$ (Problem 17(3)). Fix a prime $\ell \neq p$ and put $V_i \coloneqq T_\ell(E_i) \otimes \mathbb{Q}_\ell$.

(3) Show that a $\mathbb{Z}_\ell$-linear map $T_\ell(E_1) \to T_\ell(E_2)$ commutes with the action of $\mathrm{Gal}_{\mathbb{F}_q}$ if and only if it intertwines $\mathsf{F}_{1,\ell}$ and $\mathsf{F}_{2,\ell}$.

SOLUTION.

(4) Suppose $a^2 \neq 4q$. Show that $\mathsf{F}_{1,\ell}$ is not a scalar, that $V_1$ has a basis $v, \mathsf{F}_{1,\ell}v$, and that $V_1 \cong \mathbb{Q}_\ell[T]/(f)$ as $\mathbb{Q}_\ell[T]$-modules, with $T$ acting by $\mathsf{F}_{1,\ell}$. Deduce that there is a nonzero $\mathbb{Z}_\ell$-linear map $T_\ell(E_1) \to T_\ell(E_2)$ intertwining the two Frobenius maps.

SOLUTION.

(5) Suppose $a^2 = 4q$. Show that $q$ is a square, that $a/2 \in \mathbb{Z}$, and that $\mathsf{F}_i = [a/2]$ in $\mathrm{End}(E_i)$ for $i = 1, 2$.[a] Deduce that every $\mathbb{Z}_\ell$-linear map $T_\ell(E_1) \to T_\ell(E_2)$ intertwines the two Frobenius maps.

SOLUTION.

(6) Use Theorem 15.2 to conclude that $E_1$ and $E_2$ are isogenous over $\mathbb{F}_q$.

SOLUTION.

[a] *Hint:* compute $\deg(\mathsf{F}_i - [a/2])$ with Proposition 14.4.

**Remark 15.4.** Together with Problem 17(4), Problem 21 says that for elliptic curves $E_1$, $E_2$ over $\mathbb{F}_q$ the following are equivalent: $E_1$ and $E_2$ are isogenous over $\mathbb{F}_q$; $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$; $Z(E_1/\mathbb{F}_q;T) = Z(E_2/\mathbb{F}_q;T)$; and $E_1$ and $E_2$ have the same number of closed points of each degree (Problem 15(4)).

**Looking ahead.** By Problem 18(1), $\alpha = \sqrt{q}\,e^{i\theta}$ for a unique $\theta \in [0,\pi]$, and $a = 2\sqrt{q}\cos\theta$. Fix an elliptic curve over $\mathbb{Q}$, reduce it modulo every prime $p$ of good reduction, and let $\theta_p$ be the angle of the reduction over $\mathbb{F}_p$. How are the $\theta_p$ distributed in $[0,\pi]$ as $p$ varies? That is the Sato–Tate conjecture. The product over the primes of good reduction of the factors $(1 - a_p p^{-s} + p^{1-2s})^{-1}$, the reciprocals of the numerators of the $\zeta$-functions of the reductions, is, up to finitely many factors, the $L$-function of the curve, the subject of the Birch and Swinnerton-Dyer conjecture.

## Part 5. Supersingular curves and the trace of Frobenius

Section 16 proves that an elliptic curve over $\mathbb{F}_q$ is supersingular if and only if $p$ divides its trace of Frobenius, and works out what this says over $\mathbb{F}_p$, over $\mathbb{F}_2$ and $\mathbb{F}_3$, and over $\mathbb{F}_8$. Section 17 shows that $E(\mathbb{F}_q) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/mn\mathbb{Z}$ with $m \mid q-1$, and pins down $m$ for a supersingular curve over $\mathbb{F}_p$. Section 18 shows that an elliptic curve over $\mathbb{Q}$ is ordinary modulo infinitely many primes.

**Reading.** [Sil09, §§V.3–V.4], with [Sil09, §V.2] for the trace of Frobenius, [Sil09, §III.6] for the torsion subgroups and [Sil09, §III.8] for the Weil pairing. The three problems are exercises from [Sil09, Chapter V]: Problem 22 is Exercise 5.10, Problem 23 is Exercise 5.6 and Problem 24 is Exercise 5.11.

`Magma`. If you have no local installation, every block below runs in the free online calculator.

**Standing notation.**

- A power $q = p^e$ of a prime $p$.
- The $p^r$-power Frobenius morphism $\mathsf{F}^r \colon C \to C^{(p^r)}$ of a curve $C$ over a field of characteristic $p$, for $r \geqslant 1$, which raises coordinates to the $p^r$-th power. For an elliptic curve $E$, its dual isogeny $\mathsf{V}^r \colon E^{(p^r)} \to E$ is the Verschiebung.
- The $q$-power Frobenius *endomorphism* of an elliptic curve $E$ over $\mathbb{F}_q$. Since $E^{(q)} = E$, the morphism
$$\mathsf{F}^e \colon E \to E, \qquad (x, y) \mapsto (x^q, y^q),$$
is an endomorphism of $E$, with dual $\mathsf{V}^e \in \mathrm{End}(E)$. Here $\mathrm{End}(E)$ consists of the endomorphisms defined over $\mathbb{F}_q$.
- The group $E[m](\overline{\mathbb{F}}_q)$ of $m$-torsion points of $E$ with coordinates in $\overline{\mathbb{F}}_q$, for $m \geqslant 1$.
- The supersingular elliptic curves $E$ over $\mathbb{F}_q$: those with $E[p^r](\overline{\mathbb{F}}_q) = 0$ for one, equivalently every, $r \geqslant 1$ [Sil09, Theorem V.3.1 and the definition after it]. The others are ordinary. Silverman also says that a supersingular curve has "Hasse invariant 0" and an ordinary one "Hasse invariant 1". That is the phrasing of Exercise 5.11.

## 16. Supersingularity and the trace of Frobenius

This section is [Sil09, Exercise 5.10]. Part (5) is a little stronger than [Sil09, Exercise 5.10(e)], and part (6) needs the stronger form. By Deuring's theorem [Sil09, Theorem V.3.1], a supersingular curve is one whose Verschiebung is purely inseparable, whose endomorphism ring over $\overline{\mathbb{F}}_q$ is an order in a quaternion algebra, and whose formal group has height 2. None of these is easy to read off from a Weierstrass equation. Over a finite field there is a criterion that is: count the rational points.

**Definition 16.1** (Trace of Frobenius)**.** Let $E$ be an elliptic curve over $\mathbb{F}_q$. The trace of Frobenius of $E$ is
$$a \coloneqq q + 1 - \#E(\mathbb{F}_q).$$

**Definition 16.2** (Separable degree)**.** For an isogeny $\psi$, $\deg_s \psi$ denotes its separable degree.

In the following results, $E$ is an elliptic curve over $\mathbb{F}_q$, with $q = p^e$, and $a$ is its trace of Frobenius.

**Proposition 16.3** (Frobenius is purely inseparable [Sil09, Proposition II.2.11])**.** *For $r \geqslant 1$, the Frobenius morphism $\mathsf{F}^r$ is purely inseparable of degree $p^r$.*

**Corollary 16.4** (Purely inseparable isogenies [Sil09, Corollary II.2.12])**.** *A purely inseparable isogeny $\psi \colon E \to E'$ of degree $p^r$ factors as $\psi = \lambda \circ \mathsf{F}^r$ with $\lambda \colon E^{(p^r)} \to E'$ an isomorphism.*

**Theorem 16.5** (Kernels and separable degree [Sil09, Theorem III.4.10(a)])**.** *A nonzero isogeny $\psi$ satisfies $\#\ker\psi = \deg_s \psi$, the kernel taken on points with coordinates in $\overline{\mathbb{F}}_q$.*

**Corollary 16.6** (Separability of $[m] + [n]\mathsf{F}^e$ [Sil09, Corollary III.5.5])**.** *For $m, n \in \mathbb{Z}$, the endomorphism $[m] + [n]\mathsf{F}^e$ is separable if and only if $p \nmid m$.*

**Theorem 16.7** (Dual isogenies [Sil09, Theorem III.6.2(a)])**.** *Every isogeny $\psi$ satisfies $\hat{\psi} \circ \psi = [\deg\psi]$ and $\psi \circ \hat{\psi} = [\deg\psi]$. So $\mathsf{V}^e \circ \mathsf{F}^e = \mathsf{F}^e \circ \mathsf{V}^e = [q]$.*

**Corollary 16.8** (The degree of $[m]$ [Sil09, Corollary III.6.4(a)]). *For $m \in \mathbb{Z}$, the multiplication-by-$m$ map has degree* $\deg[m] = m^2$.

**Corollary 16.9** (The $p$-power torsion [Sil09, Corollary III.6.4(c)]). *Either $E[p^r](\overline{\mathbb{F}}_q) = 0$ for every $r \geqslant 1$, or $E[p^r](\overline{\mathbb{F}}_q) \cong \mathbb{Z}/p^r\mathbb{Z}$ for every $r \geqslant 1$.*

**Theorem 16.10** (The characteristic polynomial of Frobenius [Sil09, Theorem V.2.3.1(b)]). *The Frobenius endomorphism satisfies $(\mathsf{F}^e)^2 - [a]\mathsf{F}^e + [q] = 0$ in* $\mathrm{End}(E)$.

**Theorem 16.11** (Hasse's bound [Sil09, Theorem V.1.1]). *The trace of Frobenius satisfies $|a| \leqslant 2\sqrt{q}$.*

**Problem 22** (Supersingularity and the trace of Frobenius).
Let $E$ be an elliptic curve over $\mathbb{F}_q$.
(1) Show that $\mathsf{V}^e = [a] - \mathsf{F}^e$ in $\mathrm{End}(E)$, and deduce that $\mathsf{V}^e$ is separable if and only if $p \nmid a$. Show that $\ker \mathsf{V}^e$ has $\#E[q](\overline{\mathbb{F}}_q)$ elements, so that $\deg_s \mathsf{V}^e \in \{1, q\}$. Conclude that

$$E \text{ is supersingular} \iff a \equiv 0 \pmod{p}.$$

Solution.

(2) Suppose $q = p \geqslant 5$. Show that $E$ is supersingular if and only if $\#E(\mathbb{F}_p) = p + 1$. Where is $p \geqslant 5$ used?

Solution.

(3) Every elliptic curve over $\mathbb{F}_3$ is isomorphic to one given by $y^2 = x^3 + a_2x^2 + a_4x + a_6$. `Magma` lists the isomorphism classes, with $\#E(\mathbb{F}_3)$, $a$ and whether $E$ is supersingular:

```
K := GF(3);  reps := [];
for a2, a4, a6 in K do
  ok, E := IsEllipticCurve([K | 0, a2, 0, a4, a6]);
  if ok and not exists{F : F in reps | IsIsomorphic(E, F)} then
    Append(~reps, E);
  end if;
end for;
#reps;                                                          // 8
for E in reps do
  printf "%o  %o  %o  %o\n", aInvariants(E), #E,
         TraceOfFrobenius(E), IsSupersingular(E);
end for;
// [ 0, 0, 0, 1, 0 ]  4  0  true
// [ 0, 0, 0, 2, 0 ]  4  0  true
// [ 0, 0, 0, 2, 1 ]  7  -3  true
// [ 0, 0, 0, 2, 2 ]  1  3  true
// [ 0, 1, 0, 0, 1 ]  6  -2  false
// [ 0, 1, 0, 0, 2 ]  3  1  false
// [ 0, 2, 0, 0, 1 ]  5  -1  false
// [ 0, 2, 0, 0, 2 ]  2  2  false
```

Which column decides supersingularity, by part (1)? Count the rational points of $y^2 = x^3 - x + 1$ and of $y^2 = x^3 - x - 1$ by hand, and explain why part (2) fails for $p = 3$.

Solution.

(4) The same over $\mathbb{F}_2$, where a Weierstrass equation needs all five coefficients:

```
K := GF(2);  reps := [];
for a1, a2, a3, a4, a6 in K do
  ok, E := IsEllipticCurve([K | a1, a2, a3, a4, a6]);
  if ok and not exists{F : F in reps | IsIsomorphic(E, F)} then
    Append(~reps, E);
  end if;
end for;
#reps;                                                          // 5
for E in reps do
  printf "%o  %o  %o  %o\n", aInvariants(E), #E,
         TraceOfFrobenius(E), IsSupersingular(E);
end for;
// [ 0, 0, 1, 0, 0 ]  3  0  true
// [ 0, 0, 1, 1, 0 ]  5  -2  true
// [ 0, 0, 1, 1, 1 ]  1  2  true
// [ 1, 0, 0, 0, 1 ]  4  -1  false
// [ 1, 0, 1, 0, 1 ]  2  1  false
```

Count the rational points of the three supersingular curves by hand. Which coefficient decides supersingularity over $\mathbb{F}_2$, and what does the analogue of part (2) say for $p = 2$?

Solution.

(5) Suppose $p \mid a$. Use Corollary 16.4 to write $\mathsf{V}^e = \lambda \circ \mathsf{F}^e$ with $\lambda \in \mathrm{Aut}(E)$, deduce that $[a] = ([1] + \lambda) \circ \mathsf{F}^e$, and conclude that $q \mid a^2$. So

$$a \equiv 0 \pmod{p} \iff q \mid a^2 \iff p^{\lceil e/2 \rceil} \mid a.$$

Solution.

(6) Show that no elliptic curve over $\mathbb{F}_8$ has exactly 7 or exactly 11 rational points. `Magma` collects the values of $\#E(\mathbb{F}_8)$ over all Weierstrass equations, and the corresponding traces:

```
K := GF(8);  counts := {};
for a1, a2, a3, a4, a6 in K do
  ok, E := IsEllipticCurve([a1, a2, a3, a4, a6]);
  if ok then Include(~counts, #E); end if;
end for;
counts;                          // { 4, 5, 6, 8, 9, 10, 12, 13, 14 }
{ 9 - n : n in counts };         // { -5, -4, -3, -1, 0, 1, 3, 4, 5 }
```

Which integers in the Hasse interval are missing, and which traces belong to supersingular curves? Predict both from parts (1) and (5) before you look.

Solution.

**Remark 16.12** (The traces of supersingular curves)**.** The identity $a^2 = \deg([1] + \lambda) \cdot q$ of Problem 22(5) says more than $q \mid a^2$. Since $a^2 \leqslant 4q$, the integer $d := \deg([1] + \lambda)$ is at most 4, so a supersingular curve over $\mathbb{F}_q$ has

$$a^2 \in \{0,\ q,\ 2q,\ 3q,\ 4q\}\,,$$

and only the values with $dq$ a perfect square can occur. Over $\mathbb{F}_8$ these are $a^2 \in \{0, 16\}$, as found above; over $\mathbb{F}_p$ with $p \geqslant 5$ only $a = 0$; over $\mathbb{F}_3$ also $a^2 = 9 = 3q$, and over $\mathbb{F}_2$ also $a^2 = 4 = 2q$, the cases of parts (3) and (4). Which of the five values occur, over which $q$, is a theorem of Waterhouse. We do not need it.

## 17. The group structure of $E(\mathbb{F}_q)$

This section is [Sil09, Exercise 5.6].

In the following results, $E$ is an elliptic curve over a field $K$ and $m \geqslant 1$ is an integer with $\operatorname{char} K \nmid m$.

**Corollary 17.1** (The $m$-torsion points [Sil09, Corollary III.6.4(b)]). *The group $E[m](\overline{K})$ is isomorphic to $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$.*

**Corollary 17.2** (Rational torsion and roots of unity [Sil09, Corollary III.8.1.1]). *If every $m$-torsion point of $E(\overline{K})$ is $K$-rational, then $K$ contains the $m$-th roots of unity.*

**Problem 23** (The group structure of $E(\mathbb{F}_q)$).
Let $E$ be an elliptic curve over $\mathbb{F}_q$.

(1) Show that there are integers $m, n \geqslant 1$ with $p \nmid m$ and $E(\mathbb{F}_q) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/mn\mathbb{Z}$.[a]

Solution.

(2) With $m$ as in part (1), show that $q \equiv 1 \pmod{m}$.

Solution.

(3) Suppose $q = p \geqslant 5$ and $E$ is supersingular. Show that $m = 1$ or $m = 2$, and that $m = 1$ if $p \equiv 1 \pmod{4}$.

Solution.

(4) Both values of $m$ occur when $p \equiv 3 \pmod{4}$. Over $\mathbb{F}_7$ the curves $y^2 = x^3 + x$ and $y^2 = x^3 - x$, both with $j = 1728$, are supersingular. Here `Invariants(AbelianGroup(E))` returns $[m, mn]$, or $[mn]$ when $m = 1$:

```
E1 := EllipticCurve([GF(7) | 1, 0]);  E2 := EllipticCurve([GF(7) | -1, 0]);
IsSupersingular(E1), IsSupersingular(E2);                    // true true
Invariants(AbelianGroup(E1));                                // [ 8 ]
Invariants(AbelianGroup(E2));                                // [ 2, 4 ]
[ j : j in GF(13) | IsSupersingular(EllipticCurveFromjInvariant(j)) ];  // [ 5 ]
E3 := EllipticCurveFromjInvariant(GF(13)!5);  aInvariants(E3);   // [ 1, 0, 0, 7, 2 ]
Invariants(AbelianGroup(E3));                                // [ 14 ]
Invariants(AbelianGroup(QuadraticTwist(E3)));                // [ 14 ]
```

Explain the two values of $m$ over $\mathbb{F}_7$ from the rational points of order 2 of each curve, and the value over $\mathbb{F}_{13}$ from part (3).

Solution.

[a] *Hint:* decompose the finite abelian group $E(\mathbb{F}_q)$ into its $\ell$-primary parts, and count the cyclic factors of each with Corollary 17.1 for $\ell \neq p$ and Corollary 16.9 for $\ell = p$.

## 18. Infinitely many ordinary primes

This section is [Sil09, Exercise 5.11], with "ordinary" for "Hasse invariant 1". Silverman's exercise asks for infinitely many ordinary primes. The curve of part (4) and its supersingular primes are [Sil09, Example V.4.6]. The proof uses results that we have not proved.

**Definition 18.1** (Reduction modulo $p$). Let $E$ be an elliptic curve over $\mathbb{Q}$, given by a Weierstrass equation with coefficients in $\mathbb{Z}$ and discriminant $\Delta \in \mathbb{Z}$. For a prime $p \nmid \Delta$, reducing the coefficients

modulo $p$ gives a Weierstrass equation over $\mathbb{F}_p$ with nonzero discriminant, hence an elliptic curve $E_p$ over $\mathbb{F}_p$.

**Definition 18.2** (Supersingular and ordinary primes)**.** A prime $p \nmid \Delta$ is supersingular or ordinary for $E$ according as $E_p$ is.

**Definition 18.3** (Primes that split completely)**.** Let $L/\mathbb{Q}$ be a finite Galois extension. A prime $p$ splits completely in $L$ if $p\mathcal{O}_L$ is a product of $[L:\mathbb{Q}]$ distinct prime ideals of the ring of integers $\mathcal{O}_L$.

In the following results, $E$, $\Delta$ and $E_p$ are as in Definition 18.1, and $L/\mathbb{Q}$ is a finite Galois extension.

**Proposition 18.4** (Reduction modulo $p$ [Sil09, Proposition VII.2.1])**.** *Let $p \nmid \Delta$. Every point of $E(\mathbb{Q}_p)$ can be written $[X:Y:Z]$ with $X, Y, Z \in \mathbb{Z}_p$ not all divisible by $p$, and reducing the coordinates modulo $p$ gives a group homomorphism $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$.*

**Proposition 18.5** (Reduction of torsion [Sil09, Proposition VII.3.1(b)])**.** *Let $p \nmid \Delta$ and let $m \geqslant 1$ with $p \nmid m$. The restriction of the reduction homomorphism $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ to the $m$-torsion points of $E(\mathbb{Q}_p)$ is injective.*

**Proposition 18.6** (Completions at a completely split prime [Neu99])**.** *Let $p$ be a prime that splits completely in $L$. For each prime $\mathfrak{p}$ of $L$ above $p$ the completion $L_\mathfrak{p}$ is $\mathbb{Q}_p$, so that $L$ embeds in $\mathbb{Q}_p$.*

**Theorem 18.7** (Chebotarev density theorem [Neu99])**.** *Infinitely many primes split completely in $L$: they have density $1/[L:\mathbb{Q}]$ among all primes.*

**Problem 24** (Infinitely many ordinary primes)**.**
Fix a prime $\ell \geqslant 3$, and let $L \coloneqq \mathbb{Q}(E[\ell])$ be the field generated over $\mathbb{Q}$ by the affine coordinates of the nonzero $\ell$-torsion points of $E(\overline{\mathbb{Q}})$.

(1) Show that $L$ is a finite Galois extension of $\mathbb{Q}$, and that $L$ contains the $\ell$-th roots of unity.

Solution.

(2) Let $p \nmid \ell\Delta$ be a prime that splits completely in $L$. Show that $\ell^2 \mid \#E_p(\mathbb{F}_p)$ and that $p \equiv 1 \pmod{\ell}$.

Solution.

(3) Show that every prime $p \geqslant 5$ with $p \nmid \ell\Delta$ that splits completely in $L$ is ordinary for $E$, and conclude. Where did you use $\ell \geqslant 3$?

Solution.

(4) Take the curve $E\colon y^2 + y = x^3 - x^2 - 10x - 20$, of discriminant $-11^5$, and $\ell = 3$. By part (2), a prime $p \neq 3, 11$ that splits completely in $\mathbb{Q}(E[3])$ has all nine 3-torsion points of $E_p(\overline{\mathbb{F}}_p)$ rational. `Magma` lists the supersingular primes below 100 and the primes below 3000 with that property:

```
E := EllipticCurve([0, -1, 1, -10, -20]);          // y^2 + y = x^3 - x^2 - 10x - 20
Discriminant(E);                                    // -161051
good := [p : p in PrimesUpTo(3000) | p ne 11];
[p : p in good | p lt 100 and IsSupersingular(ChangeRing(E, GF(p)))];
// [ 2, 19, 29 ]
full := function(p, l)       // are all the l-torsion points of E_p rational?
  I := Invariants(AbelianGroup(ChangeRing(E, GF(p))));
  return #I eq 2 and I[1] mod l eq 0;
end function;
split := [p : p in good | p ne 3 and full(p, 3)];  split;
```

```
// [ 337, 523, 1087, 2437, 2719, 2749 ]
[p mod 3 : p in split];                                    // [ 1, 1, 1, 1, 1, 1 ]
[IsSupersingular(ChangeRing(E, GF(p))) : p in split];
// [ false, false, false, false, false, false ]
[p : p in good | p lt 300 and p ne 5 and full(p, 5)];      // and the same for l = 5:
// [ 31, 41, 61, 71, 101, 131, 151, 181, 191, 211, 241, 251, 271, 281 ]
```

Check the output against parts (2) and (3), and compare the list of supersingular primes with the list in the example cited before this problem. The last line runs the same test for $\ell = 5$: which primes below 300 appear, and what does that suggest about the field $\mathbb{Q}(E[5])$? (The rational point $(5, 5)$ of $E$ has order 5.)

Solution.

**Remark 18.8** (Supersingular primes)**.** The other direction is much harder. For the curve of Problem 24(4), D. H. Lehmer found exactly 27 odd supersingular primes below 31500 [Sil09, Example V.4.6]. For an elliptic curve over $\mathbb{Q}$ without complex multiplication, the supersingular primes have density 0 (Serre, Elkies), and yet there are infinitely many of them (Elkies) [Sil09, Theorems V.4.7 and V.4.9]. The Lang–Trotter conjecture predicts that about $c\sqrt{x}/\log x$ of them lie below $x$ [Sil09, Conjecture V.4.8]. For a curve with complex multiplication, such as $y^2 = x^3 + x$, the supersingular primes have density 1/2 [Sil09, Examples V.4.4 and V.4.5].

## Part 6. Reduction modulo $p$ and a bound on the torsion

The aim is a `Magma` script, written from scratch, that bounds the torsion subgroup of an elliptic curve over $\mathbb{Q}$ using only the local theory of [Sil09, §§VII.1–VII.3]. Section 19 shows that reduction modulo an odd prime of good reduction is injective on the rational torsion points, Section 20 turns this into two bounds, and Section 21 tests them against the LMFDB. Each problem prints a piece of the script with its output. You prove the mathematics behind it.

**Standing notation.**

- An elliptic curve $E$ over $\mathbb{Q}$, given by a Weierstrass equation with coefficients $a_1, \dots, a_6 \in \mathbb{Z}$ and discriminant $\Delta$.
- The torsion subgroup $T \coloneqq E(\mathbb{Q})_{\mathrm{tors}}$.
- A prime $p$ is good if $p \nmid \Delta$. Then reducing the equation modulo $p$ gives an elliptic curve $E_p$ over $\mathbb{F}_p$.
- For an abelian group $G$, $G[m]$ is the subgroup killed by $m$.
- A finite abelian group $G$ is isomorphic to $\mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}$ for at most one pair of integers $1 \leqslant n_1 \mid n_2$. When it is, we write $n_1(G)$ and $n_2(G)$ for them.

### 19. Reduction is injective on torsion

Everything in this section is in [Sil09, §§VII.1–VII.3]. The curve of Problem 25 is from Application VII.3.5 there.

In the following definition and results, $p$ is a good prime for $E$.

**Definition 19.1** (Reduction modulo $p$)**.** We regard $E(\mathbb{Q})$ as a subgroup of $E(\mathbb{Q}_p)$, and write $P \mapsto \bar{P}$ for the reduction map $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ and $E_1(\mathbb{Q}_p)$ for its kernel.

**Proposition 19.2** (Minimality [Sil09, Remark VII.1.1])**.** *The equation is minimal at $p$, because $v_p(\Delta) = 0 < 12$. So $E_p$ is the reduction of $E$ modulo $p$.*

**Proposition 19.3** (Reduction is a homomorphism [Sil09, Proposition VII.2.1 and the discussion after it]). *The reduction map $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ is a surjective group homomorphism. Its kernel $E_1(\mathbb{Q}_p)$ consists of $O$ and the $(x, y) \in E(\mathbb{Q}_p)$ with $v_p(x) < 0$.*

**Proposition 19.4** (Torsion prime to $p$ [Sil09, Proposition VII.3.1]). *If $p \nmid m$, then $E_1(\mathbb{Q}_p)[m] = \{O\}$, and the reduction map above restricts to an injection $E(\mathbb{Q}_p)[m] \hookrightarrow E_p(\mathbb{F}_p)$.*

**Theorem 19.5** (Torsion of $p$-power order [Sil09, Theorem VII.3.4(b)]). *Let $P \in E(\mathbb{Q}_p)$ have exact order $p^n$ with $n \geqslant 1$, and let $r := \lfloor 1/(p^n - p^{n-1}) \rfloor$. Then $p^{2r} x(P) \in \mathbb{Z}_p$. In particular $x(P) \in \mathbb{Z}_p$ unless $p = 2$ and $n = 1$.*

**Problem 25** (Injectivity of reduction on torsion).

(1) Let $p \geqslant 3$ be a good prime. Show that the reduction map

$$E(\mathbb{Q}) \hookrightarrow E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$$

is injective on $T$. In particular $T$ is finite, and isomorphic to a subgroup of $E_p(\mathbb{F}_p)$.[a]

Solution.

(2) The prime 2 is different: even if $p = 2$ is good, $T \to E_2(\mathbb{F}_2)$ need not be injective. Take for instance the curve $E\colon y^2 + xy = x^3 + 4x + 1$:

```
E := EllipticCurve([1, 0, 0, 4, 1]);            // y^2 + xy = x^3 + 4x + 1
Factorization(Integers()!Discriminant(E));      // [ <5, 2>, <13, 2> ]
P := E![-1/4, 1/8];  2*P eq E!0;                // true
```

So 2 is a good prime and $P = (-1/4, 1/8)$ is a rational point of order 2. Show that $\bar{P} = O$ in $E_2(\mathbb{F}_2)$. Which hypothesis of Proposition 19.4 and Theorem 19.5 fails, and what does Proposition 19.4 still give at $p = 2$?

Solution.

[a] *Hint:* a nonzero torsion point in $E_1(\mathbb{Q}_p)$ has a multiple of prime order in $E_1(\mathbb{Q}_p)$.

## 20. Two bounds

This section is [Sil09, Application VII.3.2], and its three curves are [Sil09, Examples VII.3.3.1–VII.3.3.3].

**Problem 26** (The order bound).
The first piece of the script computes the greatest common divisor $b$ of the orders $\#E_p(\mathbb{F}_p)$ over the good odd primes $p \leqslant N$:

```
// b = gcd of #E_p(F_p) over the good odd primes p <= N (there must be one)
OrderBound := function(E, N)
  Delta := Integers()!Discriminant(E);
  good := [p : p in PrimesUpTo(N) | p gt 2 and Delta mod p ne 0];
  return GCD([#ChangeRing(E, GF(p)) : p in good]);
end function;
E1 := EllipticCurve([0, 0, 1, -1, 1]);          // y^2 + y = x^3 - x + 1
Discriminant(E1);                               // -611
[#ChangeRing(E1, GF(p)) : p in [3, 5, 7]];      // [ 1, 8, 6 ]
OrderBound(E1, 30);                             // 1
```

```
E2 := EllipticCurve([0, 3]);                          // y^2 = x^3 + 3
Discriminant(E2);                                     // -3888
[#ChangeRing(E2, GF(p)) : p in [5, 7, 11, 13]];       // [ 6, 13, 12, 9 ]
OrderBound(E2, 30);                                   // 1
```

(1) Deduce that both curves have trivial torsion, and that $y^2 = x^3 + 3$ has infinitely many rational points.

Solution.

(2) `OrderBound` skips the primes dividing the discriminant of the equation it is given. Compare two equations for the first curve, with $N = 7$:

```
E4 := EllipticCurve([0, 0, 27, -81, 729]);        // y^2 + 27y = x^3 - 81x + 729
Discriminant(E1), Discriminant(E4);               // -611 -324710451
IsIsomorphic(E1, E4);                             // true
OrderBound(E1, 7), OrderBound(E4, 7);             // 1 2
```

Here $-324710451 = -3^{12} \cdot 611$. Find a change of variables between the two equations. Explain why a non-minimal equation can make the script skip a prime it could have used, but never makes it use one it should not.

Solution.

**Corollary 20.1** (The $m$-torsion [Sil09, Corollary III.6.4])**.** *Let $E$ be an elliptic curve over a field $K$, and $m \geqslant 1$. If $\operatorname{char} K \nmid m$, then $E[m](\overline{K}) \cong (\mathbb{Z}/m\mathbb{Z})^2$. If $\operatorname{char} K = p > 0$, then $E[p^e](\overline{K})$ is cyclic for every $e \geqslant 1$.*

**Problem 27** (The structure bound)**.**
The curve $y^2 = x^3 + x$ shows that the orders are not the whole story. `Invariants` returns $[n_1(G), n_2(G)]$ for $G = E_p(\mathbb{F}_p)$, omitting $n_1(G)$ when it is 1, and returns $[\,]$ when $G$ is trivial:

```
E3 := EllipticCurve([1, 0]);                          // y^2 = x^3 + x
Discriminant(E3);                                     // -64
OrderBound(E3, 30);                                   // 4
[Invariants(AbelianGroup(ChangeRing(E3, GF(p)))) : p in [3, 5, 7, 11, 13]];
// [ [ 4 ], [ 2, 2 ], [ 8 ], [ 12 ], [ 2, 10 ] ]
```

(1) Let $G$ be a finite subgroup of $E(\overline{K})$, for an elliptic curve $E$ over a field $K$. Show that $n_1(G)$ and $n_2(G)$ are defined, and that $n_2(G)$ is the exponent of $G$. Show that $\#E[n_1(G)](\overline{K}) = n_1(G)^2$, and that an integer $m \geqslant 1$ with $\#E[m](\overline{K}) = m^2$ satisfies $E[m](\overline{K}) \subseteq G$ if and only if $m \mid n_1(G)$. Deduce that $n_1(H) \mid n_1(G)$ and $n_2(H) \mid n_2(G)$ for every subgroup $H \subseteq G$.[a]

Solution.

(2) Let $b_1$ and $b_2$ be the greatest common divisors of $n_1(E_p(\mathbb{F}_p))$ and of $n_2(E_p(\mathbb{F}_p))$ over the good odd primes $p \leqslant N$. Show that $n_1(T) \mid b_1$, $n_2(T) \mid b_2$ and $b_1 b_2 \mid b$. This is the second piece of the script:

```
// [b1, b2]: n1(T) | b1 and n2(T) | b2
```

```
StructureBound := function(E, N)
  Delta := Integers()!Discriminant(E);
  good := [p : p in PrimesUpTo(N) | p gt 2 and Delta mod p ne 0];
  b1 := 0;  b2 := 0;
  for p in good do
    I := Invariants(AbelianGroup(ChangeRing(E, GF(p))));
    b1 := GCD(b1, #I eq 2 select I[1] else 1);
    b2 := GCD(b2, #I eq 0 select 1 else I[#I]);
  end for;
  return [b1, b2];
end function;
StructureBound(E3, 30);                          // [ 1, 2 ]
P := E3![0, 0];  2*P eq E3!0;                    // true
```

Conclude that $T = \{O, (0,0)\} \cong \mathbb{Z}/2\mathbb{Z}$ for $y^2 = x^3 + x$. Why does $b$ stay at 4 however large $N$ is?

Solution.

(3) Show that a rational point $P$ is a torsion point if and only if $b_2 P = O$, and that rational torsion points generating a subgroup of order $b_1 b_2$ generate $T$.

Solution.

[a] *Hint:* for a prime $\ell$, the group $G[\ell]$ lies in $E[\ell](\overline{K})$. Use Corollary 20.1.

## 21. How good are the bounds? The LMFDB

The L-functions and modular forms database [LMF25] lists millions of elliptic curves over $\mathbb{Q}$ with their torsion subgroups, computed by methods beyond [Sil09, Chapter VII]. We use it as the answer key.

**Proposition 21.1** (Reduction of isogenies [Mil20, proof of Theorem V.4.1])**.** *An isogeny over $\mathbb{Q}$ reduces to an isogeny over $\mathbb{F}_p$ at every good prime $p$ of both equations.*

**Proposition 21.2** (Isogenous curves over $\mathbb{F}_p$ [Sil09, Exercise 5.4(a)])**.** *Isogenous curves over $\mathbb{F}_p$ have the same number of rational points.*

**Problem 28** (Testing the bounds)**.**

(1) Here are the curves of smallest conductor with each torsion structure that occurs over $\mathbb{Q}$, with their LMFDB labels, $a$-invariants and torsion structures ([1] for the trivial group):

```
curves := [<"11.a1", [0, -1, 1, -7820, -263580], [1]>,
  <"14.a2", [1, 0, 1, -171, -874], [2]>,
  <"19.a2", [0, 1, 1, -9, -15], [3]>, <"15.a4", [1, 1, 1, -80, 242], [4]>,
  <"11.a2", [0, -1, 1, -10, -20], [5]>, <"14.a6", [1, 0, 1, 4, -6], [6]>,
  <"26.b2", [1, -1, 1, -3, 3], [7]>, <"15.a8", [1, 1, 1, 35, -28], [8]>,
  <"54.b2", [1, -1, 1, -14, 29], [9]>, <"66.c3", [1, 0, 0, -45, 81], [10]>,
  <"90.c7", [1, -1, 1, -122, 1721], [12]>, <"15.a2", [1, 1, 1, -135, -660], [2, 2]>,
  <"15.a5", [1, 1, 1, -10, -10], [2, 4]>, <"30.a6", [1, 0, 1, -19, 26], [2, 6]>,
  <"210.e6", [1, 0, 0, -1070, 7812], [2, 8]>];
for c in curves do
  E := EllipticCurve(c[2]);
  printf "%-7o LMFDB %-9o order bound %-3o structure bound %o\n",
```

```
    c[1], c[3], OrderBound(E, 50), StructureBound(E, 50);
end for;
// 11.a1   LMFDB [ 1 ]     order bound 5   structure bound [ 1, 5 ]
// 14.a2   LMFDB [ 2 ]     order bound 6   structure bound [ 1, 6 ]
// 19.a2   LMFDB [ 3 ]     order bound 3   structure bound [ 1, 3 ]
// 15.a4   LMFDB [ 4 ]     order bound 8   structure bound [ 1, 4 ]
// 11.a2   LMFDB [ 5 ]     order bound 5   structure bound [ 1, 5 ]
// 14.a6   LMFDB [ 6 ]     order bound 6   structure bound [ 1, 6 ]
// 26.b2   LMFDB [ 7 ]     order bound 7   structure bound [ 1, 7 ]
// 15.a8   LMFDB [ 8 ]     order bound 8   structure bound [ 1, 8 ]
// 54.b2   LMFDB [ 9 ]     order bound 9   structure bound [ 1, 9 ]
// 66.c3   LMFDB [ 10 ]    order bound 10  structure bound [ 1, 10 ]
// 90.c7   LMFDB [ 12 ]    order bound 12  structure bound [ 1, 12 ]
// 15.a2   LMFDB [ 2, 2 ]  order bound 8   structure bound [ 2, 4 ]
// 15.a5   LMFDB [ 2, 4 ]  order bound 8   structure bound [ 2, 4 ]
// 30.a6   LMFDB [ 2, 6 ]  order bound 12  structure bound [ 2, 6 ]
// 210.e6  LMFDB [ 2, 8 ]  order bound 16  structure bound [ 2, 8 ]
```

Check that no bound contradicts the LMFDB. Where is each bound attained?

Solution.

(2) The curves 15.a4 and 15.a8 are isogenous:

```
E := EllipticCurve([1, 1, 1, -80, 242]);        // 15.a4
F := EllipticCurve([1, 1, 1, 35, -28]);          // 15.a8
good := [p : p in PrimesUpTo(50) | p ne 3 and p ne 5];
[#ChangeRing(E, GF(p)) : p in good] eq [#ChangeRing(F, GF(p)) : p in good];
// true
[<p, Invariants(AbelianGroup(ChangeRing(E, GF(p)))),
     Invariants(AbelianGroup(ChangeRing(F, GF(p))))> : p in [7, 11, 13]];
// [ <7, [ 2, 4 ], [ 8 ]>, <11, [ 2, 8 ], [ 16 ]>, <13, [ 16 ], [ 2, 8 ]> ]
```

Using Propositions 21.1 and 21.2, explain why the order bound cannot separate isogenous curves, and why the structure bound separates these two.

Solution.

(3) The LMFDB lists $(5, -2)$ as a torsion generator of 15.a4:

```
E := EllipticCurve([1, 1, 1, -80, 242]);  P := E![5, -2];        // 15.a4
[k*P eq E!0 : k in [1..4]];                       // [ false, false, false, true ]
```

Prove that $T \cong \mathbb{Z}/4\mathbb{Z}$ for 15.a4 without trusting the LMFDB. Can you do the same for 11.a1, 14.a2 or 15.a2?

Solution.

(4) Find these curves yourself: on https://www.lmfdb.org/EllipticCurve/Q/, search by "torsion structure", such as `[2,4]`, and sort by conductor. On a curve's page, find its equation, discriminant, torsion subgroup and bad primes. Pick a curve of your own and compute both bounds.

Solution.

**Remark 21.3** (Beyond the local theory)**.** Over $\mathbb{Q}$ the bounds complete to an algorithm. For $y^2 = x^3 + Ax + B$ with $A, B \in \mathbb{Z}$, a rational torsion point $P \neq O$ has integer coordinates, and either $2P = O$ or $y(P)^2 \mid 4A^3 + 27B^2$ (Nagell–Lutz, [Sil09, Corollary VIII.7.2]). This leaves finitely many candidates, and the test $b_2 P = O$ decides which are torsion points. Mazur proved that $E(\mathbb{Q})_{\mathrm{tors}}$ is one of the fifteen groups in the listing [Sil09, Theorem VIII.7.5].

## Part 7. Good and bad reduction: examples

An elliptic curve over $\mathbb{Q}_p$ has good, multiplicative or additive reduction, and multiplicative reduction is split or nonsplit [Sil09, §VII.5]. The problems below run through concrete curves of each kind: Problem 29 reads the type off an equation, Problem 30 counts the rational points of the singular reduction, and Problem 31 changes the type by twisting the curve or extending the field. Problem 32 ($\star$) goes further: the curves Frey attached to solutions of $A + B = C$ have no additive reduction at all.

**Standing notation.**

- A prime $p$, with $v = v_p$ its valuation on $\mathbb{Q}_p$, extended to $\overline{\mathbb{Q}}_p$.
- An elliptic curve $E$ over $\mathbb{Q}_p$, given by a Weierstrass equation with coefficients in $\mathbb{Z}_p$, with discriminant $\Delta$ and invariants $c_4$, $c_6$ [Sil09, §III.1].
- When the equation is minimal, $E_p$ is its reduction modulo $p$, a Weierstrass cubic over $\mathbb{F}_p$.
- We write
$$a_p \coloneqq p + 1 - \#E_p(\mathbb{F}_p),$$
counting the singular point too.

### 22. Reading off the type

The following definition and results hold for every prime $p$, and are used throughout. In them, $K$ is $\mathbb{Q}_p$ or a finite extension of it, with normalized valuation $v_K$, ring of integers $\mathcal{O}_K$ and residue field $k$, and $E$ is an elliptic curve over $K$ given by a Weierstrass equation with coefficients in $\mathcal{O}_K$.

**Definition 22.1** (Reduction types [Sil09, §VII.5])**.** Suppose that the equation is minimal, and consider its reduction, a Weierstrass cubic over $k$. Then $E$ has good reduction if the reduction is nonsingular, multiplicative reduction if the reduction has a node, and additive reduction if it has a cusp. Multiplicative reduction is split if the slopes of the tangent lines at the node lie in $k$, and nonsplit otherwise.

**Proposition 22.2** (Discriminant and $c_4$ [Sil09, §III.1])**.** *For an equation $y^2 = x^3 + a_2x^2 + a_4x + a_6$ (that is, $a_1 = a_3 = 0$),*

$$\Delta = -16\big(4a_2^3a_6 - a_2^2a_4^2 - 18a_2a_4a_6 + 4a_4^3 + 27a_6^2\big), \qquad c_4 = 16\big(a_2^2 - 3a_4\big), \tag{22.1}$$

*which follow from the formulas of.*

**Proposition 22.3** (Minimal equations [Sil09, Remark VII.1.1])**.** *Any other Weierstrass equation for $E$ has discriminant $u^{-12}\Delta$ and invariant $u^{-4}c_4$ for some $u \in K^\times$. So if $v_K(\Delta) < 12$ or $v_K(c_4) < 4$, the equation is minimal.*

**Proposition 22.4** (Singular Weierstrass cubics [Sil09, Proposition III.1.4(a)])**.** *A Weierstrass cubic over a field with $\Delta = 0$ has exactly one singular point over an algebraic closure. It is a node, with two distinct tangent lines, if $c_4 \neq 0$, and a cusp, with one tangent line, if $c_4 = 0$.*

**Proposition 22.5** (Criteria for the reduction type [Sil09, Proposition VII.5.1])**.** *Suppose that the equation is minimal. Then $E$ has good reduction if $v_K(\Delta) = 0$, multiplicative reduction if $v_K(\Delta) > 0$ and $v_K(c_4) = 0$, and additive reduction if $v_K(\Delta) > 0$ and $v_K(c_4) > 0$.*

**Proposition 22.6** (Extending the field [Sil09, Proposition VII.5.4(b)])**.** *Let $L$ be a finite extension of $K$ with ramification index $e$, so that $v_L = e\, v_K$ on $K$. If $E$ has good, respectively multiplicative, reduction, then so does $E_L$.*

**Proposition 22.7** (Potential good reduction [Sil09, Proposition VII.5.5])**.** *There is a finite extension $L$ of $K$ such that $E_L$ has good reduction if and only if $j(E) \in \mathcal{O}_K$.*

**Problem 29** (Reading off the type)**.**
Let $p \geqslant 5$, and let $\varepsilon \in \mathbb{Z}$ be a non-square modulo $p$. For each equation, decide whether it is minimal at $p$, and find the reduction type of the curve over $\mathbb{Q}_p$: good, split multiplicative, nonsplit multiplicative, or additive.

Solution.

(1) $y^2 = x^3 + x^2 + p$.

Solution.

(2) $y^2 = x^3 + \varepsilon x^2 + p$.

Solution.

(3) $y^2 = x^3 + p$.

Solution.

(4) $y^2 = x^3 + p^6$.

Solution.

(5) $y^2 = x^3 + x^2 + p^{12}$.

Solution.

(6) Here is `Magma`'s opinion at $p = 7$ and $\varepsilon = 3$. For each curve it prints the valuation of the minimal discriminant, the Kodaira symbol of the reduction ($\mathrm{I}_0$ for good reduction, $\mathrm{I}_n$ with $n \geqslant 1$ for multiplicative, anything else for additive), and the polynomial $1 - a_7 T$, or $1 - a_7 T + 7T^2$ in the case of good reduction.

```
R<T> := PolynomialRing(Integers());
p := 7; eps := 3;                          // 3 is not a square modulo 7
Info := function(a)
  E := EllipticCurve(a); L := LocalInformation(E, p);
  return <L[2], L[5], R!EulerFactor(E, p)>;
end function;
Info([0, 1, 0, 0, p]);                     // <1, I1, -T + 1>
Info([0, eps, 0, 0, p]);                   // <1, I1, T + 1>
Info([0, 0, 0, 0, p]);                     // <2, II, 1>
Info([0, 0, 0, 0, p^6]);                   // <0, I0, 7*T^2 + 4*T + 1>
Info([0, 1, 0, 0, p^12]);                  // <12, I12, -T + 1>
```

Check that it agrees with your answers. Note the values of $a_7$ at the curves with bad reduction. Problem 30 explains them.

Solution.

## 23. Counting rational points on the singular reduction

**Definition 23.1** (The nonsingular locus)**.** When the equation of $E$ is minimal, the nonsingular locus $E_p^{\mathrm{ns}}$ is $E_p$ minus its singular point, if it has one.

**Problem 30** (Counting with lines through the singular point)**.**
Let $E$ be an elliptic curve over $\mathbb{Q}_p$ with bad reduction, and let $S$ be the singular point of its reduction $E_p$.

(1) Show that $S \in E_p(\mathbb{F}_p)$, and that $S \neq O$.[a]

Solution.

(2) Show that a change of variables over $\mathbb{F}_p$ moves $S$ to $(0,0)$ and puts $E_p$ in the form

$$y^2 + a_1xy = x^3 + a_2x^2,$$

and that the tangent lines at $S$ are the lines $y = \alpha x$ with $g(\alpha) = 0$, where $g(t) \coloneqq t^2 + a_1t - a_2$.

Solution.

(3) Let $t \in \mathbb{F}_p$. Show that the line $y = tx$ meets $E_p$ at $S$ and, if $g(t) \neq 0$, at exactly one more $\mathbb{F}_p$-rational point $P_t = \big(g(t), t\,g(t)\big)$, and nowhere else in the affine plane. Show that $t \mapsto P_t$ is a bijection from $\{t \in \mathbb{F}_p : g(t) \neq 0\}$ onto $E_p^{\mathrm{ns}}(\mathbb{F}_p) - \{O\}$.

Solution.

(4) Deduce that $\#E_p^{\mathrm{ns}}(\mathbb{F}_p) = p + 1 - r$, where $r$ is the number of roots of $g$ in $\mathbb{F}_p$, and hence that

$$a_p = \begin{cases} 1 & \text{for split multiplicative reduction,} \\ -1 & \text{for nonsplit multiplicative reduction,} \\ 0 & \text{for additive reduction.} \end{cases}$$

Compare with the output in Problem 29(6).

Solution.

(5) Check (4) by hand for $y^2 = x^3 + x^2$ and $y^2 = x^3 + 2x^2$ over $\mathbb{F}_5$.

Solution.

[a] *Hint:* what does the Frobenius automorphism of $\overline{\mathbb{F}}_p$ do to a singular point?

## 24. Changing the curve, changing the field

**Problem 31** (Twists and extensions)**.**
Let $p \geqslant 5$. For $d \in {\mathbb{Q}_p}^{\times}$ and $E\colon y^2 = x^3 + a_2x^2 + a_4x + a_6$, the quadratic twist of $E$ by $d$ is

$$E^d\colon y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6.$$

(1) Show that $(x, y) \mapsto (dx, d^{3/2}y)$ is an isomorphism $E \to E^d$ over $\mathbb{Q}_p(\sqrt{d})$, and that $\Delta(E^d) = d^6\Delta(E)$, $c_4(E^d) = d^2c_4(E)$ and $j(E^d) = j(E)$.

Solution.

From now on, $E\colon y^2 = x^3 + x^2 + p$ is the curve of Problem 29(1), with split multiplicative reduction.

(2) Let $\varepsilon \in \mathbb{Z}$ be a non-square modulo $p$. Show that $E^\varepsilon$ has nonsplit multiplicative reduction, and that $a_p(E^\varepsilon) = -a_p(E)$. Since $\mathbb{Q}_p(\sqrt{\varepsilon})$ is an unramified extension of $\mathbb{Q}_p$, what does this say about which features of the reduction are preserved by unramified extensions?

Solution.

(3) Show that $E^p\colon y^2 = x^3 + px^2 + p^4$ has additive reduction. Over which quadratic extension of $\mathbb{Q}_p$ does it acquire multiplicative reduction? Show that it has good reduction over no finite extension of $\mathbb{Q}_p$.

Solution.

(4) Let $E_3\colon y^2 = x^3 + p$, and let $L$ be a finite extension of $\mathbb{Q}_p$ with ramification index $e$. Show that $E_3$ has good reduction over $L$ if and only if $6 \mid e$.[a]

Solution.

(5) Show that $y^2 = x^3 + px$ has additive reduction, and good reduction over $L$ as in (4) if and only if $4 \mid e$.

Solution.

[a] *Hint:* for "if", write $p = \varpi^e w$ with $\varpi$ a uniformizer of $L$ and $w$ a unit.

## 25. ⋆ Frey curves

This section is [Sil09, Exercise 8.23], with the conductor of [Mil20, Aside II.3.5]. The LMFDB [LMF25] is the answer key.

**Problem 32** (Frey curves are semistable)**.**
An elliptic curve over $\mathbb{Q}$ is semistable if it has good or multiplicative reduction at every prime. Let $A, B, C$ be nonzero integers with $A + B = C$ and $\gcd(A, B, C) = 1$. Let

$$E\colon y^2 = x(x + A)(x - B).$$

(1) Show that $\Delta = 16A^2B^2C^2$ and $c_4 = 16(A^2 + AB + B^2)$.

Solution.

(2) Let $p$ be an odd prime dividing $ABC$. Show that the equation is minimal at $p$ and that $E$ has multiplicative reduction at $p$. Show that the reduction is split if and only if $-B$, $A$ or $B$ is a square modulo $p$, according as $p$ divides $A$, $B$ or $C$.

Solution.

(3) Suppose that $A \equiv 0 \pmod{16}$ and $B \equiv 3 \pmod 4$. Show that the substitution $x = 4x'$, $y = 8y' + 4x'$ gives an equation with coefficients in $\mathbb{Z}$,

$$y'^2 + x'y' = x'^3 + \frac{A - B - 1}{4}\,x'^2 - \frac{AB}{16}\,x',$$

with discriminant $(ABC)^2/2^8$ and $c_4 = A^2 + AB + B^2$. Deduce that $E$ has good reduction at 2 if $v_2(A) = 4$, and multiplicative reduction at 2 if $v_2(A) \geqslant 5$. So $E$ is semistable.

Solution.

(4) The conductor of $E$ is $N = \prod_p p^{f_p}$, where $f_p = 0$ if $E$ has good reduction at $p$, $f_p = 1$ if it has multiplicative reduction, and $f_p \geqslant 2$ if it has additive reduction. Under the hypotheses of part (3), compute $N$ in terms of $A, B, C$. Find the curves for $(A, B, C) = (16, -1, 15)$ and

$(32, -5, 27)$ in the LMFDB, and check their conductors and their split and nonsplit primes against your answers.

SOLUTION.

(5) Suppose that $a^\ell + b^\ell = c^\ell$ with $\ell \geqslant 5$ prime and $a, b, c$ nonzero coprime integers. Show that, after reordering and changing signs, $(A, B, C) = (a^\ell, b^\ell, c^\ell)$ satisfies the hypotheses of part (3), and that the resulting curve has conductor $\prod_{p|abc} p$ and minimal discriminant $(abc)^{2\ell}/2^8$.

SOLUTION.

## Part 8. Compactness and inertia

Let $E$ be an elliptic curve over a local field $K$. The rational points of $E$ with nonsingular reduction form a subgroup $E_0(K)$ of finite index in $E(K)$ [Sil09, Theorem VII.6.1 and Corollary VII.6.2], and $E$ has good reduction if and only if the inertia group acts trivially on its torsion, by the criterion of Néron–Ogg–Shafarevich [Sil09, Theorem VII.7.1]. Problem 33 proves the finiteness of $E(K)/E_0(K)$ by a compactness argument. Problem 34 studies how the inertia group acts on the torsion of a curve with potential good reduction.

**Standing notation.**

- A finite extension $K$ of $\mathbb{Q}_p$, with normalized valuation $v$, ring of integers $\mathcal{O}_K$, uniformizer $\pi$ and residue field $k = \mathcal{O}_K/\pi\mathcal{O}_K$, a finite field of characteristic $p$. (Silverman's Chapter VII assumes that $K$ and $k$ are perfect, and here both are.)
- A fixed algebraic closure $\overline{K}$ of $K$.
- A bar denotes reduction modulo $\pi$.
- An elliptic curve $E$ over $K$, given by a minimal Weierstrass equation $F(X, Y, Z) = 0$, with $F \in \mathcal{O}_K[X, Y, Z]$ homogeneous of degree 3. It embeds $E$ in $\mathbb{P}^2$ as $V_+(F)$.
- The primitive coordinates $[X : Y : Z]$ of $P \in \mathbb{P}^2(K)$: coordinates in $\mathcal{O}_K$, not all in $\pi\mathcal{O}_K$. Every $P$ has them, and they are unique up to a factor in $\mathcal{O}_K^\times$.
- The reduction $\bar{P} := [\bar{X} : \bar{Y} : \bar{Z}] \in \mathbb{P}^2(k)$ of $P \in \mathbb{P}^2(K)$, computed from primitive coordinates, hence well defined.
- The reduced curve $E_\pi$, defined by $\bar{F} = 0$. If $P \in E(K)$, then $\bar{P} \in E_\pi(k)$.

### 26. Finiteness by compactness

This section is [Sil09, Exercise 7.6]. The argument uses that $k$ is finite, so it does not apply, for instance, to the completion of the maximal unramified extension of $\mathbb{Q}_p$: a field complete with respect to a discrete valuation, with residue field $\overline{\mathbb{F}}_p$ (item (6)). Yet this is the case that the criterion of Néron–Ogg–Shafarevich needs: Silverman's proof of it uses the finiteness of $E(K^{\mathrm{unr}})/E_0(K^{\mathrm{unr}})$, where $K^{\mathrm{unr}}$ is the maximal unramified extension of $K$, whose residue field is an algebraic closure of $k$ [Sil09, proof of Theorem VII.7.1].

**Definition 26.1** (The topology of $K$)**.** We give $K$ the topology defined by $v$: the sets $a + \pi^n\mathcal{O}_K$, for $a \in K$ and $n \in \mathbb{Z}$, form a basis, and each of them is both open and closed. With it, $K$ is a Hausdorff topological field.

**Definition 26.2** (The subgroup $E_0(K)$)**.** Let $E_\pi^{\mathrm{ns}}$ be $E_\pi$ minus its singular point, if it has one. Then

$$E_0(K) := \{P \in E(K) : \bar{P} \in E_\pi^{\mathrm{ns}}(k)\}.$$

**Definition 26.3** (Regular functions [Poo06, Definition 1.8.3])**.** A function $\psi \in K(E)$ is regular at $Q \in E(\overline{K})$ if $\psi = g/h$ with $g, h \in K[X, Y, Z]$ homogeneous of the same degree and $h(Q) \neq 0$. Then $\psi(Q) \coloneqq g(Q)/h(Q)$, an element of $K$ if $Q \in E(K)$. The functions regular at $Q$ form a discrete valuation ring with fraction field $K(E)$.

**Proposition 26.4** ($E_0(K)$ is a subgroup [Sil09, Proposition VII.2.1])**.** *The set $E_0(K)$ is a subgroup of $E(K)$.*

**Proposition 26.5** (Compactness [Poo17, §1.1.2])**.** *The ring $\mathcal{O}_K$ is compact: it is complete, and for every $n$ it is covered by the $(\#k)^n$ translates of $\pi^n\mathcal{O}_K$. Equivalently, $K$ is locally compact.*

**Proposition 26.6** (Morphisms $E \to E$ over $K$)**.**

*(1) The morphisms $\alpha\colon E \to E$ defined over $K$ are exactly the points $\alpha = [\psi_0 : \psi_1 : \psi_2] \in E(K(E))$, that is, $\psi_0, \psi_1, \psi_2 \in K(E)$, not all zero, with $F(\psi_0, \psi_1, \psi_2) = 0$* [Poo17, Proposition 3.6.5(b)].
*(2) If $Q \in E(\overline{K})$ and $\psi_0, \psi_1, \psi_2$ are regular at $Q$ and not all zero there, then $\alpha(Q) = [\psi_0(Q) : \psi_1(Q) : \psi_2(Q)]$* [Sil09, §I.3].
*(3) Replacing each $\psi_i$ by $\phi\psi_i$, for $\phi \in K(E)^\times$, gives the same point of $E(K(E))$, and for every $Q \in E(\overline{K})$ some $\phi$ makes $\phi\psi_0, \phi\psi_1, \phi\psi_2$ regular at $Q$ and not all zero there: a power of a uniformizer of the discrete valuation ring of Definition 26.3 will do* [Sil09, proof of Proposition II.2.1].

**Proposition 26.7** (Translations [Sil09, Theorem III.3.6 and Example III.4.7])**.** *Let $P \in E(K)$. The translation $\tau_P\colon E \to E$, $Q \mapsto Q + P$, is an isomorphism defined over $K$, with inverse $\tau_{-P}$.*

**Problem 33** ($E(K)/E_0(K)$ is finite)**.**
Let $q\colon K^3 - \{0\} \to \mathbb{P}^2(K)$ be the quotient map. Give $K^3 - \{0\}$ the product topology and $\mathbb{P}^2(K)$ the quotient topology: $U \subseteq \mathbb{P}^2(K)$ is open if and only if $q^{-1}(U)$ is open. Let

$$S \coloneqq \{\mathbf{x} = (x_0, x_1, x_2) \in \mathcal{O}_K^3 : x_i \in \mathcal{O}_K^\times \text{ for some } i\}$$

be the set of primitive coordinate vectors.

(1) Show that $S$ is compact and open in $K^3 - \{0\}$, and that $q(S) = \mathbb{P}^2(K)$. Deduce that $\mathbb{P}^2(K)$ is compact. Show that $U \subseteq \mathbb{P}^2(K)$ is open if and only if $q^{-1}(U) \cap S$ is open in $S$.[a]

Solution.

(2) Show that $E(K) = V_+(F)(K)$ is closed in $\mathbb{P}^2(K)$, and deduce that it is compact.

Solution.

(3) Let $g, h \in K[X, Y, Z]$ be homogeneous of the same degree. Show that $W \coloneqq \{Q \in \mathbb{P}^2(K) : h(Q) \neq 0\}$ is open, and that $W \to K$, $Q \mapsto g(Q)/h(Q)$, is continuous. Deduce from Propositions 26.6 and 26.7 that every translation $\tau_P\colon E(K) \to E(K)$ is a homeomorphism.

Solution.

(4) Show that the reduction map $\mathbb{P}^2(K) \to \mathbb{P}^2(k)$, $P \mapsto \bar{P}$, is locally constant. Deduce that $E_0(K)$ is open in $E(K)$, and also closed.

Solution.

(5) Prove that $E(K)/E_0(K)$ is finite.

Solution.

(6) Let $K'$ be a field complete with respect to a discrete valuation whose residue field $k'$ is infinite, for instance algebraically closed. Show that its ring of integers is not compact. Which step of the argument above fails for an elliptic curve over $K'$?

Solution.

[a] *Hint:* $q^{-1}(U)$ is stable under multiplication by $K^\times$.

## 27. Inertia acting on torsion

This section is [Sil09, Exercise 7.9], with the hypothesis $m \geqslant 3$ added to its first two parts (item (5) explains why) and a `Magma` check. We use the notation of [Sil09, §VII.4].

**Definition 27.1** (Reduction types)**.** The curve $E$ has good reduction if $E_\pi$ is nonsingular, that is, if $v(\Delta) = 0$, and bad reduction otherwise. It has potential good reduction if it has good reduction over some finite extension of $K$.

**Definition 27.2** (Inertia)**.** Let $\mathrm{Gal}_K := \mathrm{Gal}(\overline{K}/K)$, $K^{\mathrm{unr}} \subseteq \overline{K}$ the maximal unramified extension of $K$, and $I_v := \mathrm{Gal}(\overline{K}/K^{\mathrm{unr}})$ the inertia group. A set on which $\mathrm{Gal}_K$ acts is unramified if $I_v$ acts trivially on it.

**Definition 27.3** (Ramification of finite extensions)**.** For an extension $L$ of $K$ inside $\overline{K}$, let $\mathrm{Gal}_L := \mathrm{Gal}(\overline{K}/L)$. If $L/K$ is finite and Galois, the inertia group of $L/K$ is the image of $I_v$ in $\mathrm{Gal}(L/K)$, isomorphic to $I_v/(I_v \cap \mathrm{Gal}_L)$. The extension $L/K$ is unramified if its inertia group is trivial, and tamely ramified if its order is prime to $p$.

**Definition 27.4** (Torsion fields and Tate modules [Sil09, §III.7])**.** For $m \geqslant 1$ with $p \nmid m$, $K(E[m])$ is the field generated over $K$ by the coordinates of the points of $E[m](\overline{K})$, so $\mathrm{Gal}_{K(E[m])}$ is the kernel of the action of $\mathrm{Gal}_K$ on $E[m](\overline{K})$. For a prime $\ell \neq p$, the Tate module $T_\ell(E)$ is the inverse limit of the $E[\ell^n](\overline{K})$ along the surjections $[\ell]$, and $\rho_\ell\colon \mathrm{Gal}_K \to \mathrm{Aut}\, T_\ell(E)$ is the action of $\mathrm{Gal}_K$ on it.

In the following results, $\ell \neq p$ is a prime and $m \geqslant 1$ is an integer with $p \nmid m$.

**Proposition 27.5** (Minimal equations [Sil09, Remark VII.1.1 and §III.1])**.** *A Weierstrass equation with coefficients in $\mathcal{O}_K$ and $v(\Delta) < 12$ is minimal. For $y^2 = x^3 + Ax + B$, $\Delta = -16(4A^3 + 27B^2)$.*

**Corollary 27.6** (Torsion [Sil09, Corollary III.6.4(b)])**.** *The group $E[m](\overline{K})$ is isomorphic to $(\mathbb{Z}/m\mathbb{Z})^2$.*

**Proposition 27.7** (The Tate module [Sil09, Proposition III.7.1(a)])**.** *The Tate module $T_\ell(E)$ is isomorphic to $\mathbb{Z}_\ell^2$. Hence each projection $T_\ell(E) \to E[\ell^n](\overline{K})$ is a surjective homomorphism of $\mathrm{Gal}_K$-modules, and its kernel is $\ell^n T_\ell(E)$, since both have index $\ell^{2n}$.*

**Proposition 27.8** (Good reduction [Sil09, Proposition VII.4.1])**.** *If $E$ has good reduction, then $E[m](\overline{K})$ and $T_\ell(E)$ are unramified.*

**Theorem 27.9** (Néron–Ogg–Shafarevich [Sil09, Theorem VII.7.1])**.** *If $T_\ell(E)$ is unramified, then $E$ has good reduction.*

**Corollary 27.10** (Potential good reduction [Sil09, Corollary VII.7.3])**.** *The curve $E$ has potential good reduction if and only if $\rho_\ell(I_v)$ is finite.*

**Proposition 27.11** (Inertia groups of finite extensions [Sil09, proof of Corollary VII.7.3])**.** *If $H \subseteq I_v$ is a closed normal subgroup of finite index, there is a finite extension $K'$ of $K$ inside $\overline{K}$ whose inertia group $\mathrm{Gal}(\overline{K}/K'K^{\mathrm{unr}}) = I_v \cap \mathrm{Gal}_{K'}$ is $H$.*

**Problem 34** (The inertia group of $K(E[m])/K$)**.**
Suppose that $E$ has potential good reduction, and let $J$ be the subgroup of the elements of $I_v$ that act trivially on $E[m](\overline{K})$ for every $m \geqslant 1$ with $p \nmid m$.

(1) Let $\ell$ be a prime and $r \geqslant 1$ with $\ell^r \geqslant 3$, and let $g \in \mathrm{GL}_2(\mathbb{Z}_\ell)$ be an element of finite order with $g \equiv I \pmod{\ell^r}$. Show that $g = I$.[a]

Solution.

(2) Show that $J = I_v \cap \ker \rho_\ell$ for every prime $\ell \neq p$, and that $J$ has finite index in $I_v$.

Solution.

(3) Let $m \geqslant 3$ with $p \nmid m$. Show that $J = I_v \cap \mathrm{Gal}_{K(E[m])}$, and deduce that the inertia group of $K(E[m])/K$ is isomorphic to $I_v/J$, independently of $m$.

Solution.

(4) Let $m \geqslant 3$ with $p \nmid m$. Show that $K(E[m])/K$ is unramified if and only if $E$ has good reduction. Which direction holds for every $m \geqslant 1$ with $p \nmid m$, and for every elliptic curve over $K$?

Solution.

(5) Let $p \geqslant 3$ and $E\colon y^2 = x^3 - \pi^2 x$.

(a) Show that $E$ has bad reduction, potential good reduction, and $K(E[2]) = K$.

Solution.

(b) Show that $I_v/J$ has order 2, and that its nontrivial element acts on every $T_\ell(E)$ as $-1$.[b]

Solution.

(c) So (3) and (4) fail for $m = 2$. Where did your proofs use $m \geqslant 3$?

Solution.

(6) Suppose that $p \geqslant 5$. Show that $K(E[m])/K$ is tamely ramified for every $m \geqslant 1$ with $p \nmid m$.[c]

Solution.

(7) `Magma` computes $\#(I_v/J)$ for elliptic curves over $\mathbb{Q}_7$:

```
p := 7;
Inertia := func< A, B |
  #InertiaGroup(GaloisRepresentation(EllipticCurve([A, B]), p)) >;
Inertia(-1, 0);                    // 1
Inertia(-p^2, 0);                  // 2
Inertia(0, p);                     // 6
```

Check the first two values against your answers, and the third against (6).

Solution.

[a] *Hint:* reduce to $g$ of prime order $q$, write $g = I + \ell^s h$ with $h \in M_2(\mathbb{Z}_\ell)$ not divisible by $\ell$, and expand $g^q$.
[b] *Hint:* over $K(\sqrt{\pi})$, $E$ is isomorphic to $y^2 = x^3 - x$.
[c] *Hint:* $\#\,\mathrm{GL}_2(\mathbb{F}_3) = 48$.

## Part 9. Hilbert 90 and Kummer theory

Hilbert's Theorem 90 says that $H^1(k, \mathbb{G}_\mathrm{m}) = 1$. Applied to the Kummer sequence $1 \to \mu_m \to \mathbb{G}_\mathrm{m} \xrightarrow{m} \mathbb{G}_\mathrm{m} \to 1$, it gives an isomorphism $k^\times/k^{\times m} \cong H^1(k, \mu_m)$ with an explicit formula, and, when $k$ contains the $m$-th roots of unity, a classification of the abelian extensions of $k$ of exponent dividing

$m$. The problems below prove all of this from the definitions in [Sil09, Appendix B]. Problem 35 computes cohomology groups by hand, Problem 36 proves Hilbert 90, Problem 37 unwinds the Kummer sequence and Problem 38 derives Kummer theory. Problems 39 and 40 compute the groups $k^\times/k^{\times m}$ for finite, local and global fields, and cut out a finite subgroup by conditions at the primes. Later, the same sequence for an elliptic curve $E$ will give an injection $E(k)/m \hookrightarrow H^1(k, E[m])$. Then there is no Hilbert 90, and its failure is measured by $H^1(k, E)$.

**Standing notation.**

- A field $k$, not assumed perfect, a separable closure $k^{\mathrm{sep}}$ of $k$, and the absolute Galois group $\mathrm{Gal}_k := \mathrm{Gal}(k^{\mathrm{sep}}/k)$. For an extension $L$ of $k$ inside $k^{\mathrm{sep}}$, $\mathrm{Gal}_L := \mathrm{Gal}(k^{\mathrm{sep}}/L)$. (Silverman's Appendix B assumes $k$ perfect and works with an algebraic closure. Nothing in it changes with $k^{\mathrm{sep}}$ in its place.)
- A profinite group $G$, often finite, and a discrete $G$-module $A$: an abelian group with a left action of $G$, written $a \mapsto ga$, such that $G \times A \to A$ is continuous for the discrete topology on $A$. (Silverman writes right actions, $a \mapsto a^\sigma$.)
- Elements $g, h$ of $G$, and elements $\sigma, \tau, \gamma$ of a Galois group such as $\mathrm{Gal}_k$ or $\mathrm{Gal}(L/k)$. Where $\sigma$ is a generator of a cyclic group, it keeps that meaning.
- An integer $m \geqslant 1$ with $\mathrm{char}\, k \nmid m$, and the algebraic group $\mu_m$ of $m$-th roots of unity, the kernel of $x \mapsto x^m$ on $\mathbb{G}_{\mathrm{m}}$. For a field $L$ containing $k$, $\mu_m(L)$ is the group of $m$-th roots of unity in $L$.
- **An abuse of notation.** For a commutative algebraic group $A$ over $k$, such as $\mathbb{G}_{\mathrm{m}}$, $\mu_m$ or an elliptic curve, we write $H^i(k, A) := H^i(\mathrm{Gal}_k, A(k^{\mathrm{sep}}))$, the cohomology of the $\mathrm{Gal}_k$-module of $k^{\mathrm{sep}}$-points of $A$. For instance,
$$H^1(k, \mathbb{G}_{\mathrm{m}}) = H^1(\mathrm{Gal}_k, (k^{\mathrm{sep}})^\times), \qquad H^1(k, \mu_m) = H^1(\mathrm{Gal}_k, \mu_m(k^{\mathrm{sep}})).$$
This is the only place where $A$ stands for $A(k^{\mathrm{sep}})$.
- The quotient $B/m := B/mB$ of an abelian group $B$. For the multiplicative group of a field $L$ it is $L^\times/L^{\times m}$, where $L^{\times m}$ is the subgroup of $m$-th powers.

## 28. Cocycles by hand

This section uses the definitions of [Sil09, §§B.1–B.2] and [Poo02, §§2–3]. Its last item is [Sil09, Exercise B.2(a)].

**Definition 28.1** (Cohomology in degrees 0 and 1 [Sil09, §§B.1–B.2], [Poo02, §3])**.** Let $A$ be a discrete $G$-module. Then $H^0(G, A) := A^G$, the subgroup of the $a \in A$ with $ga = a$ for every $g \in G$. A left 1-cocycle, or simply 1-cocycle, is a continuous map $\xi\colon G \to A$, $g \mapsto \xi_g$, such that
$$\xi_{gh} = \xi_g + g\xi_h \qquad \text{for all } g, h \in G.$$
A left 1-coboundary, or simply 1-coboundary, is a map of the form $g \mapsto ga - a$, for some $a \in A$. Both are *left*: they are written for the left action $a \mapsto ga$. Silverman's are the right ones, for $a \mapsto a^\sigma$: $\xi_{\sigma\tau} = \xi_\sigma^\tau + \xi_\tau$ and $\xi_\sigma = a^\sigma - a$. The 1-cocycles form a group $Z^1(G, A)$ under addition of values, the 1-coboundaries a subgroup $B^1(G, A)$, and $H^1(G, A) := Z^1(G, A)/B^1(G, A)$. If $G$ is finite, every map $G \to A$ is continuous. When the group law of $A$ is written multiplicatively, as for $L^\times$ or $\mu_m(L)$, the conditions read $\xi_{gh} = \xi_g \cdot g(\xi_h)$ and $\xi_g = g(a)/a$.

**Proposition 28.2** (Trivial action [Sil09, Remarks B.1.1 and B.2.2], [Poo02, §3])**.** *If $G$ acts trivially on $A$, then $H^0(G, A) = A$ and $H^1(G, A) = \mathrm{Hom}_{\mathrm{conts}}(G, A)$, the group of continuous homomorphisms.*

**Problem 35** (Computing $H^1$ from the definition)**.**
In this problem $G$ is finite.

(1) Let $G = \{1, \sigma\}$ have order 2, acting on $\mathbb{Z}$ by $\sigma a = -a$. Compute $H^0(G, \mathbb{Z})$ and $H^1(G, \mathbb{Z})$. Compare with the trivial action of $G$ on $\mathbb{Z}$.

SOLUTION.

(2) Let $G = \langle\sigma\rangle$ be cyclic of order $n$, and $N \coloneqq 1 + \sigma + \cdots + \sigma^{n-1}$, acting on $A$. Show that $\xi \mapsto \xi_\sigma$ induces an isomorphism[a]

$$H^1(G, A) \cong \ker(N\colon A \to A)/(\sigma - 1)A.$$

SOLUTION.

(3) Let $G = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \{1, c\}$. Use (2) to compute $H^1(G, \mathbb{C}^\times)$ and $H^1(G, \mu_m(\mathbb{C}))$. Compare the order of the second group with that of $\mathbb{R}^\times/\mathbb{R}^{\times m}$.

SOLUTION.

(4) Suppose that $G$ has order $n$. Show that $nH^1(G, A) = 0$. [b]

SOLUTION.

[a] *Hint:* show that $\xi_{\sigma^i} = (1 + \sigma + \cdots + \sigma^{i-1})\xi_\sigma$ for $i \geqslant 1$.
[b] *Hint:* sum the cocycle condition $\xi_{gh} = \xi_g + g\xi_h$ over $h \in G$.

## 29. Hilbert 90

This section is [Poo17, Proposition 1.3.15(ii) and Exercises 1.10–1.11], with the proof of [Mil20, Proposition IV.1.3] (also [Mil22, Theorem 5.23]). [Sil09, Proposition B.2.5] states the result and refers to Serre for the proof. Item (2) is the form in which Hilbert stated it.

**Theorem 29.1** (Dedekind [Mil22, Theorem 5.14])**.** *Let $F$ be a field and $H$ a group. If $\chi_1, \ldots, \chi_r$ are distinct homomorphisms $H \to F^\times$, then they are linearly independent over $F$: if $a_1\chi_1 + \cdots + a_r\chi_r = 0$ as a function $H \to F$, with $a_i \in F$, then $a_1 = \cdots = a_r = 0$.*

**Definition 29.2** (Norm and trace)**.** Let $L/k$ be a finite Galois extension with group $G$. The norm and the trace of $a \in L$ are $N_{L/k}(a) \coloneqq \prod_{\sigma\in G} \sigma(a)$ and $\mathrm{Tr}_{L/k}(a) \coloneqq \sum_{\sigma\in G} \sigma(a)$. Both lie in $k$.

**Proposition 29.3** (Infinite Galois theory)**.**
*(1) The subgroups $\mathrm{Gal}_L$, for $L$ running over the finite Galois extensions of $k$ inside $k^{\mathrm{sep}}$, are open and normal in $\mathrm{Gal}_k$, and they form a basis of open neighborhoods of 1* [Sil09, §B.2].
*(2) The maps $H \mapsto (k^{\mathrm{sep}})^H$ and $L \mapsto \mathrm{Gal}_L$ are inverse bijections between the closed subgroups of $\mathrm{Gal}_k$ and the extensions of $k$ inside $k^{\mathrm{sep}}$. The subgroup $H$ is open if and only if $(k^{\mathrm{sep}})^H$ is finite over $k$, and then $[\mathrm{Gal}_k : H] = [(k^{\mathrm{sep}})^H : k]$. It is normal if and only if $(k^{\mathrm{sep}})^H$ is Galois over $k$, and then $\mathrm{Gal}((k^{\mathrm{sep}})^H/k) \cong \mathrm{Gal}_k/H$ by restriction. In particular, $(k^{\mathrm{sep}})^{\mathrm{Gal}_k} = k$* [Mil22, Theorem 7.13].

**Definition 29.4** (Restriction and inflation [Sil09, §B.2], [Poo02, §4])**.** Let $A$ be a discrete $G$-module and $H \subseteq G$ a closed subgroup. The restriction $\mathrm{Res}\colon H^1(G, A) \to H^1(H, A)$ maps the class of $\xi$ to the class of $\xi|_H$. If $H$ is normal, the inflation $\mathrm{Inf}\colon H^1(G/H, A^H) \to H^1(G, A)$ maps the class of $\xi$ to the class of the composition $G \to G/H \xrightarrow{\xi} A^H \subseteq A$. A 1-cocycle of $G$ of this form is inflated from $G/H$.

**Problem 36** (Hilbert's Theorem 90)**.**

(1) Let $L/k$ be a finite Galois extension with group $G$, and $\xi\colon G \to L^\times$ a 1-cocycle. Show that there is a $c \in L^\times$ with

$$b \coloneqq \sum_{\tau \in G} \xi_\tau \cdot \tau(c) \neq 0,$$

and that then $\xi_\sigma = b/\sigma(b)$ for every $\sigma \in G$. Deduce that $H^1(G, L^\times) = 1$. [a]

Solution.

(2) Suppose moreover that $G = \langle \sigma \rangle$ is cyclic. Show that if $a \in L^\times$ has $N_{L/k}(a) = 1$, then $a = \sigma(b)/b$ for some $b \in L^\times$.

Solution.

(3) Let $(x, y)$ be a rational point of the circle $x^2 + y^2 = 1$. Apply (2) to $\mathbb{Q}(i)/\mathbb{Q}$ to find $u, v \in \mathbb{Q}$, not both 0, with

$$(x, y) = \Big(\frac{u^2 - v^2}{u^2 + v^2}, \frac{2uv}{u^2 + v^2}\Big).$$

Solution.

(4) Let $\xi\colon \mathrm{Gal}_k \to (k^{\mathrm{sep}})^\times$ be a continuous 1-cocycle.

(a) Show that there is a finite Galois extension $L$ of $k$ inside $k^{\mathrm{sep}}$ such that $\xi$ takes values in $L^\times$ and $\xi_{\sigma\tau} = \xi_\sigma$ for all $\sigma \in \mathrm{Gal}_k$ and $\tau \in \mathrm{Gal}_L$. So $\xi$ is inflated from a 1-cocycle $\mathrm{Gal}(L/k) \to L^\times$. [b]

Solution.

(b) Deduce that $H^1(k, \mathbb{G}_{\mathrm{m}}) = 1$.

Solution.

(5) ($\star$) Let $L/k$ be finite Galois with group $G$, and $\xi\colon G \to L$ a 1-cocycle for the additive group of $L$. Show that $H^1(G, L) = 0$, by a variant of (1). [c]

Solution.

[a] *Hint:* compute $\sigma(b)$ with the cocycle condition.
[b] *Hint:* $\mathrm{Gal}_k$ is compact, and $\xi^{-1}(1)$ is open.
[c] *Hint:* take $c \in L$ with $\mathrm{Tr}_{L/k}(c) \neq 0$.

## 30. The Kummer sequence

This section is the exercise in [Poo02, §3], with the explicit formula of [Sil09, Propositions B.2.5(c) and VIII.2.2]. Hilbert's Theorem 90, $H^1(k, \mathbb{G}_{\mathrm{m}}) = 1$, is Problem 36(4).

**Proposition 30.1** (Finite multiplicative groups [Mil22, Exercise 1-3])**.** *Every finite subgroup of the multiplicative group of a field is cyclic.*

**Proposition 30.2** (The long exact sequence [Sil09, Propositions B.1.2 and B.2.3], [Poo02, Theorem 1])**.** *Let $0 \to A \to B \xrightarrow{\psi} C \to 0$ be an exact sequence of discrete $G$-modules. There is an exact sequence*

$$0 \to A^G \to B^G \to C^G \xrightarrow{\delta} H^1(G, A) \to H^1(G, B) \to H^1(G, C),$$

*where, for $c \in C^G$, $\delta(c)$ is the class of $g \mapsto gb - b$ for any $b \in B$ with $\psi(b) = c$.*

**Proposition 30.3** (Inflation–restriction [Sil09, Proposition B.2.4])**.** *Let $A$ be a discrete $\mathrm{Gal}_k$-module and $L/k$ a finite Galois extension inside $k^{\mathrm{sep}}$. The sequence*

$$0 \to H^1(\mathrm{Gal}(L/k), A^{\mathrm{Gal}_L}) \xrightarrow{\mathrm{Inf}} H^1(\mathrm{Gal}_k, A) \xrightarrow{\mathrm{Res}} H^1(\mathrm{Gal}_L, A)$$

*is exact.*

**Problem 37** (The Kummer sequence)**.**
Recall that $\operatorname{char} k \nmid m$, so $m \neq 0$ in $k$.

(1) Show that $\mu_m(k^{\mathrm{sep}})$ is cyclic of order $m$, and that

$$1 \to \mu_m(k^{\mathrm{sep}}) \to (k^{\mathrm{sep}})^\times \xrightarrow{x \mapsto x^m} (k^{\mathrm{sep}})^\times \to 1$$

is an exact sequence of discrete $\mathrm{Gal}_k$-modules. Deduce from Proposition 30.2 that its connecting map induces an isomorphism $\delta\colon k^\times/k^{\times m} \to H^1(k, \mu_m)$.

Solution.

(2) Let $a \in k^\times$ and $\alpha \in k^{\mathrm{sep}}$ with $\alpha^m = a$. Check directly, without Proposition 30.2, that $\xi_\sigma := \sigma(\alpha)/\alpha$ is a continuous 1-cocycle with values in $\mu_m(k^{\mathrm{sep}})$, whose class does not depend on the choice of $\alpha$. Show that $a \mapsto [\xi]$ is a homomorphism $k^\times \to H^1(k, \mu_m)$ with kernel $k^{\times m}$. This is the map $\delta$ of (1).

Solution.

(3) Show directly from Hilbert 90 that every continuous 1-cocycle $\xi\colon \mathrm{Gal}_k \to \mu_m(k^{\mathrm{sep}})$ is cohomologous to one of the form of (2).

Solution.

(4) Let $m = 2$, with $\operatorname{char} k \neq 2$. Show that $H^1(k, \mu_2) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \{\pm 1\})$, and that for $d \in k^\times - k^{\times 2}$, $\delta(d)$ is the homomorphism $\chi_d\colon \sigma \mapsto \sigma(\sqrt{d})/\sqrt{d}$, with kernel $\mathrm{Gal}_{k(\sqrt{d})}$. Deduce that $d \mapsto k(\sqrt{d})$ is a bijection from the nontrivial elements of $k^\times/k^{\times 2}$ to the quadratic extensions of $k$ inside $k^{\mathrm{sep}}$.

Solution.

(5) Let $L$ be an extension of $k$ inside $k^{\mathrm{sep}}$. Show that the restriction $H^1(k, \mu_m) \to H^1(L, \mu_m)$ corresponds, under $\delta$, to the map $k^\times/k^{\times m} \to L^\times/L^{\times m}$ induced by inclusion, and that its kernel corresponds to $(k^\times \cap L^{\times m})/k^{\times m}$. Compute the kernel for $\mathbb{Q}(i)/\mathbb{Q}$ and $m = 2$, and compare it with Proposition 30.3.

Solution.

## 31. Kummer theory

This section is [Sil09, Exercise 8.4], in the form of [Mil22, Theorem 5.30 and Remark 5.32], together with an example that shows why the roots of unity are needed. In it, $\mu_m(k^{\mathrm{sep}}) \subseteq k$, except in item (4).

**Proposition 31.1** (Characters of a finite abelian group [Mil22, discussion before Theorem 5.30])**.** *If $B$ is a finite abelian group of exponent dividing $m$, then $\mathrm{Hom}(B, \mu_m(k^{\mathrm{sep}}))$ has the same order as $B$.*

**Problem 38** (Abelian extensions of exponent dividing $m$)**.**

(1) Suppose that $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Show that $\delta$ is an isomorphism

$$\delta\colon k^\times/k^{\times m} \to \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \mu_m(k^{\mathrm{sep}})).$$

For $a \in k^\times$ and $\alpha^m = a$, show that $\ker \delta(a) = \mathrm{Gal}_{k(\alpha)}$, and deduce that $k(\alpha)/k$ is Galois and cyclic, of degree the order of $a$ in $k^\times/k^{\times m}$.

Solution.

(2) Suppose that $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Let $\Delta$ be a subgroup of $k^\times$ containing $k^{\times m}$, with $\Delta/k^{\times m}$ finite, and $L \coloneqq k(\Delta^{1/m})$ the extension of $k$ inside $k^{\mathrm{sep}}$ generated by the $m$-th roots of the elements of $\Delta$. Show that $L/k$ is finite and Galois, and that

$$\langle\ ,\ \rangle\colon \mathrm{Gal}(L/k) \times \Delta/k^{\times m} \to \mu_m(k^{\mathrm{sep}}), \qquad \langle\sigma, a\rangle \coloneqq \sigma(\alpha)/\alpha \quad (\alpha^m = a),$$

is a well defined bilinear pairing, and that it is perfect: the induced maps

$$\mathrm{Gal}(L/k) \to \mathrm{Hom}(\Delta/k^{\times m}, \mu_m(k^{\mathrm{sep}})), \qquad \Delta/k^{\times m} \to \mathrm{Hom}(\mathrm{Gal}(L/k), \mu_m(k^{\mathrm{sep}}))$$

are isomorphisms. Deduce that $\mathrm{Gal}(L/k)$ is abelian of exponent dividing $m$ and that $[L : k] = (\Delta : k^{\times m})$.[a]

Solution.

(3) Suppose that $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Show that $\Delta \mapsto k(\Delta^{1/m})$ is a bijection from the subgroups $\Delta$ as in (2) to the finite abelian extensions of $k$ inside $k^{\mathrm{sep}}$ of exponent dividing $m$, with inverse $L \mapsto \Delta_L \coloneqq k^\times \cap L^{\times m}$.[b]

Solution.

(4) ($\star$) Let $k = \mathbb{Q}$, $m = 3$, $\zeta \in \overline{\mathbb{Q}}$ a primitive cube root of unity, and $F \coloneqq \mathbb{Q}(\zeta)$. Here $\mu_3(\overline{\mathbb{Q}}) \not\subseteq \mathbb{Q}$.

(a) Show that $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not Galois, although $2 \notin \mathbb{Q}^{\times 3}$. Which step of (1) fails?

Solution.

(b) Show that $\mathbb{Q}^\times/\mathbb{Q}^{\times 3} \to F^\times/F^{\times 3}$ is injective.[c]

Solution.

(c) Let $a \in \mathbb{Q}^\times - \mathbb{Q}^{\times 3}$, $\alpha^3 = a$, and $L \coloneqq F(\alpha)$. Show that $L/\mathbb{Q}$ is Galois of degree 6, and that conjugation by any $\tau \in \mathrm{Gal}(L/\mathbb{Q})$ with $\tau(\zeta) = \zeta^{-1}$ acts on $\mathrm{Gal}(L/F) \cong \mu_3(F)$, the isomorphism of (1) over $F$, as inversion. Deduce that $\mathrm{Gal}(L/\mathbb{Q}) \cong S_3$. Check with `Magma`:

```
R<x> := PolynomialRing(Rationals());
G := GaloisGroup(x^3 - 2);
#G, IsAbelian(G);                    // 6 false
F<zeta> := CyclotomicField(3);
S<y> := PolynomialRing(F);
#GaloisGroup(y^3 - 2);               // 3
```

Solution.

[a] *Hint:* show that both maps are injective, then count with Proposition 31.1.
[b] *Hint:* for $L$ given, compute $\Delta_L/k^{\times m}$ with Problem 37(5) and Proposition 30.3.
[c] *Hint:* Proposition 30.3 and Problem 35(4).

## 32. Computing $k^\times/k^{\times m}$

This section computes $k^\times/k^{\times m} \cong H^1(k, \mu_m)$ for finite fields, $\mathbb{Q}_p$ and $\mathbb{Q}$, and then cuts out a finite subgroup by conditions at the primes, as in [Poo02, §§9–10]. The last item is [Sil09, proof of Proposition VIII.1.6], with a `Magma` check.

**Lemma 32.1** (Hensel's lemma [Sil09, Lemma IV.1.2])**.** *Let $F \in \mathbb{Z}_p[w]$ and $a \in \mathbb{Z}_p$ with $F(a) \in p^n\mathbb{Z}_p$ for some $n \geqslant 1$ and $F'(a) \in \mathbb{Z}_p^\times$. Then $F$ has a root $b \in \mathbb{Z}_p$ with $b \equiv a \pmod{p^n}$.*

**Definition 32.2** (Unramified classes [Poo02, §§4 and 9])**.** Let $p$ be a prime. Embed $\overline{\mathbb{Q}}$ in an algebraic closure $\overline{\mathbb{Q}}_p$ of $\mathbb{Q}_p$, so that restriction is an injection $\mathrm{Gal}_{\mathbb{Q}_p} := \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) \hookrightarrow \mathrm{Gal}_{\mathbb{Q}}$. Let ${\mathbb{Q}_p}^{\mathrm{unr}} \subseteq \overline{\mathbb{Q}}_p$ be the maximal unramified extension of $\mathbb{Q}_p$, the union of its finite unramified extensions. A finite extension of $\mathbb{Q}_p$ inside $\overline{\mathbb{Q}}_p$ is unramified if and only if it lies in ${\mathbb{Q}_p}^{\mathrm{unr}}$. The inertia group is $I_p := \mathrm{Gal}(\overline{\mathbb{Q}}_p/{\mathbb{Q}_p}^{\mathrm{unr}}) \subseteq \mathrm{Gal}_{\mathbb{Q}_p}$. A class $\xi \in H^1(\mathbb{Q}, A)$, for a commutative algebraic group $A$ over $\mathbb{Q}$, is unramified at $p$ if its image in $H^1(I_p, A(\overline{\mathbb{Q}}_p))$ is trivial. For a finite set $S$ of primes, $H^1(\mathbb{Q}, A; S)$ is the subgroup of the classes unramified at every prime $p \notin S$.

**Proposition 32.3** (Unramified Kummer extensions [Sil09, proof of Proposition VIII.1.6])**.** *Let $p$ be a prime with $p \nmid m$, $a \in {\mathbb{Q}_p}^\times$ and $\alpha \in \overline{\mathbb{Q}}_p$ with $\alpha^m = a$. Then $\mathbb{Q}_p(\alpha)/\mathbb{Q}_p$ is unramified if and only if $v_p(a) \equiv 0 \pmod{m}$.*

**Problem 39** (Finite and local fields)**.**

(1) Let $q$ be a power of a prime not dividing $m$. Show that $\mathbb{F}_q^\times/\mathbb{F}_q^{\times m}$ is cyclic of order $\gcd(m, q-1)$. If $m \mid q-1$, so that $\mu_m(\overline{\mathbb{F}}_q) \subseteq \mathbb{F}_q$, compare the subgroups of $\mathbb{F}_q^\times/\mathbb{F}_q^{\times m}$ with the extensions of $\mathbb{F}_q$ of degree dividing $m$, as Problem 38(3) predicts.

Solution.

(2) Let $p$ be odd and $u \in \mathbb{Z}_p^\times$ a unit whose reduction is not a square in $\mathbb{F}_p$. Show that ${\mathbb{Q}_p}^\times/{\mathbb{Q}_p}^{\times 2}$ has order 4, with representatives $1, u, p, up$. List the quadratic extensions of $\mathbb{Q}_p$, and say which one is unramified.[a]

Solution.

(3) Show that a unit $v \in \mathbb{Z}_2^\times$ is a square if and only if $v \equiv 1 \pmod 8$. Deduce that $\mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}$ has order 8, with representatives $\pm1, \pm2, \pm5, \pm10$. `Magma` counts the quadratic extensions of $\mathbb{Q}_p$:

```
#AllExtensions(pAdicRing(3, 20), 2);    // 3
#AllExtensions(pAdicRing(5, 20), 2);    // 3
#AllExtensions(pAdicRing(2, 20), 2);    // 7
```

Check these against your answers.[b]

Solution.

[a] *Hint:* every element of ${\mathbb{Q}_p}^\times$ is $p^n v$ with $n \in \mathbb{Z}$ and $v \in \mathbb{Z}_p^\times$, and Lemma 32.1 decides which units are squares.
[b] *Hint:* for $v \equiv 1 \pmod 8$, apply Lemma 32.1 to $F(w) = w^2 + w - (v-1)/4$, then look at $(2w+1)^2$.

**Definition 32.4** (The group $k(S, m)$ [Sil09, Theorem X.1.1(c)])**.** Let $k$ be a number field, $S$ a finite set of places of $k$ containing the archimedean ones, and

$$k(S, m) := \{a \in k^\times/k^{\times m} : v(a) \equiv 0 \pmod{m} \text{ for all } v \notin S\},$$

where $v$ runs over the normalized valuations of $k$. Let $\mathcal{O}_S$ be the ring of $S$-integers, the $a \in k$ with $v(a) \geqslant 0$ for all $v \notin S$, and $\mathrm{Cl}(\mathcal{O}_S)$ its ideal class group. For $k = \mathbb{Q}$ we list only the primes in $S$, and in [Poo02, §10], $\mathbb{Q}(S, 2)$ is written $\langle -1, S \rangle$.

**Proposition 32.5** (Finiteness [Sil09, proof of Proposition VIII.1.6])**.** *The group* $\mathrm{Cl}(\mathcal{O}_S)$ *is finite, and* $\mathcal{O}_S^\times$ *is finitely generated (Dirichlet's* $S$*-unit theorem).*

**Problem 40** (Classes unramified outside $S$)**.**

(1) Show that $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ is an $\mathbb{F}_2$-vector space with basis the classes of $-1$ and of the primes. In particular, it is infinite.

Solution.

(2) Let $S$ be a finite set of primes. Show that $\mathbb{Q}(S, 2)$ is the subgroup generated by $-1$ and the primes in $S$, of order $2^{\#S+1}$.

Solution.

(3) Let $p$ be an odd prime and $a \in \mathbb{Q}^\times$. Show that $\delta(a) \in H^1(\mathbb{Q}, \mu_2)$ is unramified at $p$ if and only if $v_p(a)$ is even. Deduce that $\mathbb{Q}(S, 2) = \delta^{-1} H^1(\mathbb{Q}, \mu_2; S)$ for every finite set $S$ of primes containing 2.

Solution.

(4) ($\star$) Let $k$ be a number field and $S$ as in Definition 32.4. Construct an exact sequence[a]

$$1 \to \mathcal{O}_S^\times/\mathcal{O}_S^{\times m} \to k(S, m) \to \mathrm{Cl}(\mathcal{O}_S)[m] \to 0,$$

and deduce that $k(S, m)$ is finite. For $k = \mathbb{Q}(\sqrt{-5})$, $m = 2$ and $S$ the archimedean place, show that $k(S, 2) = \langle -1, 2 \rangle$, of order 4, and check with `Magma`:

```
K<s> := QuadraticField(-5);
OK := Integers(K);
#ClassGroup(OK);                        // 2
G, m := pSelmerGroup(2, {PowerIdeal(OK)|});
#G;                                     // 4
[K | G.i @@ m : i in [1..Ngens(G)]];  // [ -1, 2 ]
```

Solution.

[a] *Hint:* the prime ideals of $\mathcal{O}_S$ are those of the places outside $S$. If $a$ represents an element of $k(S, m)$, then $a\mathcal{O}_S = \mathfrak{a}^m$ for a fractional ideal $\mathfrak{a}$ of $\mathcal{O}_S$.

## Part 10. The Kummer pairing and weak Mordell–Weil

The weak Mordell–Weil theorem says that $E(K)/mE(K)$ is finite for an elliptic curve $E$ over a number field $K$. The proof in [Sil09, §VIII.1] has three steps. The Kummer pairing turns the finiteness of $E(K)/mE(K)$ into the finiteness of the field $L$ generated by the $m$-th divisors of the rational points. This field is abelian of exponent $m$ and unramified outside a finite set of places $S$. And Kummer theory shows that such fields are finite. This worksheet runs the argument for $m = 2$ and $K = \mathbb{Q}$, on curves whose points of order two are rational. Then the class group and the $S$-unit theorem disappear, and the injectivity of reduction on the 2-torsion becomes elementary.

Problem 41 sets up the pairing. Problem 42, the heart of the worksheet, proves that $L$ is unramified outside $S$ and deduces a bound for $E(\mathbb{Q})/2E(\mathbb{Q})$. Problem 43 tests the bound on $y^2 = x^3 - x$.

**Standing notation.**

- An elliptic curve $E$ over $\mathbb{Q}$, given by
$$y^2 = (x - e_1)(x - e_2)(x - e_3)$$
with distinct $e_1, e_2, e_3 \in \mathbb{Z}$, and its points of order two $T_i := (e_i, 0)$. So $E[2](\overline{\mathbb{Q}}) = \{O, T_1, T_2, T_3\} \subseteq E(\mathbb{Q})$.
- The finite set of primes $S := \{2\} \cup \{p : p \mid (e_1 - e_2)(e_1 - e_3)(e_2 - e_3)\}$, and $s := \#S$.
- The absolute Galois group $\mathrm{Gal}_{\mathbb{Q}} := \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, with elements $\sigma, \tau, \gamma$. For an extension $F$ of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$, $\mathrm{Gal}_F := \mathrm{Gal}(\overline{\mathbb{Q}}/F)$.
- For $Q \in E(\overline{\mathbb{Q}})$, the field $\mathbb{Q}(Q)$ generated by the coordinates of $Q$, and the point $\sigma(Q)$, obtained by applying $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$ to the coordinates.
- The compositum $L$ of the fields $\mathbb{Q}(Q)$ for $Q \in E(\overline{\mathbb{Q}})$ with $2Q \in E(\mathbb{Q})$. Silverman writes $L = \mathbb{Q}([2]^{-1}E(\mathbb{Q}))$.
- The quotient $B/2 := B/2B$ of an abelian group $B$, and the group $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ of classes of rational numbers modulo squares.

## 33. The Kummer pairing

This section is [Sil09, Proposition VIII.1.2], for $m = 2$ and $K = \mathbb{Q}$, and its reinterpretation in [Sil09, §VIII.2], with cohomology as in [Sil09, §§B.1–B.2].

**Definition 33.1** (The Kummer pairing [Sil09, §VIII.1])**.** The Kummer pairing is the map
$$\kappa\colon E(\mathbb{Q}) \times \mathrm{Gal}_{\mathbb{Q}} \to E[2](\overline{\mathbb{Q}})$$
defined as follows. Let $P \in E(\mathbb{Q})$ and $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$. Choose $Q \in E(\overline{\mathbb{Q}})$ with $2Q = P$, and set $\kappa(P, \sigma) := \sigma(Q) - Q$. That this lands in $E[2](\overline{\mathbb{Q}})$, and does not depend on $Q$, is Problem 41(1).

**Proposition 33.2** (Infinite Galois theory [Mil22, Theorem 7.13])**.** *For a closed subgroup $H$ of $\mathrm{Gal}_{\mathbb{Q}}$, let $\overline{\mathbb{Q}}^H$ be its fixed field. The maps $H \mapsto \overline{\mathbb{Q}}^H$ and $F \mapsto \mathrm{Gal}_F$ are inverse bijections between the closed subgroups of $\mathrm{Gal}_{\mathbb{Q}}$ and the extensions of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$. The subgroup $H$ is normal if and only if $\overline{\mathbb{Q}}^H$ is Galois over $\mathbb{Q}$, and then $\mathrm{Gal}(\overline{\mathbb{Q}}^H/\mathbb{Q}) \cong \mathrm{Gal}_{\mathbb{Q}}/H$ by restriction. In particular, $\overline{\mathbb{Q}}^{\mathrm{Gal}_{\mathbb{Q}}} = \mathbb{Q}$.*

**Definition 33.3** (Cohomology in degree 1 [Sil09, §§B.1–B.2])**.** Let $A$ be a discrete $\mathrm{Gal}_{\mathbb{Q}}$-module. A 1-cocycle is a continuous map $\xi\colon \mathrm{Gal}_{\mathbb{Q}} \to A$, $\sigma \mapsto \xi_\sigma$, with $\xi_{\sigma\tau} = \xi_\sigma + \sigma\xi_\tau$ for all $\sigma, \tau$. For $a \in A$, the map $\sigma \mapsto \sigma a - a$ is a 1-cocycle, and the 1-cocycles of this form are the 1-coboundaries. Then $H^1(\mathrm{Gal}_{\mathbb{Q}}, A)$ is the group of 1-cocycles modulo the 1-coboundaries. For an elliptic curve or its 2-torsion, we abuse notation and write $H^1(\mathbb{Q}, E[2]) := H^1(\mathrm{Gal}_{\mathbb{Q}}, E[2](\overline{\mathbb{Q}}))$.

**Proposition 33.4** (The long exact sequence [Sil09, Propositions B.1.2 and B.2.3])**.** *Let $0 \to A \to B \xrightarrow{\psi} C \to 0$ be an exact sequence of discrete $\mathrm{Gal}_{\mathbb{Q}}$-modules. For $c \in C^{\mathrm{Gal}_{\mathbb{Q}}}$, choose $b \in B$ with $\psi(b) = c$, and let $\delta(c) \in H^1(\mathrm{Gal}_{\mathbb{Q}}, A)$ be the class of $\sigma \mapsto \sigma b - b$. This does not depend on $b$, and the sequence*
$$0 \to A^{\mathrm{Gal}_{\mathbb{Q}}} \to B^{\mathrm{Gal}_{\mathbb{Q}}} \to C^{\mathrm{Gal}_{\mathbb{Q}}} \xrightarrow{\delta} H^1(\mathrm{Gal}_{\mathbb{Q}}, A) \to H^1(\mathrm{Gal}_{\mathbb{Q}}, B)$$
*is exact.*

**Proposition 33.5** (Trivial action [Sil09, Remark B.2.2])**.** *If $\mathrm{Gal}_{\mathbb{Q}}$ acts trivially on $A$, then*
$$H^1(\mathrm{Gal}_{\mathbb{Q}}, A) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_{\mathbb{Q}}, A),$$
*the group of continuous homomorphisms.*

**Problem 41** (The Kummer pairing)**.**

(1) Show that $\kappa(P,\sigma) \in E[2](\overline{\mathbb{Q}})$, and that it does not depend on the choice of $Q$.

Solution.

(2) Let $P, P' \in E(\mathbb{Q})$ and $\sigma, \tau \in \mathrm{Gal}_{\mathbb{Q}}$. Show that $\kappa(P+P',\sigma) = \kappa(P,\sigma) + \kappa(P',\sigma)$ and

$$\kappa(P,\sigma\tau) = \kappa(P,\sigma) + \sigma\big(\kappa(P,\tau)\big).$$

Deduce that $\kappa$ is bilinear. Where does the hypothesis $E[2](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q})$ enter?

Solution.

(3) Show that $\kappa(P,\cdot) = O$ if and only if $P \in 2E(\mathbb{Q})$, and that $\kappa(\cdot,\sigma) = O$ if and only if $\sigma \in \mathrm{Gal}_L$. Deduce that $L/\mathbb{Q}$ is Galois and that $\kappa$ induces a perfect pairing

$$E(\mathbb{Q})/2 \times \mathrm{Gal}(L/\mathbb{Q}) \to E[2](\overline{\mathbb{Q}}),$$

that is, both $E(\mathbb{Q})/2 \to \mathrm{Hom}(\mathrm{Gal}(L/\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$ and $\mathrm{Gal}(L/\mathbb{Q}) \to \mathrm{Hom}(E(\mathbb{Q})/2, E[2](\overline{\mathbb{Q}}))$ are injective. Conclude that $E(\mathbb{Q})/2$ is finite if and only if $L/\mathbb{Q}$ is finite.

Solution.

(4) Show that $0 \to E[2](\overline{\mathbb{Q}}) \to E(\overline{\mathbb{Q}}) \xrightarrow{2} E(\overline{\mathbb{Q}}) \to 0$ is an exact sequence of discrete $\mathrm{Gal}_{\mathbb{Q}}$-modules, and that its connecting map $\delta\colon E(\mathbb{Q}) \to H^1(\mathbb{Q}, E[2])$ sends $P$ to the class of $\kappa(P,\cdot)$. Show that $H^1(\mathbb{Q}, E[2]) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_{\mathbb{Q}}, E[2](\overline{\mathbb{Q}}))$, and recover from Proposition 33.4 the injectivity of $E(\mathbb{Q})/2 \to \mathrm{Hom}(\mathrm{Gal}_{\mathbb{Q}}, E[2](\overline{\mathbb{Q}}))$.

Solution.

## 34. Unramified outside $S$

This section is [Sil09, Proposition VIII.1.5], for $m = 2$ and $K = \mathbb{Q}$. Its argument is that of [Sil09, Proposition VII.4.1(a)], applied to the halves of rational points instead of torsion points. The last item adds Kummer theory, as in [Sil09, proof of Proposition VIII.1.6]. In this section $p$ is a prime, and $\overline{\mathbb{Q}}$ is embedded in an algebraic closure $\overline{\mathbb{Q}}_p$ of $\mathbb{Q}_p$, so that restriction is an injection $\mathrm{Gal}_{\mathbb{Q}_p} \coloneqq \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) \hookrightarrow \mathrm{Gal}_{\mathbb{Q}}$. The residue field of the valuation ring of $\overline{\mathbb{Q}}_p$ is an algebraic closure $\overline{\mathbb{F}}_p$ of $\mathbb{F}_p$, and a bar denotes reduction modulo its maximal ideal.

**Definition 34.1** (Inertia [Sil09, §VII.4])**.** An element $\sigma \in \mathrm{Gal}_{\mathbb{Q}_p}$ preserves the valuation ring of $\overline{\mathbb{Q}}_p$, so it induces an automorphism $\bar{\sigma}$ of $\overline{\mathbb{F}}_p$. The inertia group $I_p \subseteq \mathrm{Gal}_{\mathbb{Q}_p}$ is the subgroup of the $\sigma$ with $\bar{\sigma} = 1$. If ${\mathbb{Q}_p}^{\mathrm{unr}} \subseteq \overline{\mathbb{Q}}_p$ is the maximal unramified extension of $\mathbb{Q}_p$, then $I_p = \mathrm{Gal}(\overline{\mathbb{Q}}_p/{\mathbb{Q}_p}^{\mathrm{unr}})$. A Galois extension $F$ of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$ is unramified at $p$ if $I_p \subseteq \mathrm{Gal}_F$, that is, if $I_p$ fixes $F$.

**Proposition 34.2** (Good reduction [Sil09, §III.1 and Remark VIII.1.3])**.** *The Weierstrass equation of $E$ has discriminant*

$$\Delta = 16(e_1-e_2)^2(e_1-e_3)^2(e_2-e_3)^2.$$

*If $p \nmid \Delta$, the equation is minimal at $p$ and its reduction $E_p$ is an elliptic curve over $\mathbb{F}_p$.*

**Proposition 34.3** (Reduction [Sil09, Proposition VII.2.1])**.** *Suppose that $p \nmid \Delta$. Every $Q \in E(\overline{\mathbb{Q}}_p)$ has projective coordinates $[X:Y:Z]$ in the valuation ring of $\overline{\mathbb{Q}}_p$, not all in its maximal ideal, and $\bar{Q} \coloneqq [\bar{X}:\bar{Y}:\bar{Z}] \in E_p(\overline{\mathbb{F}}_p)$. The reduction map $E(\overline{\mathbb{Q}}_p) \to E_p(\overline{\mathbb{F}}_p)$, $Q \mapsto \bar{Q}$, is a homomorphism, and $\overline{\sigma(Q)} = \bar{\sigma}(\bar{Q})$ for $\sigma \in \mathrm{Gal}_{\mathbb{Q}_p}$.*

**Proposition 34.4** (Kummer theory [Sil09, Exercise 8.4]). *Let $F$ be a Galois extension of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$ with $\mathrm{Gal}(F/\mathbb{Q})$ abelian of exponent dividing 2. Let $D \subseteq \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ be the subgroup of the classes of the $d \in \mathbb{Q}^\times$ that are squares in $F$. Then $F = \mathbb{Q}(\sqrt{d} : d \in D)$.*

**Proposition 34.5** (Unramified quadratic extensions [Sil09, proof of Proposition VIII.1.6]). *Let $p$ be an odd prime and $d \in \mathbb{Q}^\times$. Then $\mathbb{Q}(\sqrt{d})$ is unramified at $p$ if and only if $v_p(d)$ is even.*

**Definition 34.6** (The group $\mathbb{Q}(S,2)$ [Sil09, Theorem X.1.1(c)]). The subgroup $\mathbb{Q}(S,2) \subseteq \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ consists of the classes of the $d \in \mathbb{Q}^\times$ with $v_p(d)$ even for every prime $p \notin S$. It is generated by the classes of $-1$ and of the primes in $S$, so it has order $2^{s+1}$.

**Problem 42** ($L/\mathbb{Q}$ is unramified outside $S$).

(1) Show that $\mathrm{Gal}(L/\mathbb{Q})$ is abelian of exponent dividing 2. Let $D \subseteq \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ be the subgroup of the classes of the $d \in \mathbb{Q}^\times$ that are squares in $L$. Deduce that $L = \mathbb{Q}(\sqrt{d} : d \in D)$.

Solution.

(2) Let $p \notin S$. Show that $E$ has good reduction at $p$, and that $O, T_1, T_2, T_3$ reduce to four distinct points of $E_p(\mathbb{F}_p)$. Deduce that reduction is injective on $E[2](\overline{\mathbb{Q}})$.

Solution.

(3) Let $p \notin S$, $Q \in E(\overline{\mathbb{Q}})$ with $2Q \in E(\mathbb{Q})$, and $\sigma \in I_p$. Show that $\sigma(Q) - Q$ reduces to $\bar{O}$, and deduce that $\sigma(Q) = Q$. Conclude that $L/\mathbb{Q}$ is unramified at $p$.

Solution.

(4) Show that $D \subseteq \mathbb{Q}(S,2)$. Deduce that $\#\,\mathrm{Gal}(L/\mathbb{Q}) \leqslant 2^{s+1}$ and $\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant 4^{s+1}$.

Solution.

## 35. An example

This section runs Problem 42 on $y^2 = x^3 - x$, that is, $(e_1, e_2, e_3) = (0, 1, -1)$. The images of its points of order two under the explicit form of the Kummer map are [Sil09, Proposition X.1.4].

**Problem 43** (The curve $y^2 = x^3 - x$).
In this problem, $E$ is the curve $y^2 = x^3 - x$, with $T_1 = (0,0)$, $T_2 = (1,0)$ and $T_3 = (-1,0)$. `Magma` finds its rank and its torsion subgroup, and checks that the halves of each $T_i$ are defined over $\mathbb{Q}(\zeta_8)$:

```
E := EllipticCurve([-1, 0]);
Rank(E);                          // 0 true
TorsionSubgroup(E);               // Z/2 + Z/2
K<z> := CyclotomicField(8);
EK := ChangeRing(E, K);
[#DivisionPoints(EK ! [e, 0], 2) : e in [0, 1, -1]];   // [ 4, 4, 4 ]
```

So $E(\mathbb{Q}) = E[2](\overline{\mathbb{Q}})$.

(1) Compute $S$ and the bound of Problem 42(4).

Solution.

(2) For $i = 1, 2, 3$, find the four $Q \in E(\overline{\mathbb{Q}})$ with $2Q = T_i$, and show that $L = \mathbb{Q}(i, \sqrt{2}) = \mathbb{Q}(\zeta_8)$.[a]

Solution.

(3) Let $\sigma, \tau \in \mathrm{Gal}(L/\mathbb{Q})$ with $\sigma(i) = -i$, $\sigma(\sqrt{2}) = \sqrt{2}$, $\tau(i) = i$ and $\tau(\sqrt{2}) = -\sqrt{2}$. Compute $\kappa(T_i, \sigma)$ and $\kappa(T_i, \tau)$ for $i = 1, 2, 3$, and deduce that no $T_i$ lies in $2E(\mathbb{Q})$. Compute $\#E(\mathbb{Q})/2E(\mathbb{Q})$ and compare it with the bound of (1).

Solution.

[a] *Hint:* for $Q = (x, y)$ with $y \neq 0$, the duplication formula on this curve reads $x(2Q) = (x^2 + 1)^2/(4y^2)$.

## Part 11. Heights on abelian groups

The Mordell–Weil theorem has two steps. The weak Mordell–Weil theorem says that $E(K)/mE(K)$ is finite, and a height function turns this into finite generation [Sil09, §VIII.3]. The second step does not use the curve: it works for any abelian group with a height that is quadratic up to a bounded error. This worksheet is about such heights on an abstract abelian group: which functions are heights of which kind, what their $O(1)$ constants are, and how a quadratic height behaves. Problem 44 sets up the definitions and the first estimates. Problem 45, the heart of the worksheet, proves the descent theorem and shows, on $\mathbb{Q}$ and on a direct sum of copies of $\mathbb{Z}$, that its hypotheses are needed. Problem 46 is the algebra of quadratic heights.

**Standing notation.**

- An abelian group $A$, written additively, with elements $P, Q, R$, torsion subgroup $A_{\mathrm{tors}}$, and, for an integer $m \geqslant 2$, the quotient $A/m \coloneqq A/mA$.
- For real functions $f, g$ on a set $X$, the notation $f = g + O(1)$, meaning that $f - g$ is bounded on $X$. An inequality $f \leqslant g + c$ that holds for one constant $c$ holds for every larger one, so the constants that bound heights are taken non-negative, and a lower bound is written $-c$ with $c \geqslant 0$.

### 36. Heights and the parallelogram law

This section is the vocabulary of [Sil09, §VIII.3], and the estimates [Sil09, Corollary VIII.6.4] that the heights on an elliptic curve satisfy, proved here for any height that satisfies the parallelogram law up to a bounded error [HS00, §B.4].

**Definition 36.1** (Heights [Sil09, §§VIII.3, VIII.6])**.** A height on a set $X$ is a function $h\colon X \to \mathbb{R}$. Two heights $h, h'$ on $X$ are equivalent, written $h = h' + O(1)$, if $h - h'$ is bounded on $X$. A height $h$ has the Northcott property if $\{x \in X : h(x) \leqslant C\}$ is finite for every $C \in \mathbb{R}$. The defect of a height $h$ on $A$ is the function

$$\delta_h(P, Q) \coloneqq h(P + Q) + h(P - Q) - 2h(P) - 2h(Q)$$

on $A \times A$. The height $h$ is quasi-quadratic if $\delta_h$ is bounded, and quadratic if $\delta_h = 0$ (the parallelogram law).

**Problem 44** (Heights and the parallelogram law)**.**

(1) Show that equivalence of heights on a set $X$ is an equivalence relation. Show that the Northcott property depends only on the equivalence class, and that a height with the Northcott property is bounded below. Give a height on $\mathbb{Z}$ that is bounded below and does not have the Northcott property.

Solution.

(2) Let $h$ be a quasi-quadratic height on $A$, and let $c \geqslant 0$ be a constant with $|\delta_h| \leqslant c$. Show that $|h(0)| \leqslant c/2$, that $|h(-P) - h(P)| \leqslant 2c$, and that $|h(2P) - 4h(P)| \leqslant 3c/2$ for every $P \in A$.

Solution.

(3) Let $h$ be a quasi-quadratic height on $A$, and let $c \geqslant 0$ be a constant with $|\delta_h| \leqslant c$. Show that for every $n \in \mathbb{Z}$ there is a constant $d_n \geqslant 0$ with $|h(nP) - n^2 h(P)| \leqslant d_n$ for all $P \in A$, and give a recursion for $d_n$ in terms of $c$.[a] Suppose moreover that $h \geqslant -c'$ for a constant $c' \geqslant 0$. Show that for every $Q \in A$ there is a constant $C(Q) \geqslant 0$ with $h(P+Q) \leqslant 2h(P) + C(Q)$ for all $P \in A$, and give $C(Q)$ explicitly.

Solution.

(4) Let $f\colon A \to \mathbb{R}$ be a homomorphism. Show that the height $f^2$ on $A$ is quadratic. Show that a sum of quadratic heights on $A$ is quadratic.

Solution.

(5) Let $A = \mathbb{Z}^2$, $q(x,y) \coloneqq x^2+y^2$ and $q'(x,y) \coloneqq x^2+(x+y)^2$. Show that $q$ and $q'$ are quadratic heights with the Northcott property, that $q \leqslant 3q'$ and $q' \leqslant 3q$, and that they are not equivalent.

Solution.

[a] *Hint:* apply the parallelogram law to the pair $(nP, P)$ and induct on $n \geqslant 0$.

## 37. The descent theorem

This section is [Sil09, Theorem VIII.3.1] and its proof, for an abstract abelian group, and the examples of groups that fail one of its hypotheses. The count of rational numbers of bounded height is [HS00, Theorem B.2.3].

**Theorem 37.1** (Descent theorem [Sil09, Theorem VIII.3.1])**.** *Let $h$ be a height on $A$ with the following properties.*

*(1) For every $Q \in A$ there is a constant $C_1(Q)$ with $h(P+Q) \leqslant 2h(P) + C_1(Q)$ for all $P \in A$.*
*(2) There are an integer $m \geqslant 2$ and a constant $C_2$ with $h(mP) \geqslant m^2 h(P) - C_2$ for all $P \in A$.*
*(3) The height $h$ has the Northcott property.*

*If $A/m$ is finite for the $m$ of (2), then $A$ is finitely generated.*

**Problem 45** (The descent theorem)**.**

(1) Let $h$ be a quasi-quadratic height on $A$ with the Northcott property. Show that $h$ satisfies the hypotheses of Theorem 37.1 for every $m \geqslant 2$.

Solution.

(2) Let $h$ be a height on $A$ with the Northcott property, let $m \geqslant 2$ be an integer, and let $0 < a < b$ be real numbers. Suppose that for every $Q \in A$ there is a constant $C_1(Q) \geqslant 0$ with $h(P+Q) \leqslant a\,h(P) + C_1(Q)$ for all $P$, and that there is a constant $C_2 \geqslant 0$ with $h(mP) \geqslant b\,h(P) - C_2$ for all $P$. Suppose also that $A/m$ is finite, let $Q_1, \dots, Q_s$ represent its cosets, and let $C \coloneqq \max_i C_1(-Q_i) + C_2$. Show that $A$ is generated by $Q_1, \dots, Q_s$ and the finitely many $P \in A$ with $h(P) \leqslant 1 + C/(b-a)$.

Solution.

(3) Let $A = \mathbb{Q}$, as an additive group, with the logarithmic Weil height

$$h(u/w) \coloneqq \log \max(|u|, |w|),$$

for $u/w$ in lowest terms. Show that $h$ has the Northcott property, that $h(x+y) \leqslant h(x)+h(y)+\log 2$, with equality for $x = y = 1$, that $h(mx) \leqslant h(x) + \log m$ and $\mathbb{Q}/m = 0$ for every integer $m \geqslant 2$. Which hypothesis of (2) fails, for every $m \geqslant 2$ and every $b > 1$? Is $h$ quasi-quadratic?

Solution.

(4) Let $A = \bigoplus_{i\geqslant 1} \mathbb{Z}$, let $w_1, w_2, \dots$ be positive real numbers, and $h(x) \coloneqq \sum_i w_i x_i^2$. Show that $h$ is quadratic. For which weights $w_i$ does $h$ have the Northcott property? For such weights, which hypothesis of Theorem 37.1 fails for $A$?

Solution.

## 38. Quadratic heights and their pairing

This section is [HS00, Lemma B.5.2], the algebra of a height that satisfies the parallelogram law exactly, which [Sil09, Theorem VIII.9.3(c)] states for the canonical height on an elliptic curve.

**Definition 38.1** (The pairing of a quadratic height [HS00, §B.5])**.** Let $q$ be a quadratic height on $A$. Its pairing is

$$\langle P, Q\rangle_q \coloneqq q(P+Q) - q(P) - q(Q).$$

**Problem 46** (Quadratic heights and their pairing)**.**
Let $q$ be a quadratic height on $A$.

(1) Show that $q(0) = 0$, $q(-P) = q(P)$ and $q(2P) = 4q(P)$, and that $\langle\ ,\ \rangle_q$ is symmetric and bilinear.[a] Deduce that $\langle P, P\rangle_q = 2q(P)$ and $q(nP) = n^2 q(P)$ for every $n \in \mathbb{Z}$. Conversely, show that for a symmetric bilinear map $B\colon A \times A \to \mathbb{R}$, the function $P \mapsto \frac{1}{2}B(P, P)$ is a quadratic height with pairing $B$.

Solution.

(2) Let $T \in A_{\text{tors}}$. Show that $q(T) = 0$, that $\langle P, T\rangle_q = 0$ for every $P \in A$, and that $q(P+T) = q(P)$. So $q$ and its pairing factor through $A/A_{\text{tors}}$.

Solution.

(3) Let $A = \mathbb{Z}^r$ with standard basis $e_1, \dots, e_r$. Show that every quadratic height on $A$ is $q(x) = \frac{1}{2}x^{\mathsf{T}}Gx$ for a unique real symmetric $r \times r$ matrix $G$, and that every symmetric $G$ arises. Show that if $G$ is positive definite, then $q$ has the Northcott property.

Solution.

[a] *Hint:* write the parallelogram law for the pairs $(P+R, Q)$, $(P, R-Q)$, $(P+Q, R)$ and $(R, Q)$.

# Part 12. Heights on projective space

To apply the descent theorem to $E(K)$, one needs a height on $E(K)$ with the Northcott property. Silverman builds it from a height on projective space [Sil09, §VIII.5]: a point of $\mathbb{P}^N(\overline{\mathbb{Q}})$ is large when its coordinates are large at the archimedean places or have large denominators at the non-archimedean ones. This worksheet works out the first properties of this height. Problem 47, the heart of the worksheet, proves Kronecker's theorem: the algebraic numbers of height 1 are 0 and the roots of unity. Problem 48 proves the inequalities for products and sums, and the behavior of the height under the Segre and $d$-uple embeddings. Problem 49 computes the height through the ideal

generated by the coordinates, and shows on $\mathbb{Q}(\sqrt{-5})$ why the recipe of coprime integer coordinates does not extend to every number field.

**Standing notation.**

- An algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$, and its subring $\overline{\mathbb{Z}}$ of algebraic integers.
- A number field $K$, its ring of integers $\mathcal{O}_K$, its set of places $\mathcal{P}_K$, and the subsets $\mathcal{P}_K^\infty$ and $\mathcal{P}_K^0$ of archimedean and non-archimedean places.
- For $v \in \mathcal{P}_K$, the completion $K_v$, the absolute value $|\ |_v$ on $K_v$ extending the usual absolute value on $\mathbb{Q}_p$ or $\mathbb{R}$, and the local degree $n_v \coloneqq [K_v : \mathbb{Q}_v]$, where $\mathbb{Q}_v$ is the completion of $\mathbb{Q}$ at the restriction of $v$.
- For a point $P = [x_0 : \cdots : x_N]$ of $\mathbb{P}^N(K)$ and $v \in \mathcal{P}_K$, the number $|P|_v \coloneqq \max_i |x_i|_v$. It depends on the choice of coordinates.

## 39. Kronecker's theorem

This section is [Sil09, Exercise 8.6], with the proof of [HS00, Corollary B.2.3.1]. The definitions and the first properties of the height are [Sil09, §VIII.5]. The last two items compare Kronecker's theorem with the enumeration of polynomials that proves it in small degree, and with algebraic numbers of small height.

**Proposition 39.1** (Extension and product formulas [Sil09, (VIII.5.2), (VIII.5.3)])**.** *(1) Let $L$ be a finite extension of $K$, and let $v \in \mathcal{P}_K$. Then $\sum_{w|v} n_w = [L : K]\, n_v$, the sum over the places $w \in \mathcal{P}_L$ that restrict to $v$.*
*(2) Let $x \in K^\times$. Then $\prod_{v \in \mathcal{P}_K} |x|_v^{n_v} = 1$.*

**Definition 39.2** (Heights on projective space [Sil09, §§VIII.5, VIII.6])**.** The height relative to $K$ of a point $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(K)$ is

$$H_K(P) \coloneqq \prod_{v \in \mathcal{P}_K} |P|_v^{n_v}.$$

Let $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$, and let $K$ be a number field with $P \in \mathbb{P}^N(K)$. The height of $P$ is $H(P) \coloneqq H_K(P)^{1/[K:\mathbb{Q}]}$, and its logarithmic height is $h(P) \coloneqq \log H(P)$. For $x \in \overline{\mathbb{Q}}$, let $H(x) \coloneqq H([x : 1])$ and $h(x) \coloneqq \log H(x)$.

**Proposition 39.3** (Properties of the height [Sil09, Proposition VIII.5.4])**.** *Let $P \in \mathbb{P}^N(K)$.*
*(1) The height $H_K(P)$ does not depend on the choice of coordinates of $P$.*
*(2) The height satisfies $H_K(P) \geqslant 1$.*
*(3) Let $L$ be a finite extension of $K$. Then $H_L(P) = H_K(P)^{[L:K]}$.*
*In particular, $H(P)$ does not depend on the choice of $K$, and $H(P) \geqslant 1$.*

**Theorem 39.4** (Northcott's theorem [Sil09, Theorem VIII.5.11])**.** *Let $C$ and $d$ be real numbers. Then the set of points $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$ with $H(P) \leqslant C$ and $[\mathbb{Q}(P) : \mathbb{Q}] \leqslant d$ is finite, where $\mathbb{Q}(P)$ is the field generated by the ratios $x_i/x_j$ with $x_j \neq 0$ of the coordinates of $P$. In particular, the set $\{P \in \mathbb{P}^N(K) : H_K(P) \leqslant C\}$ is finite.*

**Problem 47** (Kronecker's theorem)**.**
This problem proves Kronecker's theorem: a nonzero algebraic number $x \in \overline{\mathbb{Q}}^\times$ has $H(x) = 1$ if and only if $x$ is a root of unity. By (2), it is equivalent to Kronecker's original form: a nonzero algebraic integer $x \in \overline{\mathbb{Z}}$ with $|\tau(x)| \leqslant 1$ for every embedding $\tau\colon \mathbb{Q}(x) \to \mathbb{C}$ is a root of unity.
(1) Show that every root of unity has height 1. Show that $H(0) = 1$, so the hypothesis $x \neq 0$ cannot be dropped.

Solution.

(2) Let $x \in \overline{\mathbb{Q}}$ and $K := \mathbb{Q}(x)$. Show that $H(x) = 1$ if and only if $|x|_v \leqslant 1$ for every $v \in \mathcal{P}_K$, and that this holds if and only if $x \in \overline{\mathbb{Z}}$ and $|\tau(x)| \leqslant 1$ for every embedding $\tau\colon K \to \mathbb{C}$. Show that $x = (3+4i)/5$ satisfies $|\tau(x)| = 1$ for both embeddings of $\mathbb{Q}(i)$, and compute $H(x)$.

Solution.

(3) Let $x \in \overline{\mathbb{Q}}$. Show that $H(x^n) = H(x)^n$ for every integer $n \geqslant 1$, and that $H(1/x) = H(x)$ if $x \neq 0$. Prove Kronecker's theorem.[a]

Solution.

(4) Let $f = t^d + a_1 t^{d-1} + \cdots + a_d \in \mathbb{Z}[t]$ be monic of degree $d \geqslant 1$, with all complex roots in the closed unit disc. Show that $|a_i| \leqslant \binom{d}{i}$ for every $i$.[b] So there are finitely many such $f$ of each degree. `Magma` lists those of degree 4 with $a_4 \neq 0$, and checks that each is a product of cyclotomic polynomials:

```
R<t> := PolynomialRing(Integers());
d := 4;
B := [[-Binomial(d, i)..Binomial(d, i)] : i in [1..d]];
S := [];
for c in CartesianProduct(B) do
  f := t^d + &+[c[i]*t^(d - i) : i in [1..d]];
  if c[d] ne 0 and forall{r : r in Roots(f, ComplexField(30)) |
                          Abs(r[1]) le 1 + 10^-10} then
    Append(~S, f);
  end if;
end for;
#CartesianProduct(B), #S;                       // 3159 24
forall{f : f in S | forall{g : g in Factorization(f) |
                           IsCyclotomicPolynomial(g[1])}};       // true
```

Use this output to prove Kronecker's theorem for every $x \in \overline{\mathbb{Q}}^\times$ with $[\mathbb{Q}(x) : \mathbb{Q}] \leqslant 4$, without Theorem 39.4.

Solution.

(5) Let $d \geqslant 1$, and let $2^{1/d}$ be the real root of $t^d - 2$. Show that $h(2^{1/d}) = (\log 2)/d$. So $h$ takes arbitrarily small positive values on $\overline{\mathbb{Q}}^\times$. `Magma` checks this, and computes the logarithmic height of a root $b$ of Lehmer's polynomial:

```
R<t> := PolynomialRing(Integers());
[RealField(10) | d*AbsoluteLogarithmicHeight(NumberField(t^d - 2).1)
                 : d in [2..5]];       // [ 0.6931471805, 0.6931471805, ... ]
RealField(10)!Log(2);                  // 0.6931471805
f := t^10 + t^9 - t^7 - t^6 - t^5 - t^4 - t^3 + t + 1;
IsIrreducible(f), IsCyclotomicPolynomial(f);          // true false
K<b> := NumberField(f);
RealField(10)!AbsoluteLogarithmicHeight(b);           // 0.01623576120
```

Explain why $h(b) > 0$, and compute $[\mathbb{Q}(b) : \mathbb{Q}]\, h(b)$.

Solution.

[a] *Hint:* if $H(x) = 1$, consider the set $\{x, x^2, x^3, \dots\}$.
[b] *Hint:* the coefficient $a_i$ is, up to sign, the $i$-th elementary symmetric function of the roots.

## 40. Basic properties of the height

This section is [Sil09, Exercise 8.8]. The last two items are [HS00, Proposition B.2.4(b), (c)], for the Segre embedding [Har77, Exercise I.2.14] and the $d$-uple embedding [Har77, Exercise I.2.12]. The heights are those of Definition 39.2.

**Problem 48** (Basic properties of the height)**.**
In (1) and (2), let $N \geqslant 1$ and $x_1, \dots, x_N \in \overline{\mathbb{Q}}$.

(1) Show that $H(x_1 x_2 \cdots x_N) \leqslant H(x_1)H(x_2)\cdots H(x_N)$. Show that the inequality can be strict.

Solution.

(2) Show that $H(x_1 + x_2 + \cdots + x_N) \leqslant N\, H(x_1)H(x_2)\cdots H(x_N)$, and that the factor $N$ cannot be replaced by a smaller one.[a]

Solution.

(3) Let $N, M \geqslant 0$, and let $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(\overline{\mathbb{Q}})$ and $Q = [y_0 : \cdots : y_M] \in \mathbb{P}^M(\overline{\mathbb{Q}})$. The Segre embedding sends $(P, Q)$ to the point

$$S(P,Q) := [x_0y_0 : x_0y_1 : \cdots : x_iy_j : \cdots : x_Ny_M] \in \mathbb{P}^{MN+M+N}(\overline{\mathbb{Q}}),$$

with the products $x_iy_j$ in lexicographic order of $(i, j)$. Show that $S(P, Q)$ does not depend on the choice of coordinates, and that $H(S(P, Q)) = H(P)H(Q)$.

Solution.

(4) Let $N \geqslant 0$ and $d \geqslant 1$, let $M := \binom{N+d}{N} - 1$, and let $f_0, \dots, f_M$ be the distinct monomials of degree $d$ in $X_0, \dots, X_N$. The $d$-uple embedding sends $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(\overline{\mathbb{Q}})$ to $P^{(d)} := [f_0(P) : \cdots : f_M(P)] \in \mathbb{P}^M(\overline{\mathbb{Q}})$. Show that

$$H(P^{(d)}) = H(P)^d = H([x_0^d : \cdots : x_N^d]).$$

Solution.

[a] *Hint:* the triangle inequality reads $|t_1 + \cdots + t_N|_v \leqslant N^{\epsilon(v)} \max_i |t_i|_v$, with $\epsilon(v) = 1$ if $v$ is archimedean and $\epsilon(v) = 0$ otherwise.

## 41. Heights and ideals

This section is [Sil09, Exercise 8.9], which extends [Sil09, Remark VIII.5.5] from $\mathbb{Q}$ to a number field. The description of the non-archimedean places is [HS00, Proposition B.1.3] and the discussion before it. For a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$, let $\mathrm{ord}_{\mathfrak{p}} \colon K^\times \to \mathbb{Z}$ be the exponent of $\mathfrak{p}$ in the factorization of the fractional ideal $x\mathcal{O}_K$, and $\mathrm{Nm}\,\mathfrak{p} := \#\mathcal{O}_K/\mathfrak{p}$. The norm extends multiplicatively to fractional ideals: $\mathrm{Nm}_{K/\mathbb{Q}}\,\mathfrak{a} := \prod_{\mathfrak{p}} (\mathrm{Nm}\,\mathfrak{p})^{\mathrm{ord}_{\mathfrak{p}}\,\mathfrak{a}}$.

**Proposition 41.1** (Non-archimedean places [HS00, Proposition B.1.3])**.** *The non-archimedean places of $K$ correspond to the nonzero prime ideals of $\mathcal{O}_K$. If $v \in \mathcal{P}_K^0$ corresponds to $\mathfrak{p}$, then $|x|_v^{n_v} = (\mathrm{Nm}\,\mathfrak{p})^{-\mathrm{ord}_{\mathfrak{p}}(x)}$ for every $x \in K^\times$.*

**Problem 49** (Heights and ideals)**.**

(1) Let $x_0, \dots, x_N \in K$, not all zero, let $P := [x_0 : \cdots : x_N] \in \mathbb{P}^N(K)$, and let $\mathfrak{b}$ be the fractional ideal of $K$ generated by $x_0, \dots, x_N$. Show that

$$H_K(P) = \left(\mathrm{Nm}_{K/\mathbb{Q}}\, \mathfrak{b}\right)^{-1} \prod_{v \in \mathcal{P}_K^\infty} |P|_v^{n_v}.$$

Solution.

(2) Let $K = \mathbb{Q}$. Deduce that if $x_0, \dots, x_N$ are coprime integers, then $H_{\mathbb{Q}}(P) = \max_i |x_i|$. More generally, show that if $\mathfrak{b}$ is principal, then $P$ has coordinates in $\mathcal{O}_K$ that generate the unit ideal, and for such coordinates $H_K(P) = \prod_{v \in \mathcal{P}_K^\infty} |P|_v^{n_v}$.

Solution.

(3) Let $K := \mathbb{Q}(\sqrt{-5})$, with $a := \sqrt{-5}$, and let $P := [2 : 1 + a] \in \mathbb{P}^1(K)$. `Magma` computes both sides of (1), and compares with its own logarithmic height of $(1 + a)/2$:

```
K<a> := QuadraticField(-5);
O := Integers(K);
xs := [K | 2, 1 + a];
b := ideal<O | xs>;
IsPrincipal(b), Norm(b);                                    // false 2
arch := &*[Max([Abs(Evaluate(x, v)) : x in xs])^LocalDegree(v)
           : v in InfinitePlaces(K)];
primes := &join[Support(x*O) : x in xs | x ne 0];
fin := &*[Norm(p)^(-Min([Valuation(x, p) : x in xs])) : p in primes];
RealField(10)!arch, fin;                                    // 6.000000000 1/2
[RealField(10) | AbsoluteLogarithmicHeight((1 + a)/2), Log(3)/2];
                                          // [ 0.5493061444, 0.5493061444 ]
```

Check the output by hand: show that $\mathfrak{b}$ has norm 2 and is not principal, and compute $H_K(P)$ and $H(P)$. Show that $P$ has no coordinates in $\mathcal{O}_K$ that generate the unit ideal. Explain the last line of the output.

Solution.

## Part 13. Heights and torsion on elliptic curves

Kronecker's theorem says that the elements of $\overline{\mathbb{Q}}^\times$ of height 1 are the roots of unity, the torsion of $\mathbb{G}_{\mathrm{m}}$. This worksheet proves its analogue for an elliptic curve, for the height $h_x$ of the $x$-coordinate, assuming the duplication bound of [Sil09, Corollary VIII.6.4(b)]. The torsion points have bounded height, not height 0, and that is enough for finiteness. First come the valuations of a sum of several terms, and the shape of the coordinates of a rational point, over $\mathbb{Q}$ and over a number field: the first step of Silverman's proof of the Mordell–Weil theorem over $\mathbb{Q}$.

**Standing notation.**

- A number field $k$, elements $a, b \in k$ with $\Delta := -16(4a^3 + 27b^2) \neq 0$, and the elliptic curve $E_{a,b}\colon y^2 = x^3 + ax + b$ over $k$.
- The height $H$ on $\mathbb{P}^N(\overline{\mathbb{Q}})$, and $h := \log H$. For $t \in \mathbb{Q}$, with $t = u/w$ in lowest terms, $H([t : 1]) = \max(|u|, |w|)$. See [Sil09, §VIII.5].

- The $x$-coordinate $x\colon E_{a,b} \to \mathbb{P}^1$, with $x(O) = [1:0]$, and the height $h_x(P) \coloneqq h(x(P))$ of a point $P \in E_{a,b}(\overline{\mathbb{Q}})$. So $h_x(O) = 0$.

## 42. Valuations of sums

The proofs below compare the valuations of the terms of a sum. This section makes the ultrametric inequality precise: when it is an equality, and what happens when it is not.

**Definition 42.1** (Discrete valuations)**.** A discrete valuation on a field $F$ is a surjective map $v\colon F \to \mathbb{Z} \cup \{\infty\}$ with $v(\alpha) = \infty$ only for $\alpha = 0$, $v(\alpha\beta) = v(\alpha) + v(\beta)$ and $v(\alpha + \beta) \geqslant \min(v(\alpha), v(\beta))$. Its valuation ring is $\mathcal{O}_v \coloneqq \{\alpha \in F : v(\alpha) \geqslant 0\}$, a uniformizer is an element $\pi \in F$ with $v(\pi) = 1$, and the residue field is $\kappa \coloneqq \mathcal{O}_v/\pi\mathcal{O}_v$, with reduction $\alpha \mapsto \bar{\alpha}$. Examples are $v_p$ on $\mathbb{Q}$, and $v_{\mathfrak{p}}$ on a number field $k$, for a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_k$.

**Problem 50** (The unique minimizer criterion)**.**
Let $v$ be a discrete valuation on a field $F$, with uniformizer $\pi$ and residue field $\kappa$. Let $\alpha_1, \dots, \alpha_n \in F$, not all 0, $m \coloneqq \min_i v(\alpha_i)$, and $I \coloneqq \{i : v(\alpha_i) = m\}$.

(1) Show that $v(\alpha_1 + \cdots + \alpha_n) \geqslant m$, with equality if $I$ has one element.

Solution.

(2) Show that $v(\alpha_1 + \cdots + \alpha_n) = m$ if and only if $\sum_{i \in I} \overline{\alpha_i/\pi^m} \neq 0$ in $\kappa$. Give an example with $\#I = 2$ where the inequality is strict.

Solution.

## 43. The shape of a rational point

This section is the first step of the proof of [Sil09, Lemma VIII.4.2], which proves that $h_x$ satisfies the hypotheses of the descent theorem over $\mathbb{Q}$. It turns the rational points of bounded height into a finite search over integers. The second part is the same argument over a number field, where ideals take the place of the integer $d$. The integrality of $a, b$ is harmless: if $u^4a, u^6b$ are integral, then $(x, y) \mapsto (u^2x, u^3y)$ is an isomorphism onto $E_{u^4a,u^6b}$, and it changes $h_x$ by at most $2\log|u|$ for $u \in \mathbb{Z}$.

**Definition 43.1** (Heights over a number field [Sil09, §VIII.5])**.** For $t \in k$,

$$H([t:1])^{[k:\mathbb{Q}]} = \prod_{v \in \mathcal{P}_k} \max(1, |t|_v)^{n_v},$$

with $n_v \coloneqq [k_v : \mathbb{Q}_p]$ or $[k_v : \mathbb{R}]$. For the place of a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_k$, $|t|_v^{n_v} = N(\mathfrak{p})^{-v_{\mathfrak{p}}(t)}$, where $N(\mathfrak{p})$ is the order of $\mathcal{O}_k/\mathfrak{p}$. The denominator ideal of $t$ is $\mathfrak{d}(t) \coloneqq \prod_{\mathfrak{p}} \mathfrak{p}^{\max(0, -v_{\mathfrak{p}}(t))}$, an ideal of $\mathcal{O}_k$.

**Problem 51** (The shape of a rational point)**.**
(1) Let $a, b \in \mathbb{Z}$, and $P \in E_{a,b}(\mathbb{Q}) \smallsetminus \{O\}$. Show that $x(P) = n/d^2$ and $y(P) = m/d^3$ with $n, m, d \in \mathbb{Z}$, $d \geqslant 1$ and $\gcd(n, d) = \gcd(m, d) = 1$, and that $H(x(P)) = \max(|n|, d^2)$.[a]

Solution.

(2) Let $a, b \in \mathcal{O}_k$, and $P \in E_{a,b}(k) \smallsetminus \{O\}$. Show that there is an ideal $\mathfrak{d}$ of $\mathcal{O}_k$ with $\mathfrak{d}(x(P)) = \mathfrak{d}^2$ and $\mathfrak{d}(y(P)) = \mathfrak{d}^3$, and that

$$H(x(P))^{[k:\mathbb{Q}]} = N(\mathfrak{d})^2 \prod_{v|\infty} \max(1, |x(P)|_v)^{n_v}.$$

Check that for $k = \mathbb{Q}$ this is (1).

Solution.

[a] *Hint:* compare $v_p(x)$ and $v_p(y)$ at a prime $p$ with $v_p(x) < 0$.

## 44. Kronecker's theorem for elliptic curves

Kronecker's theorem [Sil09, Exercise 8.6] says that $\alpha \in \overline{\mathbb{Q}}^\times$ has $H(\alpha) = 1$ if and only if $\alpha$ is a root of unity. This section proves the analogue for $E_{a,b}$ and $h_x$, from the results below.

**Proposition 44.1** (The duplication formula [Sil09, Algorithm III.2.3(d)]). *Let $P \in E_{a,b}(\overline{\mathbb{Q}})$ with $2P \neq O$, and $x := x(P)$. Then*

$$x(2P) = \frac{x^4 - 2ax^2 - 8bx + a^2}{4(x^3 + ax + b)}.$$

**Proposition 44.2** (The duplication bound [Sil09, Corollary VIII.6.4(b)]). *There is a constant $C \geqslant 0$, depending on $a$ and $b$, with $|h_x(2P) - 4h_x(P)| \leqslant C$ for all $P \in E_{a,b}(\overline{\mathbb{Q}})$.*

**Theorem 44.3** (Northcott [Sil09, Theorem VIII.5.11]). *For all constants $B$ and $D$, the set of $R \in \mathbb{P}^N(\overline{\mathbb{Q}})$ with $H(R) \leqslant B$ and $[\mathbb{Q}(R) : \mathbb{Q}] \leqslant D$ is finite. Here $\mathbb{Q}(R)$ is the field of definition of $R$.*

**Problem 52** (Kronecker for elliptic curves).
Let $C$ be the constant of Proposition 44.2. Only its lower bound, $h_x(2P) \geqslant 4h_x(P) - C$, is used.

(1) Show that every torsion point $T \in E_{a,b}(\overline{\mathbb{Q}})$ has $h_x(T) \leqslant C/3$.[a]

Solution.

(2) Show that for every $D \geqslant 1$ there are only finitely many torsion points $T \in E_{a,b}(\overline{\mathbb{Q}})$ with $[\mathbb{Q}(T) : \mathbb{Q}] \leqslant D$. Deduce that the torsion subgroup of $E_{a,b}(K)$ is finite for every number field $K$ containing $k$.

Solution.

(3) Show that the naive analogue of Kronecker's theorem fails in both directions: a torsion point need not have $h_x = 0$, and a rational point with $h_x = 0$ need not be a torsion point.

(a) On $y^2 = x^3 + 1$, show that $T = (2, 3)$ is a torsion point of order 6, with $h_x(T) = \log 2$.

Solution.

(b) On $y^2 = x^3 + x + 1$, the rational point $P = (0, 1)$ has $h_x(P) = 0$. For this curve over $\mathbb{Q}$, one can take $C = \log 619$ in the lower bound $h_x(2P) \geqslant 4h_x(P) - C$. Show that $P$ has infinite order.[b]

Solution.

[a] *Hint:* the multiples of $T$ form a finite set that contains the double of each of its elements. Look at the one of largest height.
[b] *Hint:* compute $x(2P)$ and $x(4P)$.

## Part 14. The Kubert–Tate normal form and good reduction everywhere

This worksheet is about two sections of [Sil09]: torsion points (§VIII.7), and minimal equations and good reduction (§VIII.8). Mazur's theorem lists the fifteen groups that occur as the torsion subgroup of an elliptic curve over $\mathbb{Q}$. The Kubert–Tate normal form puts an elliptic curve with a $k$-point $P$ of order $m \geqslant 4$ in a family with two parameters, in which $m$ is cut out by a polynomial condition on the parameters. These conditions define curves, the modular curves $Y_1(m)$, whose $k$-points parametrize such pairs $(E, P)$. Problem 53 derives the normal form, and finds the conditions for $m = 4, 5, 6, 7, 9, 11, 12$. For $m = 11$ the curve has genus one. Problem 54 shows that every elliptic curve over $\mathbb{Q}$ has bad reduction at some prime.

**Standing notation.**

- A field $k$, and an elliptic curve $E$ over $k$ given by a Weierstrass equation
$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$
with coefficients in $k$, discriminant $\Delta$ and point at infinity $O$ [Sil09, §III.1].
- A change of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$, with $u \in k^\times$ and $r, s, t \in k$. It gives a Weierstrass equation for $E$ in $x', y'$, with discriminant $u^{-12}\Delta$. Any two Weierstrass equations for $E$ are related by a change of variables [Sil09, Proposition III.3.1(b)].
- Coordinates $x(Q)$ and $y(Q)$ of a $\overline{k}$-point $Q \neq O$ of $E$.

### 45. The Kubert–Tate normal form and Mazur's list

Problem 53 is [Sil09, Exercise 8.13]. Its item (1) is Exercise 8.13(a), the orders 6, 7, 9 and 12 are Exercise 8.13(b), and the order 11 is Exercise 8.13(c). The normal form, and the conditions for all the orders allowed by Theorem 45.1, are in [Kub76]. The elliptic curves with a $k$-point of order $m \geqslant 4$ are parametrized by the $k$-points of a modular curve [Sil09, Remark VIII.7.8].

**Theorem 45.1** (Mazur [Sil09, Theorem VIII.7.5])**.** *Let $E$ be an elliptic curve over $\mathbb{Q}$. Then $E(\mathbb{Q})_{\text{tors}}$ is isomorphic to one of the fifteen groups*
$$\mathbb{Z}/N\mathbb{Z} \text{ with } 1 \leqslant N \leqslant 10 \text{ or } N = 12, \qquad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2N\mathbb{Z} \text{ with } 1 \leqslant N \leqslant 4.$$
*Each of them occurs as $E(\mathbb{Q})_{\text{tors}}$ for some elliptic curve $E$ over $\mathbb{Q}$.*

**Proposition 45.2** (The group law [Sil09, Proposition III.2.2(a) and Algorithm III.2.3(a)])**.** *If a line in $\mathbb{P}^2$ meets $E$ at the $\overline{k}$-points $P_1, P_2, P_3$, counted with multiplicity, then $P_1 + P_2 + P_3 = O$. The negative of a $\overline{k}$-point $(x_0, y_0)$ is $(x_0, -y_0 - a_1x_0 - a_3)$.*

**Definition 45.3** (Kubert–Tate normal form)**.** A Weierstrass equation is in Kubert–Tate normal form if it is
$$y^2 + (1 - c)xy - by = x^3 - bx^2 \tag{45.1}$$
for some $b, c \in k$.

**Problem 53** (The Kubert–Tate normal form and Mazur's list)**.**
(1) Let $P \in E(k)$ be a $k$-point of order $m \geqslant 4$, possibly infinite. In three steps, show that a change of variables gives a Weierstrass equation for $E$ in Kubert–Tate normal form, in which $P = (0, 0)$. After each change of variables, write again $x, y$ for the coordinates and $a_1, \ldots, a_6$ for the coefficients.
(a) Show that a change of variables with $u = 1$ and $s = 0$ moves $P$ to $(0, 0)$, and that then $a_6 = 0$.

Solution.

(b) Suppose that $P = (0,0)$, so that $a_6 = 0$. Show that $a_3 \neq 0$, and that the change of variables with $u = 1$, $r = t = 0$ and $s = a_4/a_3$ makes $a_4 = 0$. Show that then the tangent line to $E$ at $P$ is $y = 0$.[a]

Solution.

(c) Suppose that $P = (0,0)$ and $a_4 = a_6 = 0$, with $a_3 \neq 0$. Show that $a_2 \neq 0$, and that the change of variables with $u = a_3/a_2$ and $r = s = t = 0$ gives a Weierstrass equation in Kubert–Tate normal form. Find $b$ and $c$.[b]

Solution.

(2) Conversely, suppose that $E$ is given by an equation in Kubert–Tate normal form. Let $P := (0,0) \in E(k)$, and let $m$ be its order, possibly infinite. Show that $b \neq 0$, that $2P = (b, bc)$ and $3P = (c, b-c)$, and that $m \geqslant 4$.[c]

Solution.

(3) With $E$ and $P$ as in (2), suppose that a change of variables gives another equation for $E$ in Kubert–Tate normal form, with parameters $b'$ and $c'$, in which $P$ is again $(0,0)$. Show that $b' = b$ and $c' = c$. So the pair $(E, P)$ determines $b$ and $c$.[d]

Solution.

(4) With $E$, $P$ and $m$ as in (2), show that $m = 4$ if and only if $c = 0$, and $m = 5$ if and only if $b = c$.

Solution.

(5) With $E$, $P$ and $m$ as in (2), find the condition on $b$ and $c$ for $m = 6$.

Solution.

(6) With $E$, $P$ and $m$ as in (2), suppose that $m \geqslant 5$, so that $c \neq 0$, and put $d := b/c$. Show that[e]

$$4P = \big(d^2 - d,\ d^2(c - d + 1)\big).$$

Solution.

(7) With $m \geqslant 5$ and $d$ as in (6), show that $m = 7$ if and only if $P$, $2P$ and $4P$ are collinear, and that this holds if and only if $c = d^2 - d$. Deduce a one-parameter family of elliptic curves with a $k$-point of order 7.

Solution.

(8) With $d$ as in (6), suppose that $m \geqslant 6$, so that $d \neq 1$, and put $f := c/(d-1)$. Show that $m = 9$ if and only if $2P$, $3P$ and $4P$ are collinear, and that this holds if and only if

$$c^2 - (d-1)c - (d-1)^3 = 0, \quad \text{that is,} \quad d = f^2 - f + 1.$$

Deduce a one-parameter family of elliptic curves with a $k$-point of order 9.[f]

Solution.

(9) With $d$ and $f$ as in (8), suppose that $m \geqslant 9$, so that $5P, \ldots, 8P$ are not $O$. The following code takes $b = cd$, works over the field $\mathbb{Q}(c,d)$ of rational functions, and factors the numerators of $x(4P) - x(8P)$ and $x(5P) - x(6P)$ after the substitution $c = f(d-1)$.

```
K<c, d> := FunctionField(Rationals(), 2);
E := EllipticCurve([1 - c, -c*d, -c*d, 0, 0]);    // b = c*d
P := E![0, 0];
```

```
4*P;                    // (d^2 - d : c*d^2 - d^3 + d^2 : 1)
R<f, d> := PolynomialRing(Rationals(), 2);
// the irreducible factors of the numerator of x(iP) - x(jP), with c = f(d - 1)
Factors := func<i, j | [g[1] : g in Factorization(
  Evaluate(Numerator((i*P)[1] - (j*P)[1]), [f*(d - 1), d]))]>;
Factors(4, 8);
// [ d, d - 1, f - 1, f^2 - f*d^2 - f*d + 3*d^2 - 3*d + 1 ]
Factors(5, 6);
// [ d - 1, f, f^3*d - 3*f^2*d + 4*f*d - f - d^2 ]
```

SOLUTION.

(a) Show that $m = 12$ if and only if

$$f^2 - (d^2 + d)f + 3d^2 - 3d + 1 = 0.$$

Suppose that $\operatorname{char} k \neq 2$. Find a one-parameter family of elliptic curves with a $k$-point of order 12.[g]

SOLUTION.

(b) Show that $m = 11$ if and only if

$$d^2 - (f^3 - 3f^2 + 4f)d + f = 0.$$

Suppose that $\operatorname{char} k \notin \{2, 11\}$. Show that this curve is birational to the curve

$$w^2 = f(f^3 - 4f^2 + 8f - 4),$$

and that the latter has genus one.[h]

SOLUTION.

[a] *Hint:* the tangent line to $E$ at $P$ is $a_3 y = a_4 x$. If $a_3 = 0$, it passes through $O$.
[b] *Hint:* if $a_2 = 0$, the tangent line $y = 0$ meets $E$ only at $P$.
[c] *Hint:* find the tangent line to $E$ at $P$, and the line through $P$ and $2P$. Use Proposition 45.2.
[d] *Hint:* the change of variables takes the $\overline{k}$-point $(x', y') = (0, 0)$ to $(x, y) = (r, t)$. For $r = t = 0$, compute the new coefficients $a_1', \ldots, a_4'$.
[e] *Hint:* the line through $P$ and $3P$ is $y = (d - 1)x$.
[f] *Hint:* the line through $2P$ and $3P$ has slope $(cd - d + 1)/(d - 1)$.
[g] *Hint:* as a polynomial in $f$, its discriminant is $(d - 1)^2(d^2 + 4d - 4)$. Parametrize the conic $(d + 2)^2 - w^2 = 8$ by $d + 2 - w = 2v$.
[h] *Hint:* complete the square in $d$. The discriminant of $f^3 - 4f^2 + 8f - 4$ is $-176$.

## 46. No good reduction everywhere

This section is [Sil09, Exercise 8.15], and follows its hint. A Weierstrass equation over $\mathbb{Z}$ with $\Delta = \pm 1$ would have good reduction at every prime, and congruences modulo 8 rule it out.

In the following definitions and results, $E$ is an elliptic curve over $\mathbb{Q}$ and $p$ is a prime number, with valuation $v_p$. Write $\mathbb{Z}_{(p)}$ for the ring of rational numbers whose denominators are prime to $p$.

**Definition 46.1** (Minimal equations [Sil09, §VII.1])**.** A Weierstrass equation for $E$ is minimal at $p$ if its coefficients lie in $\mathbb{Z}_{(p)}$ and $v_p(\Delta)$ is as small as possible among such equations for $E$.

**Proposition 46.2** (Good reduction [Sil09, Proposition VII.5.1(a)])**.** *The curve $E$ has good reduction at $p$ if and only if a Weierstrass equation for $E$ minimal at $p$ has $v_p(\Delta) = 0$.*

**Corollary 46.3** (Global minimal equations [Sil09, Corollary VIII.8.3])**.** *The curve $E$ has a Weierstrass equation with coefficients in $\mathbb{Z}$ that is minimal at every prime $p$.*

**Proposition 46.4** (Weierstrass formulas [Sil09, §III.1]). *Put*

$$b_2 = a_1^2 + 4a_2, \qquad b_4 = 2a_4 + a_1a_3, \qquad b_6 = a_3^2 + 4a_6,$$
$$b_8 = a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2, \qquad c_4 = b_2^2 - 24b_4, \qquad c_6 = -b_2^3 + 36b_2b_4 - 216b_6.$$

*Then* $\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6$, *and* $1728\Delta = c_4^3 - c_6^2$.

**Problem 54** (No good reduction everywhere).
Let $E$ be an elliptic curve over $\mathbb{Q}$. Suppose, for a contradiction, that $E$ has good reduction at every prime $p$.

(1) Show that $E$ has a Weierstrass equation with coefficients in $\mathbb{Z}$ and $\Delta = \pm 1$. Fix such an equation, and put $\varepsilon := \Delta$.

SOLUTION.

(2) Show that $a_1$ is odd.[a]

SOLUTION.

(3) Show that $c_4 \equiv 1 \pmod 8$.

SOLUTION.

(4) Put $u := c_4 - 12\varepsilon$. Show that

$$c_6^2 = u\,(u^2 + 36\varepsilon u + 432),$$

and deduce that $u$ is a positive square or three times a positive square.[b]

SOLUTION.

(5) Show that $u \equiv 5 \pmod 8$, and conclude that $E$ has bad reduction at some prime $p$.

SOLUTION.

[a] *Hint:* if $a_1$ is even, show that $\Delta \equiv -27b_6^2 \pmod 8$.
[b] *Hint:* the integer $u$ is odd, and every common divisor of the two factors divides $432 = 2^4 \cdot 3^3$.

## Part 15. The canonical height and the regulator

In this worksheet we see how the canonical height $\hat{h}$ on an elliptic curve $E$ allows us to think about the Mordell–Weil group $E(\mathbb{Q})$ as a lattice $\Lambda$ in an euclidean space $V$. As a nice application of this, we have Néron's count of the rational points of bounded height. We use `Magma` to double check our computations.

### Standing notation.

- An elliptic curve $E$ over $\mathbb{Q}$, given by a Weierstrass equation.
- The height $h_x(P) := \log H(x(P))$ of a rational point $P \neq O$, where $H(u/w) := \max(|u|, |w|)$ for $u/w$ in lowest terms, and $h_x(O) := 0$. `Magma` calls it `NaiveHeight`.
- The canonical height $\hat{h}(P) := \frac{1}{2}\lim_{n\to\infty} 4^{-n}h_x(2^nP)$ of a rational point $P$, and the height pairing $\langle P, Q\rangle := \hat{h}(P+Q) - \hat{h}(P) - \hat{h}(Q)$, so that $\langle P, P\rangle = 2\hat{h}(P)$ [Sil09, §VIII.9]. The limit exists by Problem 55(2). `Magma` computes them as `CanonicalHeight(P)`, which is $2\hat{h}(P)$, not $\hat{h}(P)$, and `HeightPairing(P, Q)`.

## 47. Tate's construction of the canonical height

This section is Tate's construction of the canonical height [Sil09, Proposition VIII.9.1], first on an abelian group, as in [HS00, Theorem B.4.1], then on the rational points of an elliptic curve. Problem 55 asks for the proofs, which need only the following ingredients.

**Definition 47.1** (Quadratic heights)**.** Let $A$ be an abelian group. A height on $A$ is a function $h\colon A \to \mathbb{R}$. Two heights satisfy $h = h' + O(1)$ if $|h - h'|$ is bounded on $A$. A height $h$ is quasi-quadratic if

$$h(P+Q) + h(P-Q) = 2h(P) + 2h(Q) + O(1),$$

that is, if the difference of the two sides is bounded on $A \times A$. It is quadratic if the two sides are equal for all $P, Q \in A$ (the parallelogram law).

**Definition 47.2** (Heights on an elliptic curve [Sil09, §VIII.6])**.** Let $K$ be a number field, $h$ the absolute logarithmic height on $\mathbb{P}^1(K)$ [Sil09, §VIII.5], $E$ an elliptic curve over $K$, and $f \in K(E)$ a nonconstant function. The height relative to $f$ is the height $h_f(P) := h(f(P))$ on $E(K)$. The function $f$ is even if $f(-P) = f(P)$ for all $P$, as $x$ is. Over $\mathbb{Q}$, $h_x$ is the height of the Standing notation.

**Theorem 47.3** (Quasi-parallelogram law [Sil09, Theorem VIII.6.2])**.** *In Definition 47.2, if $f$ is even, then $h_f$ is a quasi-quadratic height on $E(K)$.*

**Lemma 47.4** (Comparison of heights [Sil09, Lemma VIII.6.3])**.** *In Definition 47.2, let $f, g \in K(E)$ be nonconstant even functions. Then $(\deg g)\, h_f = (\deg f)\, h_g + O(1)$ on $E(K)$.*

**Problem 55** (Tate's construction)**.**

(1) Let $A$ be an abelian group, and $h$ a quasi-quadratic height on $A$. Prove that for every $P \in A$ the limit

$$\hat{h}(P) := \lim_{n\to\infty} 4^{-n} h(2^n P)$$

exists, and that $\hat{h}$ is the unique quadratic height on $A$ with $\hat{h} = h + O(1)$.

Solution.

(2) Let $K$, $E$ and $f$ be as in Definition 47.2, with $f$ even. Deduce that for every $P \in E(K)$ the limit

$$\frac{1}{\deg f} \lim_{N\to\infty} 4^{-N} h_f(2^N P)$$

exists, that it is a quadratic height on $E(K)$, and that it does not depend on $f$.

Solution.

## 48. The lattice of rational points

This section draws [Sil09, Proposition VIII.9.6] for the curve $y^2 = x^3 + 17$ of [Sil09, Example III.2.4], after [HS00, §B.5]. The rational points $P_1$ and $P_3$ of Problem 56 carry Silverman's names. He notes that they generate the Mordell–Weil group, and that this is true but not so easy to prove. Problem 56 proves it with `Magma`'s `CPSHeightBounds`, which returns real numbers $l$ and $u$ with $l \leqslant h_x(P) - 2\hat{h}(P) \leqslant u$ for every rational point $P$ [BCP97]. First, two facts about lattices in general.

**Definition 48.1** (Lattices and covolume [HS00, §B.5])**.** A lattice in a real vector space $V$ of dimension $r$ is a subgroup $\Lambda$ generated by a basis $\lambda_1, \dots, \lambda_r$ of $V$. If $V$ is Euclidean, the covolume $\operatorname{covol}(\Lambda)$ is the volume of $\{t_1\lambda_1 + \cdots + t_r\lambda_r : 0 \leqslant t_i < 1\}$.

**Proposition 48.2** (The covolume [HS00, Exercise B.14(b)]). *In Definition 48.1,* $\operatorname{covol}(\Lambda)^2 = \det(\lambda_i \cdot \lambda_j)_{i,j}$.

From now on, $V$ and $\Lambda$ denote the following.

**Proposition 48.3** (The height lattice [Sil09, Proposition VIII.9.6]). *Let* $V \coloneqq E(\mathbb{Q}) \otimes_{\mathbb{Z}} \mathbb{R}$, *let* $v_P \coloneqq P \otimes 1$ *for a rational point* $P$, *and let* $\Lambda$ *be the image of* $E(\mathbb{Q})$ *in* $V$. *The pairing extends to an inner product on* $V$, *with* $|v_P|^2 = 2\hat{h}(P)$. *Then* $\Lambda$ *is a lattice in* $V$, *and the kernel of* $E(\mathbb{Q}) \to \Lambda$ *is* $E(\mathbb{Q})_{\mathrm{tors}}$.

**Definition 48.4** (The elliptic regulator [Sil09, §VIII.9]). Let $r$ be the rank of $E$, and $Q_1, \dots, Q_r$ rational points whose images form a basis of $\Lambda$. The elliptic regulator of $E$ is $R_E \coloneqq \det\big(\langle Q_i, Q_j\rangle\big)_{1\leqslant i,j\leqslant r}$. By Proposition 48.2, $\sqrt{R_E} = \operatorname{covol}(\Lambda)$ is the covolume of $\Lambda$ in $V$.

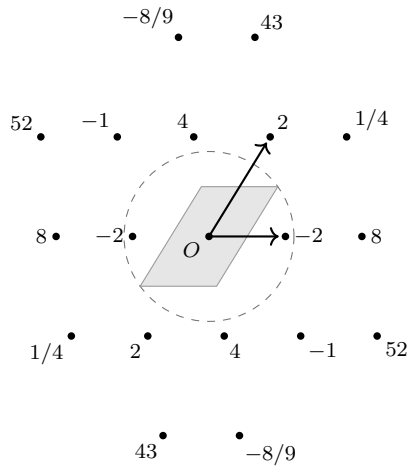



FIGURE 1. The rational points of the curve $E$ of Problem 56 near the origin of the plane $V$. Each dot is the vector $v_P$ of a rational point $P = aP_1 + bP_3$ with $2\hat{h}(P) \leqslant 3.4$, labeled by its $x$-coordinate. Since $|v_P|^2 = 2\hat{h}(P)$, the labels grow more complicated outward: integers near $O$, and the first denominators, $1/4$ and $-8/9$, farther out. The rational points $P$ and $-P$ have the same $x$-coordinate and sit opposite each other. The arrows are $v_{P_1}$ and $v_{P_3}$. The shaded parallelogram $\{sv_{P_1} + tv_{P_3} : |s|, |t| \leqslant \frac{1}{2}\}$ has area $\sqrt{\det G} = 0.592076$, and every vector of $V$ is a vector of $\mathbb{Z}v_{P_1} + \mathbb{Z}v_{P_3}$ plus a vector in it. The dashed circle $2\hat{h} = 0.561255$ passes through its two farthest corners $\pm\frac{1}{2}(v_{P_1} + v_{P_3})$, so it contains the parallelogram.

**Problem 56** (Two rational points generate $E(\mathbb{Q})$).
Let $E\colon y^2 = x^3 + 17$ over $\mathbb{Q}$, with LMFDB label 10404.f2, and let $P_1 = (-2, 3)$ and $P_3 = (2, 5)$ be two of its rational points. Let $g_{ij} \coloneqq \langle P_i, P_j\rangle$ for $i, j \in \{1, 3\}$, and let $G \coloneqq (g_{ij})$ be the Gram matrix of $P_1$ and $P_3$. `Magma` computes the rank and the torsion of $E$, the matrix $G$ and its determinant, the bounds $l$ and $u$, and the $x$-coordinates of the rational points $P \neq O$ with $H(x(P)) \leqslant 5$:

```
R := RealField(6);
E := EllipticCurve([0, 0, 0, 0, 17]);
Rank(E); #TorsionSubgroup(E);                    // 2 true
```

```
                                                    // 1
P1 := E![-2, 3];  P3 := E![2, 5];
G := HeightPairingMatrix([P1, P3]);  ChangeRing(G, R);
// [0.454617 0.363961]
// [0.363961 1.06248]
R!Determinant(G), R!Regulator(E);                   // 0.350554 0.350554
l, u := CPSHeightBounds(E);  R!l, R!u;              // -1.63999 1.18106
X := {a/d : a in [-5..5], d in [1..5] | GCD(a, d) eq 1 and IsPoint(E, a/d)};
X;                                                  // { -2, -1, 1/4, 2, 4 }
X subset {(m*P1 + n*P3)[1] : m, n in [-2..2] | [m, n] ne [0, 0]};   // true
```

The items below show that $P_1$ and $P_3$ generate $E(\mathbb{Q})$.

(1) Show that $v_{P_1}$ and $v_{P_3}$ form a basis of $V$.

Solution.

(2) Let $P$ be a rational point. Show that there are $m, n \in \mathbb{Z}$ such that $Q := P - mP_1 - nP_3$ has $v_Q$ in the shaded parallelogram of Fig. 1. Show that $Q \in \mathbb{Z}P_1 + \mathbb{Z}P_3$ if and only if $P \in \mathbb{Z}P_1 + \mathbb{Z}P_3$.[a]

Solution.

(3) Show that every rational point $Q$ with $v_Q$ in the shaded parallelogram has

$$2\hat{h}(Q) \leqslant \tfrac{1}{4}\max(g_{11} + 2g_{13} + g_{33},\ g_{11} - 2g_{13} + g_{33}),$$

and deduce that $H(x(Q)) \leqslant 5$.[b]

Solution.

(4) Conclude that $P_1$ and $P_3$ generate $E(\mathbb{Q})$.

Solution.

[a] *Hint:* write $v_P$ in the basis of (1), and round its coordinates.

[b] *Hint:* the function $v \mapsto |v|^2$ is convex, so on a parallelogram it is largest at a corner.

## 49. Néron's counting theorem

This section is Néron's counting theorem [HS00, Theorem B.6.3] for the curve of Problem 56, with its constant, through the count of lattice points in ellipsoids [HS00, Lemma B.6.4].

**Lemma 49.1** (Lattice points in ellipsoids [HS00, Lemma B.6.4])**.** *Let $r$ be the rank of $E$, suppose that $r \geqslant 1$, and let $q$ be a positive definite quadratic form on $V$. Then, as $T \to \infty$,*

$$\#\{\lambda \in \Lambda : q(\lambda) \leqslant T\} = \frac{\mathrm{vol}\{v \in V : q(v) \leqslant 1\}}{\mathrm{covol}(\Lambda)}\, T^{r/2} + O\big(T^{(r-1)/2}\big).$$

**Theorem 49.2** (Néron's counting theorem [HS00, Theorem B.6.3])**.** *Suppose that $E$ has rank $r \geqslant 1$. There is a constant $c > 0$ such that, as $T \to \infty$,*

$$\#\{P \in E(\mathbb{Q}) : h_x(P) \leqslant T\} = c\,T^{r/2} + O\big(T^{(r-1)/2}\big).$$

**Problem 57** (Counting rational points of bounded height)**.**
Let $E$, $P_1$, $P_3$, $G$, $l$ and $u$ be as in Problem 56. Continuing its listing, Magma counts the rational points with $h_x \leqslant T$:

```
Gi := G^-1;
```

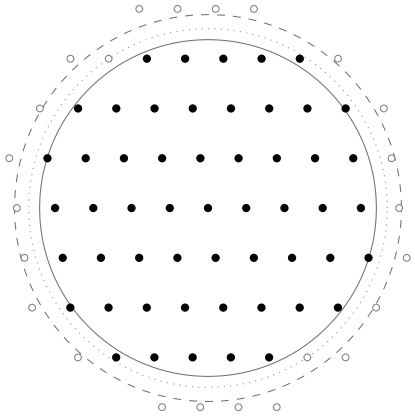


FIGURE 2. The rational points of the curve $E$ of Problem 56 counted by Néron's theorem for $T = 10$. Each dot is the vector $v_P$ of a rational point $P$ with $|v_P|^2 = 2\hat{h}(P) \leqslant 14$. It is filled if $h_x(P) \leqslant 10$, and open otherwise. There are 53 filled dots. If $h_x$ were equal to $2\hat{h}$, the filled dots would be exactly those inside the dotted circle $2\hat{h} = 10$. Instead, $h_x - 2\hat{h}$ is only known to lie between $l = -1.63999$ and $u = 1.18106$. So every rational point inside the solid circle $2\hat{h} = 10 - u$ has $h_x \leqslant 10$: all 49 dots there are filled. No rational point outside the dashed circle $2\hat{h} = 10 - l$ has $h_x \leqslant 10$: all dots there are open. In the ring between the two circles, the answer depends on $h_x - 2\hat{h}$, which has no pattern: 4 of its 10 dots are filled. The ring has area $\pi(u - l)$ for every $T$, while the disc inside it has area about $\pi T$. So, for large $T$, the count is about the area of the disc divided by the area $\sqrt{R_E}$ of a fundamental parallelogram.

```
count := function(T)
  N := [Floor(Sqrt((T - l)*Gi[i, i])) : i in [1, 2]];
  return #[0 : a in [-N[1]..N[1]], b in [-N[2]..N[2]] | NaiveHeight(a*P1 + b*P3) le T];
end function;
[<T, count(T)> : T in [10, 20, 50, 100]];
// [ <10, 53>, <20, 103>, <50, 263>, <100, 529> ]
```

The function `count` runs over a box that contains every $(a, b) \in \mathbb{Z}^2$ with $2\hat{h}(aP_1 + bP_3) \leqslant T - l$. The items below prove Theorem 49.2 for $E$, with $c = \pi/\sqrt{R_E}$.

(1) Show that $P \mapsto v_P$ is a bijection $E(\mathbb{Q}) \to \Lambda$, that $R_E = \det G$, and that $\operatorname{covol}(\Lambda) = \sqrt{R_E}$.

SOLUTION.

(2) Show that

$$\{P : 2\hat{h}(P) \leqslant T - u\} \subseteq \{P : h_x(P) \leqslant T\} \subseteq \{P : 2\hat{h}(P) \leqslant T - l\},$$

and check it against Fig. 2.

SOLUTION.

(3) Deduce that the number of rational points $P$ with $h_x(P) \leqslant T$ is $cT + O(\sqrt{T})$ with $c = \pi/\sqrt{R_E}$, and compare $cT$ with the output.

SOLUTION.

(4) Suppose that you did not know the rank of $E$. How would the counts reveal it?

Solution.

## Part 16. Szpiro's conjecture and the ABC conjecture

This worksheet is a short version of [Sil09, §VIII.11]. The minimal discriminant $\Delta_E$ of an elliptic curve $E$ over $\mathbb{Q}$ and its conductor $N_E$ are divisible by the same primes, the primes of bad reduction. In $N_E$ they appear to small powers, while in $\Delta_E$ a prime may appear to a large power. Szpiro's conjecture bounds $|\Delta_E|$ by a constant times $N_E^{6+\varepsilon}$. It would already imply Fermat's last theorem for all sufficiently large exponents [Sil09, Proposition VIII.11.2]. It is closely related to the ABC conjecture, which bounds the height of a rational point $[A : B : C]$ of the line $X+Y = Z$ by a power of the radical of $ABC$. Problem 58 computes, with the help of `Magma`, the minimal discriminant and the reduction types of the curve $y^2 = x(x + A)(x - B)$ attached to $A + B = C$. Problems 59 and 60 use it to show that each conjecture implies a form of the other.

**Standing notation.**

- An elliptic curve $E$ over $\mathbb{Q}$. A Weierstrass equation for $E$ has invariants $c_4$, $c_6$ and discriminant $\Delta$ [Sil09, §III.1], which lie in $\mathbb{Z}$ when its coefficients do.
- A prime $p$, with valuation $v_p$ on $\mathbb{Q}$, and the ring $\mathbb{Z}_{(p)}$ of rational numbers whose denominators are prime to $p$.

### 50. The minimal discriminant of a Frey curve

Problem 58 proves [Sil09, Lemma VIII.11.3]. Silverman bounds the scaling factor $u$ of a change of variables with two identities that write $288A^2$ and $288B^2$ in terms of $c_4$ and $c_6$. Here `Magma` finds identities in terms of $c_4$ and $\Delta$ instead, and they give both parts of the lemma at once.

**Definition 50.1** (Minimal equations [Sil09, §§VII.1, VII.5])**.** A Weierstrass equation for $E$ is minimal at $p$ if its coefficients lie in $\mathbb{Z}_{(p)}$ and $v_p(\Delta)$ is as small as possible among such equations. It is global minimal if its coefficients lie in $\mathbb{Z}$ and it is minimal at every prime $p$. The curve $E$ has good, multiplicative or additive reduction at $p$ if the reduction modulo $p$ of an equation minimal at $p$ is nonsingular, has a node, or has a cusp.

**Proposition 50.2** (Reduction types [Sil09, Remark VII.1.1 and Proposition VII.5.1])**.** *Consider a Weierstrass equation for $E$ with coefficients in $\mathbb{Z}_{(p)}$.*

*(1) If $v_p(\Delta) < 12$ or $v_p(c_4) < 4$, then the equation is minimal at $p$.*
*(2) Suppose that the equation is minimal at $p$. Then $E$ has good reduction at $p$ if $v_p(\Delta) = 0$, multiplicative reduction if $v_p(\Delta) > 0$ and $v_p(c_4) = 0$, and additive reduction if $v_p(\Delta) > 0$ and $v_p(c_4) > 0$.*

**Definition 50.3** (Minimal discriminant [Sil09, §VIII.11])**.** The curve $E$ has a global minimal Weierstrass equation [Sil09, Corollary VIII.8.3]. Its discriminant is the minimal discriminant $\Delta_E$ of $E$. It does not depend on the equation, since two global minimal equations are related by a change of variables with $u = \pm 1$ [Sil09, Proposition VII.1.3(b)].

**Proposition 50.4** (Changes of variables [Sil09, Proposition III.3.1(b) and Table 3.1])**.** *Any two Weierstrass equations for an elliptic curve over $\mathbb{Q}$ are related by a change of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$ with $u \in \mathbb{Q}^\times$ and $r, s, t \in \mathbb{Q}$. If the first has discriminant $\Delta$ and invariant $c_4$, then the second has discriminant $u^{-12}\Delta$ and invariant $u^{-4}c_4$.*

In this section, $A, B, C$ are nonzero integers with $A + B = C$ and $\gcd(A, B, C) = 1$, and $E$ is the elliptic curve

$$E\colon y^2 = x(x + A)(x - B). \tag{50.1}$$

The invariant $c_4$ and the discriminant $\Delta$ are those of this equation.

**Problem 58** (The minimal discriminant of a Frey curve)**.**
The following code works in the polynomial ring $\mathbb{Z}[A, B]$. It checks that $c_4 = 16F$ and $\Delta = 16G^2$ for the polynomials $F$ and $G$ below, and writes $A^3$ and $B^3$ as combinations of $F$ and $G$ with coefficients in $\mathbb{Z}[A, B]$.

```
P<A, B> := PolynomialRing(Integers(), 2);
// y^2 = x(x + A)(x - B), over the field of fractions of P
E := EllipticCurve([FieldOfFractions(P) | 0, A - B, 0, -A*B, 0]);
c4 := P!cInvariants(E)[1];
Delta := P!Discriminant(E);
F := A^2 + A*B + B^2;
G := A*B*(A + B);
c4 eq 16*F, Delta eq 16*G^2;     // true true
I := IdealWithFixedBasis([F, G]);
Coordinates(I, A^3);             // [ A, -1 ]
Coordinates(I, B^3);             // [ B, -1 ]
```

(1) Show that $A$, $B$, $C$ are pairwise coprime, and that $\gcd(c_4, \Delta) = 16$.[a]

Solution.

(2) Show that the minimal discriminant of $E$ satisfies $|\Delta_E| = 2^4(ABC)^2$ or $|\Delta_E| = 2^{-8}(ABC)^2$. In particular, $|\Delta_E| \geqslant 2^{-8}(ABC)^2$.[b]

Solution.

(3) Show that $E$ has multiplicative reduction at every odd prime $p$ dividing $ABC$.

Solution.

[a] *Hint:* a prime dividing $F(A, B)$ and $G(A, B)$ divides $A^3$ and $B^3$.
[b] *Hint:* let a global minimal equation come from (50.1) by a change of variables with scaling factor $u$, as in Proposition 50.4. Show that $u \in \mathbb{Z}$ and that $u^4$ divides $\gcd(c_4, \Delta)$.

## 51. From Szpiro to ABC

Problem 59 is the proof of [Sil09, Proposition VIII.11.5(a)]. It applies Conjecture 51.4 to the curve (50.1) of Problem 58, whose minimal discriminant is large and whose conductor is small. Szpiro made his conjecture in 1983, and Masser and Oesterlé proposed the ABC conjecture in 1985. Silverman's displayed definition of the conductor takes $f_p(E) = 2$ at every prime of additive reduction, which he notes is not quite correct at 2 and 3. At these two primes the correct exponent has a technical definition, sketched in Remark 51.2. The two conductors differ by a factor dividing $2^6 \cdot 3^3$, which the constants in the conjectures absorb.

**Definition 51.1** (Conductor [Sil09, §VIII.11])**.** The conductor of $E$ is

$$N_E := \prod_p p^{f_p(E)},$$

where the exponent $f_p(E)$ is 0 if $E$ has good reduction at $p$, 1 if it has multiplicative reduction at $p$, and 2 if it has additive reduction at $p$ and $p \geqslant 5$. If $E$ has additive reduction at $p = 2$ or $p = 3$, then $f_p(E)$ is defined in [Sil94, §IV.10], and $2 \leqslant f_2(E) \leqslant 8$ and $2 \leqslant f_3(E) \leqslant 5$.

**Remark 51.2** (The exponents $f_2$ and $f_3$ [Sil09, §C.16])**.** Let $p$ be a prime at which $E$ has additive reduction. The inertia group at $p$ acts on the Tate modules of $E$, and an integer $\delta_p(E) \geqslant 0$ measures the wild ramification in this action. It vanishes for $p \geqslant 5$, and $f_p(E) = 2 + \delta_p(E)$.

The exponent can also be computed directly. For any prime $p$, let $m_p$ be the number of irreducible components, ignoring multiplicities, of the special fiber of the minimal complete Néron model of $E$ at $p$. Then the Ogg–Saito formula [Sil09, Proposition C.16.2] gives

$$f_p(E) = v_p(\Delta_E) + 1 - m_p.$$

**Definition 51.3** (Height and radical)**.** Every rational point $P$ of the line $X + Y = Z$ in $\mathbb{P}^2$ has coordinates $P = [A : B : C]$ with $A, B, C \in \mathbb{Z}$ and $\gcd(A, B, C) = 1$, unique up to sign. Then $A + B = C$. The height of $P$ is

$$H(P) \coloneqq \max\bigl(|A|, |B|, |C|\bigr).$$

The radical of a nonzero integer $n$ is

$$R(n) \coloneqq \prod_{p \mid n} p.$$

**Conjecture 51.4** (Szpiro [Sil09, Conjecture VIII.11.1])**.** *For every $\varepsilon > 0$ there is a constant $\kappa_\varepsilon$ such that every elliptic curve $E$ over $\mathbb{Q}$ satisfies*

$$|\Delta_E| \leqslant \kappa_\varepsilon N_E^{6+\varepsilon}.$$

**Conjecture 51.5** (ABC [Sil09, Conjecture VIII.11.4])**.** *For every $\varepsilon > 0$ there is a constant $\lambda_\varepsilon$ with the following property. Let $P = [A : B : C]$ be a rational point of the line $X + Y = Z$ in $\mathbb{P}^2$, with $A, B, C \in \mathbb{Z}$ coprime and $ABC \neq 0$. Then*

$$H(P) \leqslant \lambda_\varepsilon R(ABC)^{1+\varepsilon}.$$

**Problem 59** (Szpiro implies ABC with exponent 3/2)**.**
Assume Conjecture 51.4. Let $P = [A : B : C]$ be a rational point of the line $X + Y = Z$ in $\mathbb{P}^2$, with $A, B, C \in \mathbb{Z}$ coprime and $ABC \neq 0$. You will show that for every $\varepsilon > 0$ there is a constant $\lambda_\varepsilon$, independent of $P$, with

$$H(P) \leqslant \lambda_\varepsilon R(ABC)^{3/2+\varepsilon}.$$

(1) Show that the absolute values of $A$, $B$, $C$, suitably ordered, are the coordinates of a rational point $P'$ of the same line with $H(P') = H(P)$, and with the same radical of the product of the coordinates. So you may assume that $0 < A \leqslant B < C$. Show that then $ABC \geqslant H(P)^2/2$.[a]

Solution.

(2) Let $E$ be the curve (50.1). Show that $N_E \leqslant 2^7 R(ABC)$.[b]

Solution.

(3) Let $\varepsilon > 0$. Use Conjecture 51.4 to show that $H(P)^4 \leqslant \mu_\varepsilon R(ABC)^{6+\varepsilon}$ for a constant $\mu_\varepsilon$ that depends only on $\varepsilon$, and conclude.

Solution.

[a] *Hint:* two of $A$, $B$, $-C$ have the same sign.
[b] *Hint:* one of $A$, $B$, $C$ is even.

## 52. From ABC to Szpiro

Problem 60 follows the proof of [Sil09, Proposition VIII.11.5(b)], which assumes that $\gcd(c_4, c_6) = 1$. The general case is [Sil09, Exercise 8.21]. The proof applies Conjecture 51.5 to the rational point $[c_4^3 : -c_6^2 : 1728\Delta]$ of the line $X + Y = Z$. The book has $\Delta$ in place of $1728\Delta$, a misprint corrected in its errata.

**Proposition 52.1** (Weierstrass formulas [Sil09, §III.1])**.** *The invariants and the discriminant of a Weierstrass equation satisfy* $1728\Delta = c_4^3 - c_6^2$.

**Problem 60** (ABC implies Szpiro)**.**
Assume Conjecture 51.5. Let $E$ be given by a global minimal Weierstrass equation, with invariants $c_4$, $c_6$ and discriminant $\Delta = \Delta_E$, and suppose that $\gcd(c_4, c_6) = 1$. You will show that for every $\delta > 0$ there is a constant $\kappa_\delta$, independent of $E$, with

$$|\Delta_E| \leqslant \kappa_\delta N_E^{6+\delta}.$$

(1) Show that $c_4 c_6 \neq 0$, and that $P := [c_4^3 : -c_6^2 : 1728\Delta]$ is a rational point of the line $X + Y = Z$ in $\mathbb{P}^2$ with coprime integer coordinates. Deduce that

$$H(P) = \max\bigl(|c_4|^3, |c_6|^2, 1728|\Delta|\bigr).$$

Solution.

(2) Show that $R(c_4^3 c_6^2 \cdot 1728\Delta) \leqslant 6|c_4 c_6| N_E$.[a]

Solution.

(3) Let $0 < \varepsilon < 1/5$. Use Conjecture 51.5 and the inequalities $|c_4| \leqslant H(P)^{1/3}$ and $|c_6| \leqslant H(P)^{1/2}$ to show that

$$H(P)^{(1-5\varepsilon)/6} \leqslant 6^{1+\varepsilon} \lambda_\varepsilon N_E^{1+\varepsilon}.$$

Deduce the statement of the problem.

Solution.

[a] *Hint:* every prime dividing $\Delta_E$ divides $N_E$.

## Part 17. Thue's equation

This worksheet works through [Sil09, Exercises 9.5 and 9.6(a)–(b)]. In 1909 Thue proved that an equation $F(x, y) = b$, where $F$ is a binary form of degree at least 3 with integer coefficients and nonzero discriminant and $b$ is a nonzero integer, has only finitely many integer solutions. His idea was to turn the equation into a problem of Diophantine approximation. A solution with $y$ large makes $x/y$ very close to a root of $F(T, 1)$, and algebraic numbers cannot be approximated too well by rational numbers. Problem 61 makes the first step quantitative: a polynomial is small only near its roots. Problem 62 proves Thue's theorem with Roth's theorem, and extends it to the $S$-integers of a number field for forms with an irreducible factor of degree at least 3. One part of each exercise is corrected or restricted, as explained at the start of each section.

**Standing notation.**

- A number field $K$, its set of places $\mathcal{P}_K$, and the subset $\mathcal{P}_K^\infty$ of archimedean places.

- For $v \in \mathcal{P}_K$, the completion $K_v$, the absolute value $|\ |_v$ on $K_v$ extending the usual absolute value on $\mathbb{Q}_p$ or $\mathbb{R}$, and the local degree $n_v \coloneqq [K_v : \mathbb{Q}_v]$, where $\mathbb{Q}_v$ is the completion of $\mathbb{Q}$ at the restriction of $v$.
- An algebraic closure $\overline{K}$ of $K$. For $v \in \mathcal{P}_K$, an algebraic closure $\overline{K_v}$ of $K_v$, to which $|\ |_v$ extends uniquely, as $K_v$ is complete, and a $K$-embedding $\iota_v \colon \overline{K} \to \overline{K_v}$. Then $|z|_v \coloneqq |\iota_v(z)|_v$ extends $|\ |_v$ to $\overline{K}$. The $\iota_v$ are arbitrary, and every statement below holds for every choice of them. For $K = \mathbb{Q}$, the field $\overline{\mathbb{Q}}$ is the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, $\iota_\infty$ is the inclusion $\overline{\mathbb{Q}} \subseteq \mathbb{C}$, and $|\ |$ is the usual absolute value.
- The height $H_K(P) \coloneqq \prod_{v \in \mathcal{P}_K} \max_i |x_i|_v^{n_v}$ of a point $P = [x_0 : \cdots : x_N]$ of $\mathbb{P}^N(K)$, which does not depend on the coordinates and satisfies $H_K(P) \geqslant 1$ [Sil09, §VIII.5]. For $x \in K$, $H_K(x) \coloneqq H_K([x : 1])$. For coprime integers $p$ and $q$ with $q \neq 0$, $H_{\mathbb{Q}}(p/q) = \max(|p|, |q|)$.
- The product formula $\prod_{v \in \mathcal{P}_K} |a|_v^{n_v} = 1$ for $a \in K^\times$ [Sil09, Product Formula VIII.5.3].

## 53. Polynomials are small only near their roots

Problem 61 is [Sil09, Exercise 9.5]. If $t$ is far from every root of a polynomial $f$, then $|f(t)|$ is not small. Quantitatively, $|f(t)|$ is at least a constant times the distance from $t$ to the nearest root, and the constant depends on how close the roots are to each other, which the discriminant controls. Silverman's last part asks for a constant that depends only on the degree of $f$ and on the projective height of its vector of coefficients. No such constant exists: dividing $f$ by an integer $N$ leaves this height unchanged, but at an archimedean place it divides $|f(t)|_v$ by $N$. So Problem 61(5) asks for a constant that depends on the affine height of the coefficients instead.

**Definition 53.1** (Discriminant)**.** Let $L$ be a field and $f = a_0 T^n + a_1 T^{n-1} + \cdots + a_n \in L[T]$ with $n \geqslant 1$ and $a_0 \neq 0$. Over an algebraic closure of $L$, $f = a_0 \prod_{i=1}^n (T - \xi_i)$. The discriminant of $f$ is

$$D(f) \coloneqq a_0^{2n-2} \prod_{i<j} (\xi_i - \xi_j)^2.$$

It is a polynomial with integer coefficients in $a_0, \ldots, a_n$, by the theorem on symmetric polynomials, since each $\xi_i$ occurs in $\prod_{i<j}(\xi_i - \xi_j)^2$ to degree at most $2n-2$. So $D(f) \in L$, and $D(f) \in \mathbb{Z}$ when $f \in \mathbb{Z}[T]$. It is nonzero if and only if the roots of $f$ are distinct.

**Definition 53.2** (Mahler measure [HS00, §B.7])**.** The Mahler measure of $f = a_0 \prod_{i=1}^n (T - \xi_i) \in \mathbb{C}[T]$, with $a_0 \neq 0$, is

$$M(f) \coloneqq |a_0| \prod_{i=1}^n \max(1, |\xi_i|).$$

**Lemma 53.3** (Mahler measure and coefficients [HS00, Lemma B.7.3.1 and the formula after it])**.** *Let $f = a_0 T^n + \cdots + a_n \in \mathbb{C}[T]$ with $a_0 \neq 0$. Then $M(f) \leqslant (n+1)^{1/2} \max_k |a_k|$.*

**Proposition 53.4** (Gauss's lemma [HS00, proof of Proposition B.7.3])**.** *Let $v$ be a non-archimedean place of $K$. For a polynomial $g$ with coefficients in $\overline{K}$, let $|g|_v$ be the maximum of the absolute values of its coefficients. Then $|gh|_v = |g|_v |h|_v$. In particular, if $f = a_0 \prod_{i=1}^n (T - \xi_i)$, then*

$$|f|_v = |a_0|_v \prod_{i=1}^n \max(1, |\xi_i|_v).$$

**Proposition 53.5** (Hadamard's inequality)**.** *For $z = (z_1, \ldots, z_m) \in \mathbb{C}^m$, let $\|z\| \coloneqq \left(\sum_k |z_k|^2\right)^{1/2}$. Let $V$ be an $m \times m$ complex matrix with rows $r_1, \ldots, r_m$. Then $|\det V| \leqslant \prod_{i=1}^m \|r_i\|$.*

**Problem 61** (Polynomials are small only near their roots)**.**
Let $f = a_0T^n + a_1T^{n-1} + \cdots + a_n \in K[T]$ be a polynomial of degree $n \geqslant 2$, with $a_0a_n \neq 0$ and with distinct roots $\xi_1, \dots, \xi_n \in \overline{K}$. So $f = a_0 \prod_{i=1}^n (T - \xi_i)$, and $D(f) \neq 0$. (Silverman allows $n = 1$, where $f(t) = a_0(t - \xi_1)$ and every statement below is immediate.)

(1) Let $v \in \mathcal{P}_K$, and let $\varepsilon_v \coloneqq 1/2$ if $v$ is archimedean and $\varepsilon_v \coloneqq 1$ if $v$ is non-archimedean. Let $t \in \overline{K}$, and let $\xi_i$ be a root of $f$ nearest to $t$, that is, $|t - \xi_i|_v = \min_j |t - \xi_j|_v$. Show that

$$|f(t)|_v \geqslant \varepsilon_v^{n-1} |f'(\xi_i)|_v |t - \xi_i|_v.$$

[a]

Solution.

(2) In this item and the next, $K = \mathbb{Q}$, $f \in \mathbb{Z}[T]$, and $|\ |$ is the usual absolute value on $\mathbb{C}$. Show that every root $\xi_i$ of $f$ satisfies

$$|f'(\xi_i)| \geqslant (n-1)^{-(n-1)/2} M(f)^{-(n-2)}.$$

[b]

Solution.

(3) Let $A \coloneqq \max(|a_0|, \dots, |a_n|)$. Show that every rational number $t$ satisfies

$$|f(t)| \geqslant \frac{\min\bigl(|t - \xi_1|, \dots, |t - \xi_n|\bigr)}{2^{n-1}(n+1)^{n-3/2} A^{n-2}}.$$

Solution.

(4) Let $S \subseteq \mathcal{P}_K$ be a finite set of places. Show that there is a constant $C_f > 0$, depending only on $f$, such that every $t \in K$ satisfies

$$\prod_{v \in S} \min\bigl(1, |f(t)|_v^{n_v}\bigr) \geqslant C_f \prod_{v \in S} \min_{1 \leqslant i \leqslant n} \min\bigl(1, |t - \xi_i|_v^{n_v}\bigr).$$

Solution.

(5) Let $H \coloneqq H_K([1 : a_0 : \cdots : a_n])$. Find a constant $C_f$ for (4) that depends only on $n$, $[K : \mathbb{Q}]$ and $H$.[c]

Solution.

[a] *Hint:* for $j \neq i$, compare $|\xi_i - \xi_j|_v$ with $|t - \xi_j|_v$.
[b] *Hint:* number the roots so that $i = 1$, write $D(f)$ as $f'(\xi_1)^2$ times the square of $a_0^{n-2} \prod_{2 \leqslant i < j \leqslant n} (\xi_i - \xi_j)$, and apply Hadamard's inequality to the Vandermonde matrix of $\xi_2, \dots, \xi_n$.
[c] *Hint:* for $v \in S$, bound $|f'(\xi_i)|_v$ from below as in (2), with Gauss's lemma in place of the Mahler measure at the non-archimedean places. Then bound $\prod_{v \in S} \min(1, |D(f)|_v^{n_v})$ from below by $H_K(D(f))^{-1}$.

## 54. Thue's equation

Problem 62 is [Sil09, Exercise 9.6(a)–(b)], with Roth's theorem in the form of [HS00, Theorem D.2.1]. If $y$ is large, then at each place of $S$ the ratio $x/y$ of a solution is close to a root of $F(T, 1)$, and this root may change from place to place. Theorem 54.3 has a single algebraic number $\xi$, but it holds for every choice of the embeddings $\iota_v$, so $\xi$ can stand for a different conjugate at each place. So Problem 62(3) assumes that $F$ has an irreducible factor of degree at least 3, whose roots are conjugate. Silverman states that part for every $F$ of degree at least 3 with nonzero discriminant.

**Definition 54.1** (Binary forms)**.** A binary form of degree $d$ over $K$ is a nonzero homogeneous polynomial $F \in K[X,Y]$ of degree $d$. Over $\overline{K}$ it is a product of linear forms,

$$F = \prod_{i=1}^{d} L_i, \qquad L_i = \beta_i X - \alpha_i Y.$$

Let $\Delta_{ij} \coloneqq \alpha_i\beta_j - \alpha_j\beta_i$. The discriminant of $F$ is $\prod_{i<j} \Delta_{ij}^2$. It does not depend on the factorization, and if $F(1,0) \neq 0$, it is the discriminant $D(f)$ of $f(T) \coloneqq F(T,1)$. It is nonzero if and only if no two of the $L_i$ are proportional.

**Definition 54.2** ($S$-integers [Sil09, proof of Proposition VIII.1.6])**.** Let $S \subseteq \mathcal{P}_K$ be a finite set containing $\mathcal{P}_K^\infty$. The ring of $S$-integers of $K$ is

$$\mathcal{O}_S \coloneqq \{a \in K : |a|_v \leqslant 1 \text{ for all } v \notin S\}.$$

**Theorem 54.3** (Roth [HS00, Theorem D.2.1])**.** *Let $S \subseteq \mathcal{P}_K$ be a finite set, $\xi \in \overline{K}$ and $\varepsilon > 0$. Then only finitely many $t \in K$ satisfy*

$$\prod_{v \in S} \min\bigl(1, |t-\xi|_v^{n_v}\bigr) \leqslant H_K(t)^{-(2+\varepsilon)}.$$

**Theorem 54.4** (Northcott [Sil09, Theorem VIII.5.11])**.** *For every $B > 0$, only finitely many $t \in K$ satisfy $H_K(t) \leqslant B$.*

**Problem 62** (Thue's equation)**.**
Let $F \in K[X,Y]$ be a binary form of degree $d \geqslant 3$ with nonzero discriminant, and let $b \in K^\times$.

(1) Let $K = \mathbb{Q}$, and suppose that $F \in \mathbb{Z}[X,Y]$ and $b \in \mathbb{Z}$. Show that $F(x,y) = b$ has only finitely many solutions $(x,y) \in \mathbb{Z}^2$.[a]

Solution.

(2) `Magma` solves Thue equations. The code below lists all the integer solutions of $x^3 - 2y^3 = b$ for $1 \leqslant b \leqslant 6$. Explain why the argument of (1) could not produce such a list.

```
R<x> := PolynomialRing(Integers());
t := Thue(x^3 - 2);                  // the form X^3 - 2*Y^3
[ <b, Solutions(t, b)> : b in [1..6] ];
// [ <1, [[-1, -1], [1, 0]]>, <2, [[0, -1]]>, <3, [[-5, -4], [1, -1]]>,
//   <4, []>, <5, []>, <6, [[2, 1]]> ]
```

Solution.

(3) Let $S \subseteq \mathcal{P}_K$ be a finite set containing $\mathcal{P}_K^\infty$, and suppose that $F$ has an irreducible factor $G \in K[X,Y]$ of degree $g \geqslant 3$. Show that $F(x,y) = b$ has only finitely many solutions $(x,y) \in \mathcal{O}_S \times \mathcal{O}_S$.[b]

Solution.

[a] *Hint:* first reach $F(1,0)F(0,1) \neq 0$ with changes of variables $(x,y) \mapsto (x, y+mx)$ and $(x,y) \mapsto (x+my, y)$, $m \in \mathbb{Z}$. Then let $f(T) = F(T,1)$, write $b = y^d f(x/y)$ for $y \neq 0$, and use Problem 61(3) and Roth's theorem.

[b] *Hint:* reduce to $F(1,0) \neq 0$ as in (1), and let $\xi_1, \dots, \xi_g$ be the roots of $G(T,1)$. For a solution with $y \neq 0$, use that $F/G$ evaluated at $(x,y)$ is a nonzero $S$-integer, after enlarging $S$, and the product formula to bound $\prod_{v\in S} \min_i \min(1, |x/y - \xi_i|_v^{n_v})$ by a constant times $H_K(x/y)^{-g}$. Then change the embeddings $\iota_v$ so that one root is the nearest at every $v \in S$.

## Part 18. Siegel's theorem

This worksheet works through two proofs of [Sil09, §§IX.3–IX.4]. An elliptic curve over a number field $K$ can have infinitely many rational points, but only finitely many of them have an $S$-integral $x$-coordinate. Siegel gave two proofs. In the first, a rational point $P$ of large height with $x(P) \in \mathcal{O}_S$ is very close to $O$ at some place of $S$, and Siegel's theorem says that rational points cannot approach $O$ that fast. Problem 63 introduces the distance functions that make this precise, and carries out the proof. The second proof works for every equation $y^2 = f(x)$, where $f$ is a polynomial of degree at least 3 with distinct roots. It turns an $S$-integral solution into a solution of $u + w = 1$ in units, and this equation has only finitely many solutions. Problem 64 carries it out, and then runs it on $y^2 = x^3 - 36x$ over $\mathbb{Z}[1/6]$. Both rest on Roth's theorem, and neither bounds the solutions.

**Standing notation.**

- A number field $K$ with algebraic closure $\overline{K}$, set of places $\mathcal{P}_K$ and set of archimedean places $\mathcal{P}_K^\infty$.
- For $v \in \mathcal{P}_K$, the completion $K_v$, the absolute value $|\ |_v$ on $K_v$ extending the usual absolute value on $\mathbb{Q}_p$ or $\mathbb{R}$, and the local degree $n_v := [K_v : \mathbb{Q}_v]$, where $\mathbb{Q}_v$ is the completion of $\mathbb{Q}$ at the restriction of $v$. If $w$ is a place of a finite extension of $K$ above $v$, then $|a|_w = |a|_v$ for $a \in K$.
- The height $H_K(a) := \prod_{v \in \mathcal{P}_K} \max(1, |a|_v)^{n_v}$ of $a \in K$, which satisfies $H_K(a) \geqslant 1$ [Sil09, §VIII.5].
- A finite set $S \subseteq \mathcal{P}_K$ containing $\mathcal{P}_K^\infty$. The ring of $S$-integers $\mathcal{O}_S := \{a \in K : |a|_v \leqslant 1 \text{ for all } v \notin S\}$ and its group of units $\mathcal{O}_S^\times$, the group of $S$-units.

### 55. Integral points and Siegel's theorem

Problem 63 is the proof of [Sil09, Corollary IX.3.2.1]. It uses the distance function $d_v$ of Definition 55.1, for a place $v$ of $K$. The distance $d_v(P, Q)$ is small when the $K_v$-point $P$ of a curve is $v$-adically close to the $K_v$-point $Q$. It depends on the choice of a function, but the statement of Theorem 55.3 does not [Sil09, Remark IX.2.1]. With the height $h_f$ of Definition 55.2, Siegel's theorem says that the rational points $P$ of an elliptic curve cannot approach $Q$ quickly, compared with their height: for every $\varepsilon > 0$, all but finitely many of them satisfy $d_v(P, Q) \geqslant e^{-\varepsilon h_f(P)}$.

**Definition 55.1** (The $v$-adic distance [Sil09, §IX.2])**.** Let $C$ be a smooth projective curve over $K$, let $v \in \mathcal{P}_K$, and let $Q \in C(K_v)$. Choose a function $t_Q \in K_v(C)$ that has a zero of order $e \geqslant 1$ at $Q$ and no other zeros. The *$v$-adic distance* from $P \in C(K_v)$ to $Q$ is

$$d_v(P, Q) := \min\bigl(|t_Q(P)|_v^{1/e}, 1\bigr),$$

with $d_v(P, Q) := 1$ if $P$ is a pole of $t_Q$.

**Definition 55.2** (Heights on an elliptic curve [Sil09, §VIII.6])**.** Let $E$ be an elliptic curve over $K$, and let $f \in K(E)$ be a nonconstant function. The *height relative to $f$* of a rational point $P \in E(K)$ is $h_f(P) := \frac{1}{[K:\mathbb{Q}]} \log H_K(f(P))$, with $h_f(P) := 0$ if $P$ is a pole of $f$. The function $f$ is *even* if $f(-P) = f(P)$ for every $P \in E(\overline{K})$.

**Theorem 55.3** (Siegel [Sil09, Theorem IX.3.1])**.** *Let $E$ be an elliptic curve over $K$ such that $E(K)$ is infinite. Let $Q \in E(\overline{K})$, let $f \in K(E)$ be a nonconstant even function, and let $v$ be a place of the field $K(Q)$, with $d_v(\cdot, Q)$ defined on $E$ as a curve over $K(Q)$. Then*

$$\lim_{\substack{P \in E(K) \\ h_f(P) \to \infty}} \frac{\log d_v(P, Q)}{h_f(P)} = 0.$$

**Problem 63** (Integral points on elliptic curves)**.**
Let $E$ be an elliptic curve over $K$, given by a Weierstrass equation with coordinate functions $x$ and $y$.

(1) Show that $t_O := 1/x$ has a zero of order 2 at $O$ and no other zeros. Deduce that, for this choice of $t_O$, every $v \in \mathcal{P}_K$ and every $P \in E(K_v) \setminus \{O\}$,

$$d_v(P, O) = \max\big(1, |x(P)|_v\big)^{-1/2}.$$

Solution.

(2) Let $P \in E(K) \setminus \{O\}$ be a rational point with $x(P) \in \mathcal{O}_S$. Show that

$$h_x(P) = \frac{1}{[K : \mathbb{Q}]} \sum_{v \in S} n_v \log \max\big(1, |x(P)|_v\big),$$

and that some $v \in S$ satisfies $h_x(P) \leqslant -2\,\#S \log d_v(P, O)$, with $d_v$ as in (1).

Solution.

(3) Show that the set $\{P \in E(K) \setminus \{O\} : x(P) \in \mathcal{O}_S\}$ is finite.[a]

Solution.

[a] *Hint:* if not, some place $v \in S$ serves in (2) for infinitely many rational points of the set.

## 56. The equation $y^2 = f(x)$ and the $S$-unit equation

Problem 64 is the proof of [Sil09, Theorem IX.4.3], Siegel's second proof of finiteness. It needs no Diophantine approximation on the curve. Over a large enough field, and after enlarging $S$, each factor $x - \alpha_i$ of $f(x)$ is a unit times a square. Differences of these factors are differences of roots of $f$, which are units, and three roots give two units with sum 1. Silverman reaches this situation with three reductions. A fourth is needed, that the roots of $f$ are $S$-integers, since the proof treats $x - \alpha_i$ as an element of $\mathcal{O}_S$.

**Theorem 56.1** (The $S$-unit equation [Sil09, Theorem IX.4.1])**.** *Let $\lambda, \mu \in K^\times$. Then the equation $\lambda u + \mu w = 1$ has only finitely many solutions $(u, w) \in \mathcal{O}_S^\times \times \mathcal{O}_S^\times$.*

**Definition 56.2** (Adapted sets of places)**.** Let $f \in K[T]$ be a polynomial of degree $d \geqslant 1$. The set $S$ is adapted to $f$ if the following conditions hold.

(1) The polynomial $f$ splits over $K$, as $f = c\prod_{i=1}^{d}(T - \alpha_i)$, with $c \in \mathcal{O}_S^\times$ and $\alpha_1, \dots, \alpha_d \in \mathcal{O}_S$.
(2) For all $i \neq j$, $\alpha_i - \alpha_j \in \mathcal{O}_S^\times$.
(3) The ring $\mathcal{O}_S$ is a principal ideal domain.

**Problem 64** (The equation $y^2 = f(x)$)**.**
Let $f \in K[T]$ be a polynomial of degree $d \geqslant 3$ with distinct roots in $\overline{K}$. A *solution* is a pair $(x, y) \in \mathcal{O}_S \times \mathcal{O}_S$ with $y^2 = f(x)$. The goal is to show that there are only finitely many solutions.

(1) Show that it is enough to prove this when $S$ is adapted to $f$.[a]

Solution.

(2) From now on, $S$ is adapted to $f$, with $c$ and $\alpha_i$ as in Definition 56.2. Let $(x, y)$ be a solution. Show that for each $i$ there are $b_i \in \mathcal{O}_S^\times$ and $z_i \in \mathcal{O}_S$ with $x - \alpha_i = b_i z_i^2$.[b]

Solution.

(3) Let $L \subseteq \overline{K}$ be the field obtained from $K$ by adjoining a square root of every element of $\mathcal{O}_S^\times$. Show that $L$ is a finite extension of $K$. Let $T$ be the set of places of $L$ above $S$, and $\mathcal{O}_T$ the ring of $T$-integers of $L$. Choose $\beta_i \in L$ with $\beta_i^2 = b_i$. Show that for all $i \neq j$,

$$\beta_i z_i + \beta_j z_j \in \mathcal{O}_T^\times \quad \text{and} \quad \beta_i z_i - \beta_j z_j \in \mathcal{O}_T^\times.$$

Solution.

(4) Let $\theta := \beta_1 z_1 - \beta_3 z_3$, which is nonzero by (3). Verify *Siegel's identities*

$$\frac{\beta_1 z_1 + \beta_2 z_2}{\theta} - \frac{\beta_2 z_2 + \beta_3 z_3}{\theta} = 1 \quad \text{and} \quad \frac{\beta_1 z_1 - \beta_2 z_2}{\theta} + \frac{\beta_2 z_2 - \beta_3 z_3}{\theta} = 1.$$

Deduce that there is a finite subset of $L$, depending only on $K$, $S$ and $f$, that contains $\theta$ for every solution $(x, y)$ and every choice of the $b_i$, $z_i$ and $\beta_i$.[c]

Solution.

(5) Show that there are only finitely many solutions. Where did the proof use that $d \geqslant 3$?

Solution.

(6) Let $K = \mathbb{Q}$, $S = \{\infty, 2, 3\}$ and $f = T^3 - 36T = T(T-6)(T+6)$, with $\alpha_1 = 0$, $\alpha_2 = 6$ and $\alpha_3 = -6$. Here $\mathcal{O}_S = \mathbb{Z}[1/6]$.

(a) Show that $S$ is adapted to $f$. Find $\mathcal{O}_S^\times$, representatives of $\mathcal{O}_S^\times / \mathcal{O}_S^{\times 2}$, and the field $L$.

Solution.

(b) For each of the solutions $(-3, 9)$, $(12, 36)$, $(18, 72)$ and $(294, 5040)$, write $x - \alpha_k = b_k z_k^2$ with $b_k \in \{\pm 1, \pm 2, \pm 3, \pm 6\}$ and $z_k$ a positive integer.

Solution.

(c) For the solution $(12, 36)$, take $\beta_1 = \sqrt{3}$, $\beta_2 = \sqrt{6}$ and $\beta_3 = \sqrt{2}$. The code below computes the terms $u$ and $w$ of the first of Siegel's identities, and their minimal polynomials over $\mathbb{Q}$. Explain how the output shows that $u, w \in \mathcal{O}_T^\times$.

```
R<t> := PolynomialRing(Rationals());
L<i, r2, r3> := NumberField([t^2 + 1, t^2 - 2, t^2 - 3] : Abs);
c1 := 2*r3;  c2 := r2*r3;  c3 := 3*r2;       // beta_k*z_k for (12, 36)
u := (c1 + c2)/(c1 - c3);
w := -(c2 + c3)/(c1 - c3);
u + w;                          // 1
MinimalPolynomial(u);           // t^4 + 8*t^3 + 2*t^2 - 8*t + 1
MinimalPolynomial(w);           // t^4 - 12*t^3 + 32*t^2 - 24*t + 4
```

Solution.

(d) The intrinsic `SIntegralPoints` returns all the rational points $P \neq O$ of the elliptic curve $y^2 = x^3 - 36x$ with $x(P) \in \mathbb{Z}[1/6]$, one for each pair $\pm P$. The code below lists their $x$-coordinates. One of them is not an integer. Find its $b_k$ and $z_k$. Could the proof above have produced this list?

```
E := EllipticCurve([-36, 0]);
[ Q[1] : Q in SIntegralPoints(E, [2, 3]) ];
// [ -6, -3, -2, 0, 6, 25/4, 12, 18, 294 ]
```

Solution.

[a] *Hint:* replace $K$ by a finite extension $K'$ and $S$ by a finite set of places of $K'$ containing the places above $S$. This only adds solutions.
[b] *Hint:* a prime element of $\mathcal{O}_S$ divides at most one of the $x - \alpha_i$.
[c] *Hint:* multiply the first terms of the two identities.

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